

FMCG ERP

Manufacturing Module Manual

Version: 1.0

Date: 5 June 2026

Scope: Recipes · BOM · Production · Planning · Shop Floor · NPD · Quality · Compliance

Chapters: 13

Field names and workflows reflect the actual ERP codebase. Screens may change as the ERP evolves.

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Manufacturing Module — Overview

Manual: FMCG ERP Manufacturing Module Manual

Audience: Production Managers, Quality Officers, Planning Managers, Shop Floor Supervisors, NPD Teams

Scope: All manufacturing-related screens, workflows, and import formats

What This Module Covers

The Manufacturing module in the FMCG ERP spans eight linked subsystems:

Subsystem	Route	Purpose
Recipes	<code>/dashboard/recipes</code>	Formulation management — ingredients, process parameters, lifecycle
BOM & Formula	<code>/dashboard/bom</code>	Bill of materials, formula versions, cost explosion
Production	<code>/dashboard/production</code>	Plans, work orders, scheduling, OEE, costing
Advanced Planning	<code>/dashboard/planning</code>	Finite capacity scheduling, MRP, MPS, Kanban
Shop Floor	<code>/dashboard/shop-floor</code>	Real-time terminal, supervisor console, downtime, handover
NPD	<code>/dashboard/npd</code>	New product development stage-gate pipeline
Quality	<code>/dashboard/quality</code>	QC inspections, certificates, allergen, complaints
Compliance	<code>/dashboard/compliance</code>	GS1 labels, barcodes, regulatory certificates

Navigation

Access Manufacturing modules via the left sidebar under the **Production** and **Quality** clusters. All modules require authentication. Tab-level access is controlled by permissions (see below).

Quick navigation paths:

- New Recipe: **Sidebar** → **Recipes** → **New Recipe**
- Production Plan: **Sidebar** → **Production** → **Plans tab** → **New Plan**
- QC Inspection: **Sidebar** → **Quality** → **Inspections tab** → **New Inspection**
- Shop Floor: **Sidebar** → **Shop Floor** → **Terminal tab**
- NPD Project: **Sidebar** → **NPD** → **New Project**

Permission Map

Permission	Grants Access To
<code>recipe.view</code>	Recipe list and detail pages
<code>recipe.create</code>	Create new recipes
<code>recipe.delete</code>	Delete DRAFT recipes
<code>bom.view</code>	BOM list and detail pages
<code>bom.create</code>	Create new BOMs
<code>production.view</code>	All production tabs (plans, orders, scheduling, OEE, etc.)
<code>planning.view</code>	All planning tabs (scenarios, schedule, MRP, MPS, etc.)
<code>npd.view</code>	NPD project list
<code>npd.create</code>	Create new NPD projects
<code>quality.view</code>	QC inspection list and detail
<code>gs1.view</code>	Compliance → GS1 & Labels tab

Users see only the tabs and buttons their permissions allow. Missing permissions show a "No access" message rather than an error.

Chapters in This Manual

1. [Recipes](#) — Header, BOM items, process parameters, status workflow
2. [Recipe Bulk Import](#) — CSV format, 3-tab import, validation
3. [BOM & Formula](#) — BOM types, lifecycle, formula editing
4. [Production Plans](#) — Plan creation, status, confirmation
5. [Work Orders & Scheduling](#) — Orders, scheduling, work centers, routing
6. [Batch & Lots](#) — Batch tracking, lot numbers, traceability
7. [QC Inspections](#) — Inspection types, test results, decisions
8. [Shop Floor Operations](#) — Terminal, supervisor, queue, downtime, handover
9. [Advanced Planning & MRP](#) — Scenarios, capacity, MRP, MPS
10. [New Product Development](#) — Stage-gate pipeline, project management
11. [OEE, Downtime & Yield](#) — OEE KPIs, downtime categories, waste
12. [Compliance & Labelling](#) — GS1, barcodes, regulatory certificates

Recipes

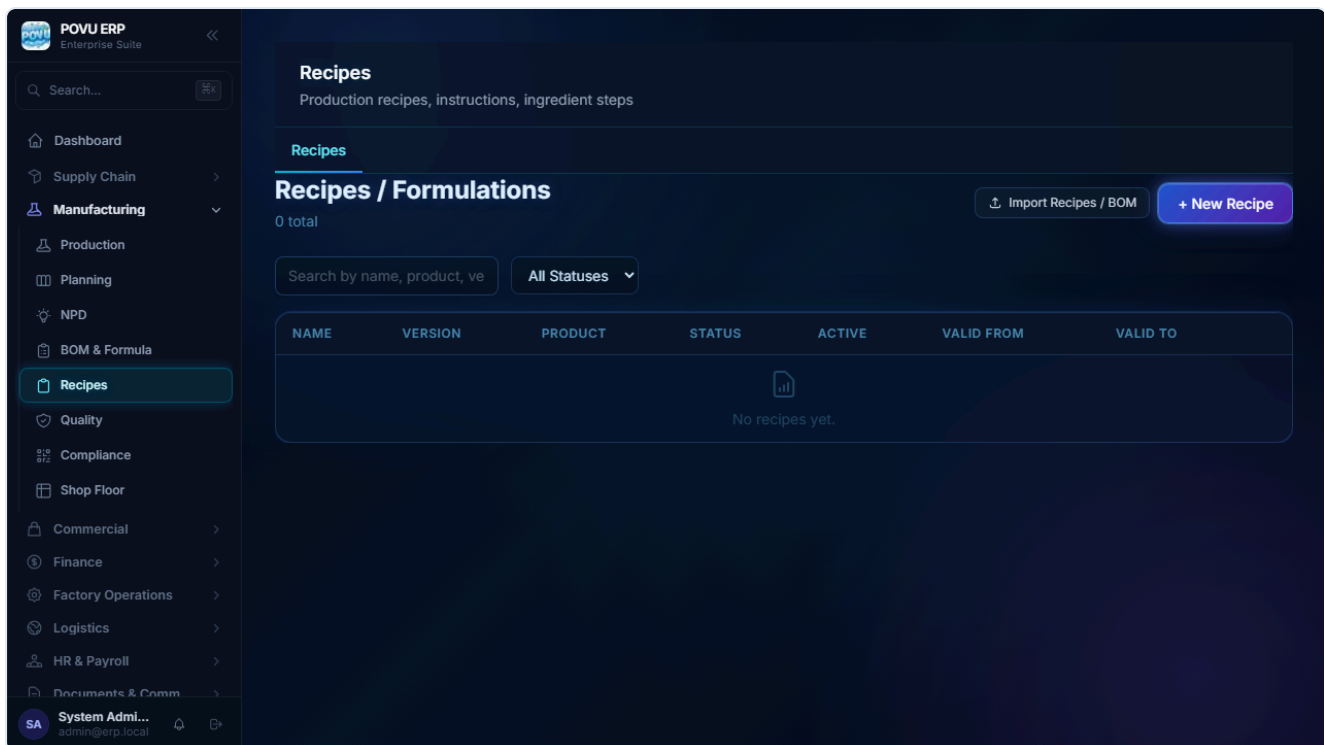
Route: `/dashboard/recipes`

Permission required: `recipe.view`

Tabs: Single-page (no workspace tabs — standalone module)

What It Does

The Recipes module manages product formulations: the exact materials (BOM items) and manufacturing process parameters (temperature, pH, mixing time, etc.) for each product version. A recipe must be **Approved** before it can be used in production work orders.



Recipes page showing recipe list with Search, Status filter, Import Recipes / BOM, and + New Recipe buttons.

Recipe List Page

Search

The search bar filters across four fields simultaneously:

- Recipe name
- Product SKU
- Product name
- Version number

Typing in the search box performs a live client-side filter. There is no server-side pagination on the default list (fetches up to the API default limit).

Table Columns

Column	Description
Name	Recipe name as a clickable link to the detail page
Version	Version string, displayed in monospace font
Product	Product SKU + product name
Status	Status badge (see Status Values below)
Active	Active/Inactive badge driven by <code>is_active</code>
Valid From	Start of validity window (date, blank if not set)
Valid To	End of validity window (date, blank if not set)
Actions	Delete button (DRAFT only, requires <code>recipe.delete</code>)

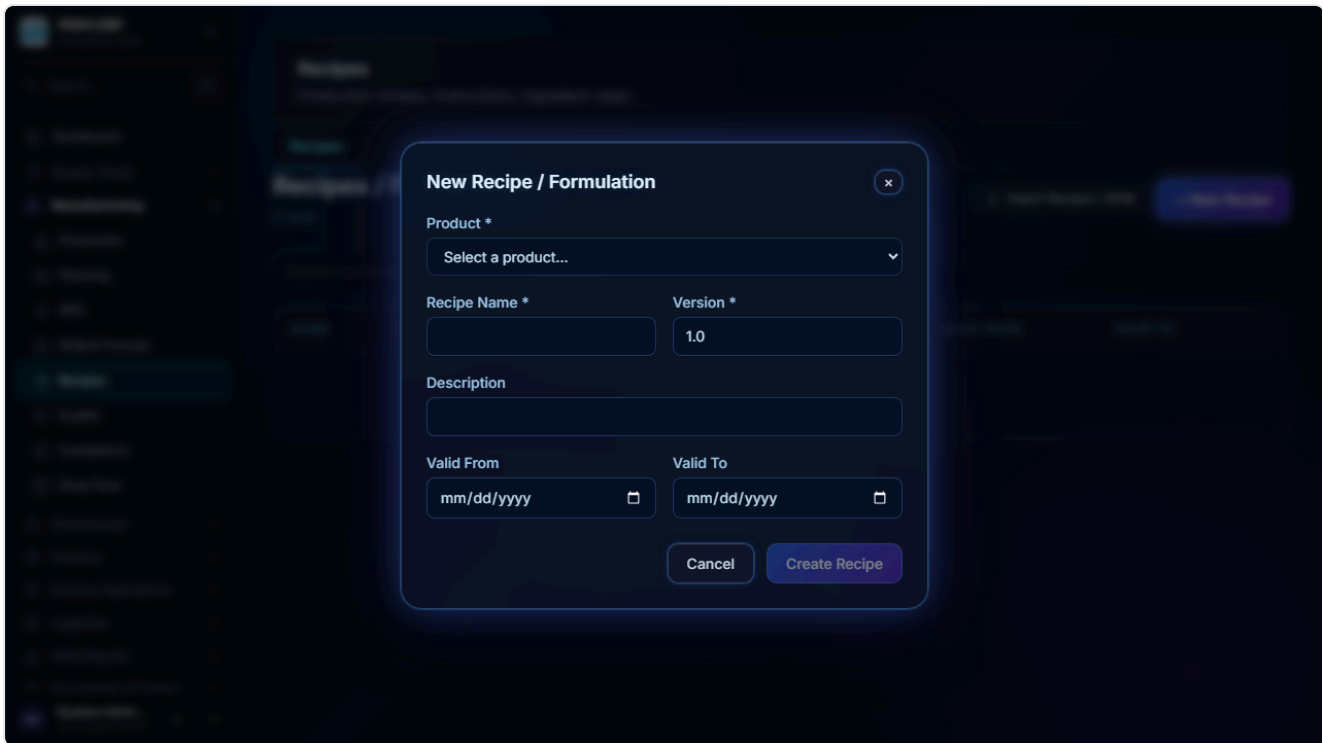
Status Values

Value	Meaning
DRAFT	Recipe under development; can be edited and deleted
APPROVED	Approved for production use; edits and deletes blocked
OBSOLETE	Retired; no longer in active use

Status transitions are managed from the recipe detail page. New recipes always start as `DRAFT` .

Creating a Recipe

Button: `+ New Recipe` (requires `recipe.create`)



New Recipe / Formulation modal opened from the + New Recipe button.

New Recipe Modal Fields

Field	Label	Required	Default	Notes
<code>product_id</code>	Product	Yes	—	Select dropdown from products list
<code>name</code>	Recipe Name	Yes	—	Free text
<code>version</code>	Version	Yes	"1.0"	e.g. "1.0", "2.0", "1.1"
<code>description</code>	Description	No	—	Free text; not shown in list
<code>valid_from</code>	Valid From	No	—	Date picker
<code>valid_to</code>	Valid To	No	—	Date picker

Submit button is disabled until both `product_id` and `name` and `version` are filled.

The recipe is always created with status `DRAFT`. The `is_active` flag defaults to `true` in the backend even though it is not exposed in the creation form.

Recipe Detail Page

Route: `/dashboard/recipes/{recipe_id}`

The detail page shows the full recipe with two sub-sections beyond the header:

BOM Items (Ingredients)

Each BOM item row maps to a `RecipeItem` record:

Backend field	Label / Column	Type	Notes
<code>material_id</code>	Material	UUID	Links to materials master
<code>line_no</code>	Line No	int	Ordering sequence; must be unique per recipe
<code>quantity</code>	Quantity	Decimal	Amount of material per batch
<code>unit</code>	Unit	str	e.g. "KG", "L", "G", "ML"
<code>loss_percentage</code>	Loss %	Decimal	Default <code>0</code> ; represents expected process loss
<code>is_optional</code>	Optional	bool	Default <code>false</code> ; marks non-mandatory ingredients
<code>alternative_group</code>	Alt Group	str or null	Groups substitutable items; null = no substitution
<code>notes</code>	Notes	str or null	Free text per line

DRAFT-only constraint: Adding, editing, or deleting BOM items is only allowed while the recipe status is `DRAFT` . The API returns HTTP 422 if attempted on APPROVED or OBSOLETE recipes.

Process Parameters (Steps)

Each step row maps to a `ProcessParameter` record:

Backend field	Label / Column	Type	Notes
<code>step_no</code>	Step No	int	Sequence; must be unique per recipe
<code>step_name</code>	Step Name	str	e.g. "Blending", "Pasteurisation"
<code>target_temperature</code>	Temp (°C)	Decimal or null	Target process temperature
<code>target_ph</code>	pH	Decimal or null	Target pH value
<code>target_viscosity</code>	Viscosity (cP)	Decimal or null	Target viscosity in centipoise
<code>mixing_time_minutes</code>	Mix Time (min)	int or null	Mixing duration
<code>rpm</code>	RPM	int or null	Mixer/agitator speed
<code>notes</code>	Notes	str or null	Free text per step

DRAFT-only constraint: Same as BOM items — process parameters can only be modified on DRAFT recipes.

Deleting a Recipe

- Delete button appears in the list table **only** if `recipe.status == "DRAFT"` AND the user has `recipe.delete` permission.
- Approved and Obsolete recipes cannot be deleted.
- Deleting a recipe also removes all its BOM items and process parameters.

Status Workflow

DRAFT → APPROVED → OBSOLETE

- `DRAFT → APPROVED` : Approve action on the detail page (requires appropriate permission).
- `APPROVED → OBSOLETE` : Obsolete action on the detail page.
- There is no reverse transition from APPROVED back to DRAFT.

Permissions Summary

Action	Permission Required
View recipe list	<code>recipe.view</code>
View recipe detail	<code>recipe.view</code>
Create recipe	<code>recipe.create</code>
Edit recipe header (DRAFT)	<code>recipe.create</code>
Add / edit BOM items (DRAFT)	<code>recipe.create</code>
Add / edit process parameters (DRAFT)	<code>recipe.create</code>
Delete recipe (DRAFT)	<code>recipe.delete</code>
Approve / Obsolete	<code>recipe.create</code> (or higher — check role config)

API Endpoints (Reference)

Method	Path	Description
GET	/api/v1/recipes/	List recipes (filters: <code>product_id</code> , <code>status</code>)
POST	/api/v1/recipes/	Create recipe (always creates as DRAFT)
GET	/api/v1/recipes/{recipe_id}	Get recipe detail with items and process params
PATCH	/api/v1/recipes/{recipe_id}	Update recipe header (DRAFT only)
DELETE	/api/v1/recipes/{recipe_id}	Delete recipe (DRAFT only)

The backend enforces DRAFT-only edits via an `_assert_draft()` guard that raises HTTP 422 with an error message if the recipe is not in DRAFT status.

Recipe Bulk CSV Import

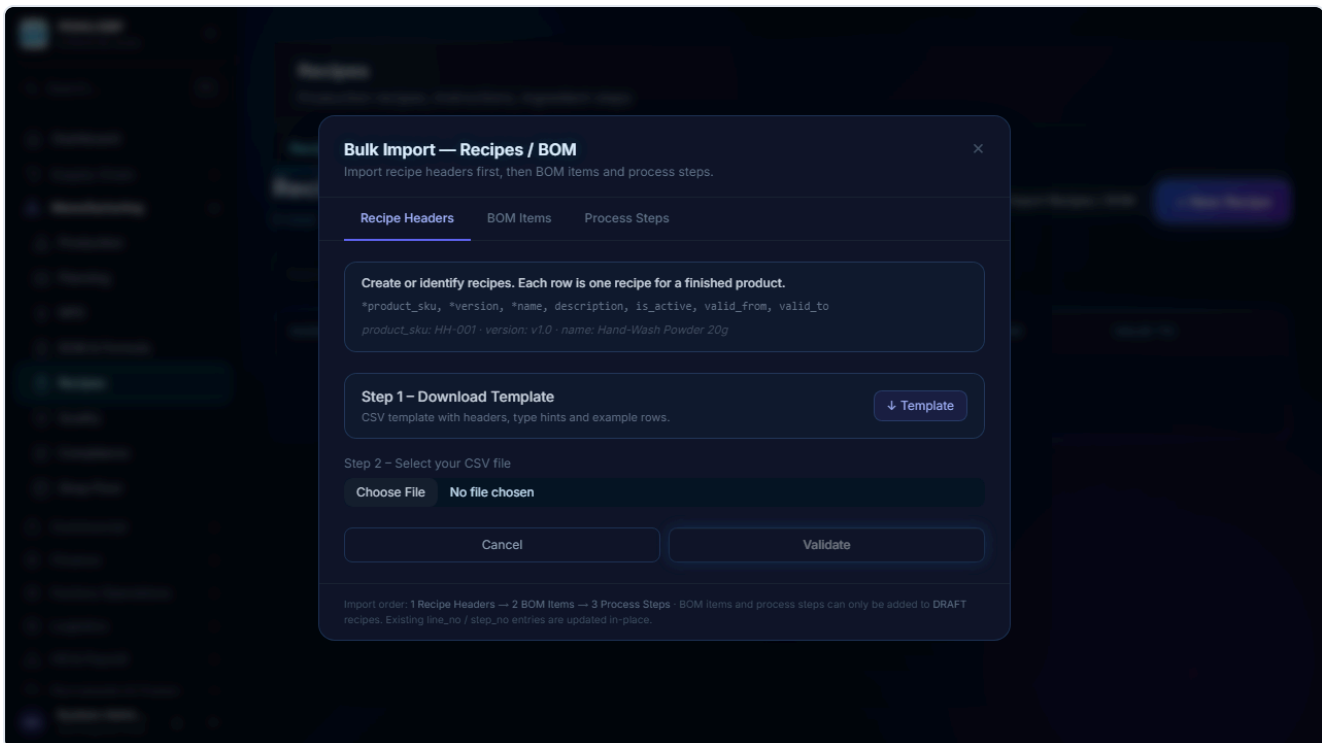
Component: RecipeBulkImportModal

Triggered from: Recipe list page → Import button

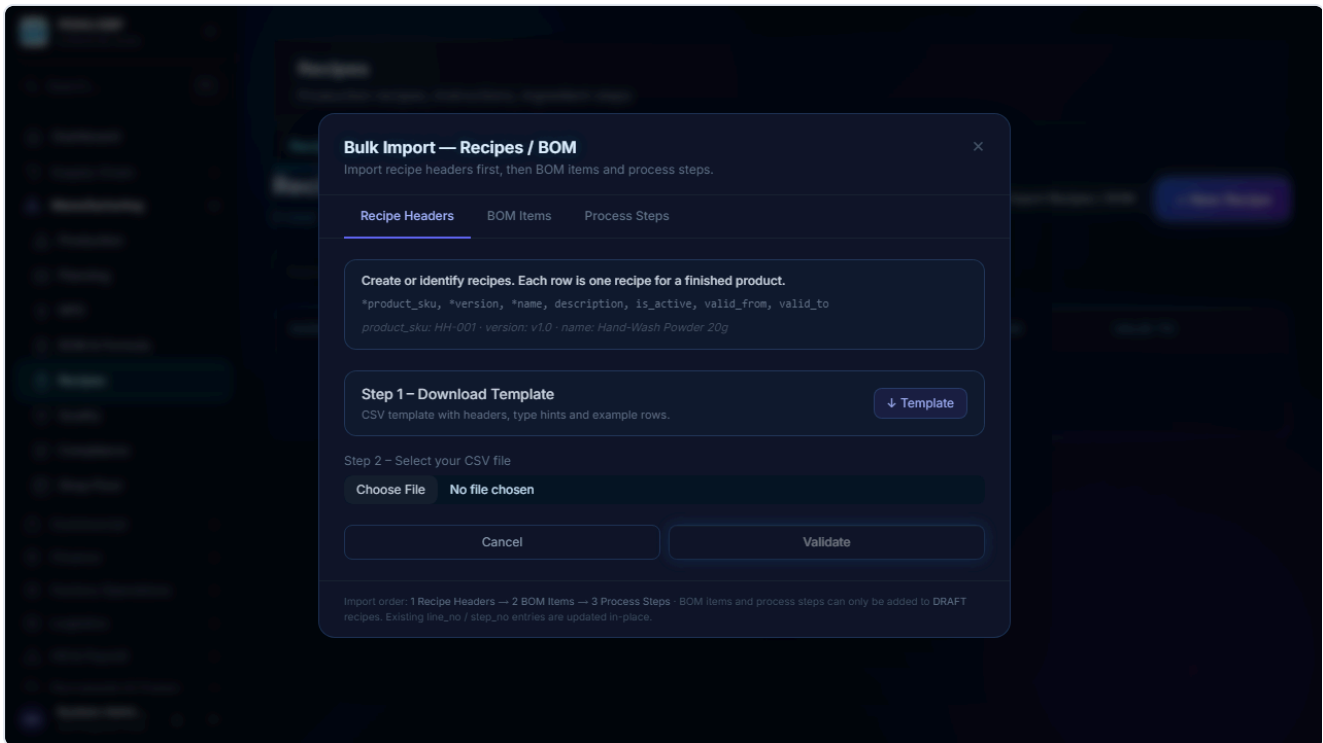
Modules involved: recipes, recipe_items, recipe_steps

Overview

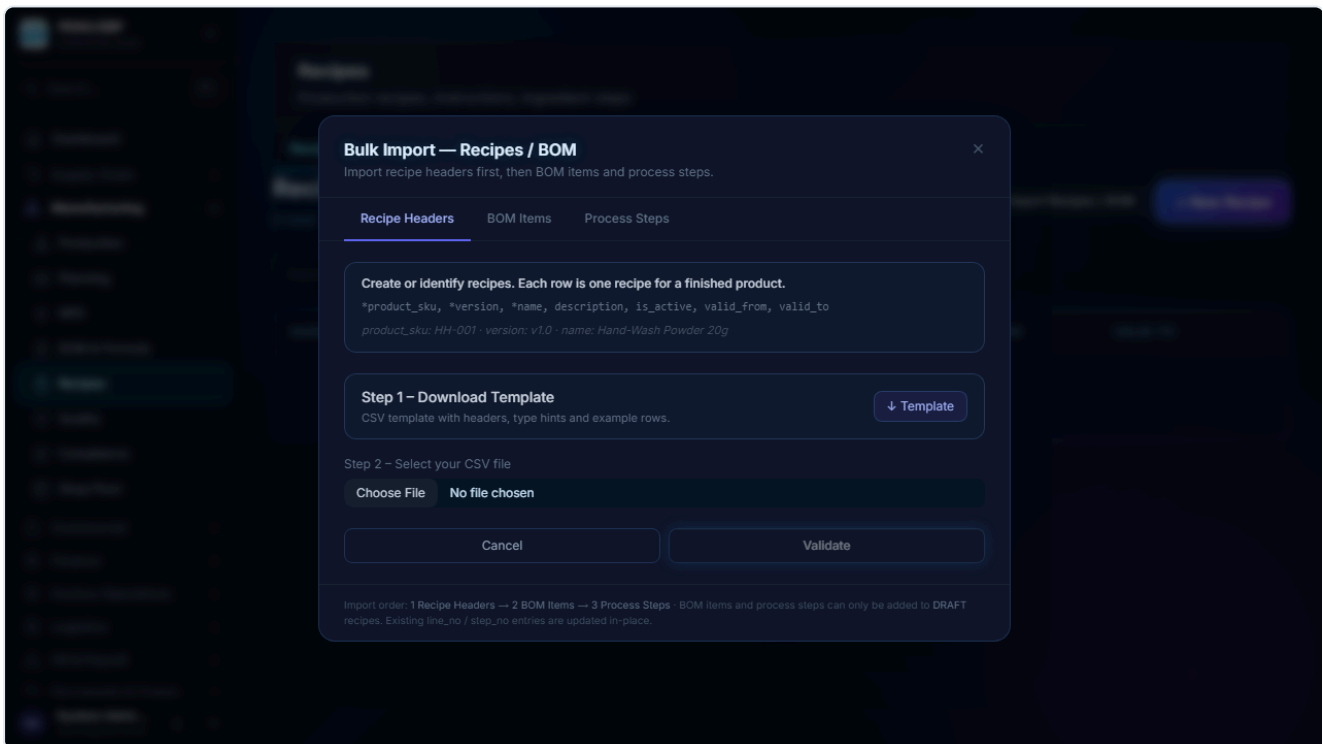
The bulk import modal allows you to upload recipe data from CSV files. It has three separate tabs — one per data type — and each tab runs an independent validate-then-import flow. You must import in the correct order:



Bulk Import — Recipe Headers tab: template download, CSV picker, Validate, and Import buttons.



Bulk Import — BOM Items tab for uploading ingredient quantities per recipe.



Bulk Import — Process Steps tab for uploading process parameters per recipe.

1. **Recipe Headers** first
2. **BOM Items** second (recipes must already exist)
3. **Process Steps** third (recipes must already exist)

Existing recipes are identified by `product_sku` + `version`. Existing BOM item `line_no` values and step `step_no` values are updated in-place if they already exist.

Tab 1: Recipe Headers

Module key: `recipes`

CSV Columns

Column	Required	Type	Notes
<code>product_sku</code>	✓ Yes (*)	string	Must match an existing product SKU in master data
<code>version</code>	✓ Yes (*)	string	e.g. <code>1.0</code> , <code>2.0</code>
<code>name</code>	✓ Yes (*)	string	Recipe name
<code>description</code>	No	string	Optional free text
<code>is_active</code>	No	<code>true / false</code>	Defaults to <code>true</code> if blank
<code>valid_from</code>	No	date (YYYY-MM-DD)	Start of validity window
<code>valid_to</code>	No	date (YYYY-MM-DD)	End of validity window

* = required column; header row must use exactly these column names.

Sample CSV

```
product_sku,version,name,description,is_active,valid_from,valid_to
SKU-001,1.0,Povu Mango Juice Standard,,true,2026-01-01,
SKU-001,2.0,Povu Mango Juice Reformulated,Lower sugar formulation,true,2026-06-01,
SKU-002,1.0,Povu Passion Juice,,true,,
```

Tab 2: BOM Items

Module key: `recipe_items`

CSV Columns

Column	Required	Type	Notes
<code>product_sku</code>	✓ Yes (*)	string	Must match an existing product SKU

Column	Required	Type	Notes
version	✓ Yes (*)	string	Must match an existing recipe version for this SKU
line_no	✓ Yes (*)	integer	Sequence number; existing line_no is updated in-place
material_code	✓ Yes (*)	string	Must match an existing material code in master data
quantity	✓ Yes (*)	decimal	Amount per batch
unit	✓ Yes (*)	string	e.g. KG , L , G , ML
loss_percent	No	decimal	Process loss %; defaults to 0
optional	No	true / false	Whether this ingredient is optional; defaults to false
alternative_group	No	string	Substitution group code; blank = not substitutable
notes	No	string	Free text per line

Constraint: The target recipe must be in `DRAFT` status. Items cannot be imported into APPROVED or OBSOLETE recipes.

Sample CSV

```
product_sku,version,line_no,material_code,quantity,unit,loss_percent,optional,alternative_group
SKU-001,1.0,10,MAT-MANGO,450,KG,2.5,false,,Fresh mango pulp
SKU-001,1.0,20,MAT-SUGAR,80,KG,0,false,,
SKU-001,1.0,30,MAT-CITRIC,2.5,KG,0,true,ACID-GRP,Can substitute with ascorbic acid
SKU-001,1.0,40,MAT-WATER,467.5,L,0,false,,Process water
```

Tab 3: Process Steps

Module key: `recipe_steps`

CSV Columns

Column	Required	Type	Notes
product_sku	✓ Yes (*)	string	Must match an existing product SKU

Column	Required	Type	Notes
version	✔ Yes (*)	string	Must match an existing recipe version for this SKU
step_no	✔ Yes (*)	integer	Step sequence; existing step_no is updated in-place
step_name	✔ Yes (*)	string	e.g. Blending , Pasteurisation , Filling
temperature_c	No	decimal	Target temperature in Celsius
target_ph	No	decimal	Target pH
viscosity_cp	No	decimal	Target viscosity in centipoise
mix_time_min	No	integer	Mixing time in minutes
rpm	No	integer	Agitator RPM
notes	No	string	Free text per step

Constraint: Same as BOM Items — recipe must be in `DRAFT` status.

Sample CSV

```
product_sku,version,step_no,step_name,temperature_c,target_ph,viscosity_cp,mix_time_min,rpm,notes
SKU-001,1.0,10,Pre-mixing,,,,,10,120,Dry blend sugars and citric
SKU-001,1.0,20,Blending,25,3.8,,20,200,Add mango pulp and water
SKU-001,1.0,30,Pasteurisation,85,,,,,0,Hold 15 min at 85°C
SKU-001,1.0,40,Cooling,25,,,,,0,Plate cooler to 25°C
SKU-001,1.0,50,Filling,,,,,0,
```

Import Flow

The same four-step flow applies to all three tabs:

1. Download Template → Get an empty CSV with correct headers
2. Pick CSV File → Browse and select your completed CSV
3. Validate → Server validates rows, returns pass/fail counts
4. Import → Confirmed import of valid (or all) rows

Validation Response

After validation, the modal shows:

- Total rows found

- Rows that passed validation
- Rows with errors (with error message per row)
- Option to download error rows as CSV for correction

Import Modes

Mode	Label	Behaviour
<code>import_valid_only</code>	Import valid rows only	Imports all rows that passed validation; skips errors
<code>strict</code>	Strict (abort on any error)	Aborts the entire import if any row has an error

Choose `strict` when data integrity is critical (e.g. costed formulas where a missing line would corrupt cost calculations).

Error Handling

- **Product SKU not found:** Row rejected; error shows the unknown SKU.
- **Material code not found:** Row rejected; error shows the unknown material code.
- **Recipe not found for SKU+version:** BOM/Steps import row rejected; import Recipe Headers first.
- **Recipe not in DRAFT:** BOM/Steps import row rejected with "Recipe is not DRAFT" message.
- **Duplicate line_no/step_no:** Existing record is updated in-place; not treated as an error.
- **Missing required field:** Row rejected with field name in error message.

Import Order Enforcement

Always import in this order:

Recipe Headers (Tab 1) → BOM Items (Tab 2) → Process Steps (Tab 3)

BOM Items and Process Steps reference the recipe by `product_sku` + `version`. If the recipe header does not exist, those rows will fail validation with "Recipe not found".

BOM & Formula Management

Route: /dashboard/bom

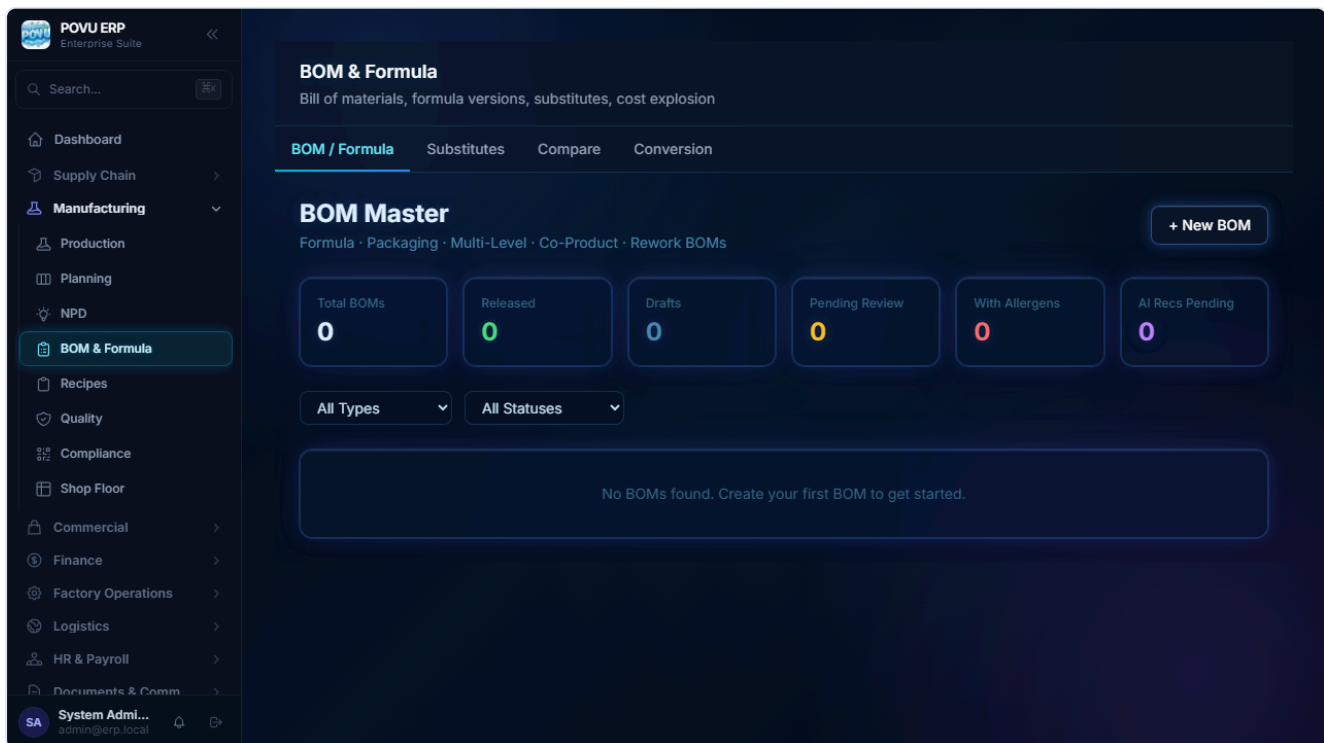
Permission required: bom.view

Workspace tabs: BOM / Formula, Substitutes, Compare, Conversion

What It Does

The BOM & Formula module manages the Bill of Materials for all manufactured products. A BOM defines the components, quantities, and structure for producing a finished good. BOMs support multiple types (formula, packaging, intermediate, etc.) and a full lifecycle from DRAFT to RELEASED.

The BOM module is separate from the Recipes module. Recipes focus on process parameters and formulation trials; BOMs focus on costing, production release, and structural explosion.



BOM workspace overview showing the formula list with lifecycle status badges.

BOM Types

Type	Use Case
FORMULA	Primary product formulation — ingredients for a finished good
INTERMEDIATE	Semi-finished goods used in another BOM
PACKAGING	Packaging materials — labels, bottles, cartons
MULTILEVEL	Complex multi-level assemblies
PHANTOM	Virtual grouping — not physically produced; explodes to its children
REWORK	Re-processing of off-spec product
COPRODUCT	By-products produced alongside the main product

BOM Lifecycle States

State	Meaning
DRAFT	Under development; editable
UNDER_REVIEW	Submitted for approval review
APPROVED	Approved; ready for release to production
RELEASED	Released to production; work orders can reference this BOM
SUPERSEDED	Replaced by a newer version; kept for traceability
ARCHIVED	No longer in use; read-only

BOM List Tab

Dashboard KPIs

KPI	Description
total_boms	Count of all BOMs
released_boms	Count in RELEASED state
draft_boms	Count in DRAFT state
pending_review	Count in UNDER_REVIEW state
boms_with_allergens	Count with allergen declarations

KPI	Description
<code>pending_ai_recs</code>	Count with pending AI optimisation recommendations

Filters

Filter	Options
Type	All Types / FORMULA / INTERMEDIATE / PACKAGING / MULTILEVEL / PHANTOM / REWORK / COPRODUCT
Status	All Statuses / DRAFT / UNDER_REVIEW / APPROVED / RELEASED / SUPERSEDED / ARCHIVED

Table Columns

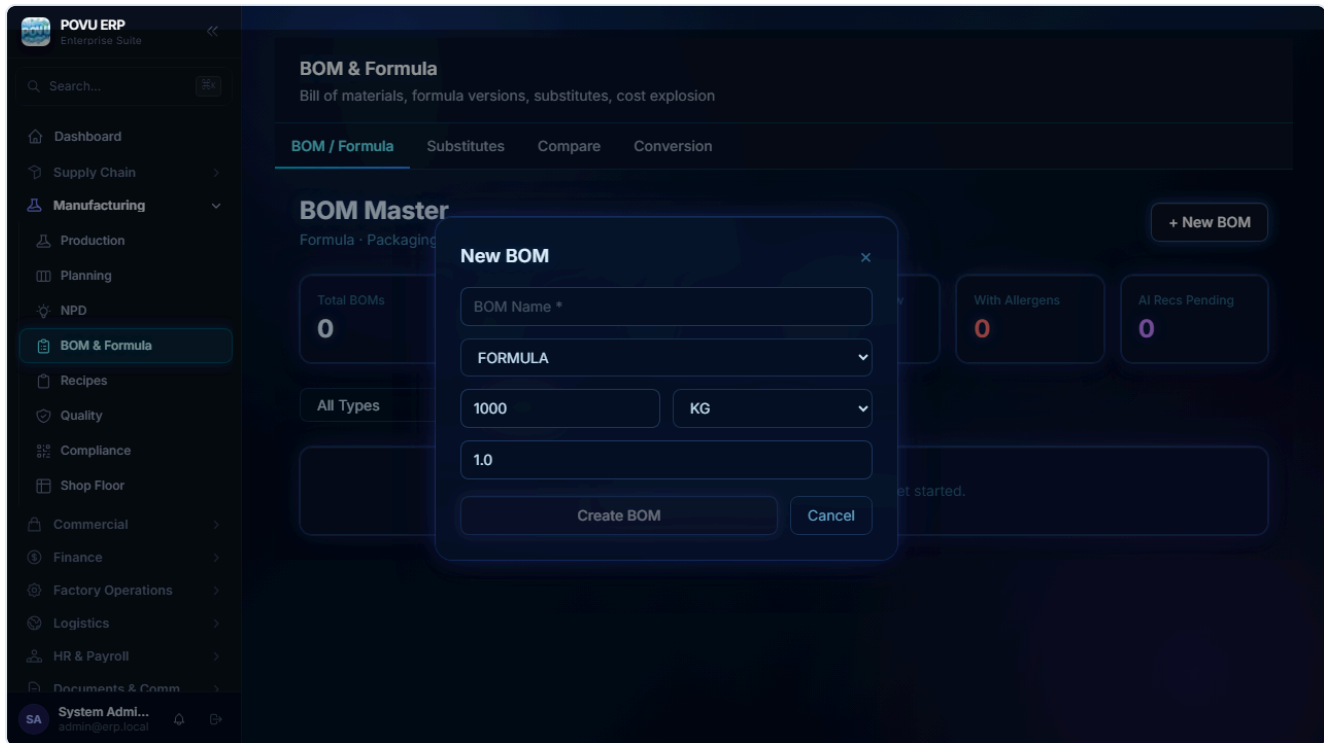
Column	Field	Notes
Code	<code>bom_code</code>	Auto-generated code; monospace font
Name	<code>bom_name</code>	Clickable link to BOM detail page; shows "default" badge if <code>is_default=true</code>
Type	<code>bom_type</code>	Coloured badge per type
Product	<code>product_name</code>	Associated finished goods product
Base Qty	<code>base_qty</code> + <code>base_uom</code>	e.g. "1,000 KG" — the batch size this BOM is written for
Version	<code>version_no</code>	Prefixed with "v", e.g. "v1.0"
Status	<code>lifecycle</code>	Lifecycle state badge
Std Cost	<code>standard_batch_cost</code>	Standard batch cost in KES; "—" if not costed
Actions	—	View / Formula quick links

`is_default` Flag

A product can have multiple BOM versions. The one marked `is_default=true` is used by production work orders when no specific BOM version is specified.

Creating a BOM

Button: + New BOM (requires `bom.create`)



New BOM modal — BOM Name, Type, Base Qty, Unit, and Version fields.

Create Form Fields

Field	Label	Required	Default	Allowed Values
bom_name	BOM Name	Yes	—	Free text
bom_type	Type	Yes	FORMULA	FORMULA / INTERMEDIATE / PACKAGING / MULTILEVEL / PHANTOM / REWORK / COPRODUCT
base_qty	Base Qty	Yes	1000	Number
base_uom	Unit	Yes	KG	KG / L / UNIT / MT / G / ML
version_no	Version	Yes	1.0	e.g. "1.0", "2.0"

Submit is disabled until `bom_name` is filled. The new BOM is created in `DRAFT` lifecycle state.

BOM Detail Page

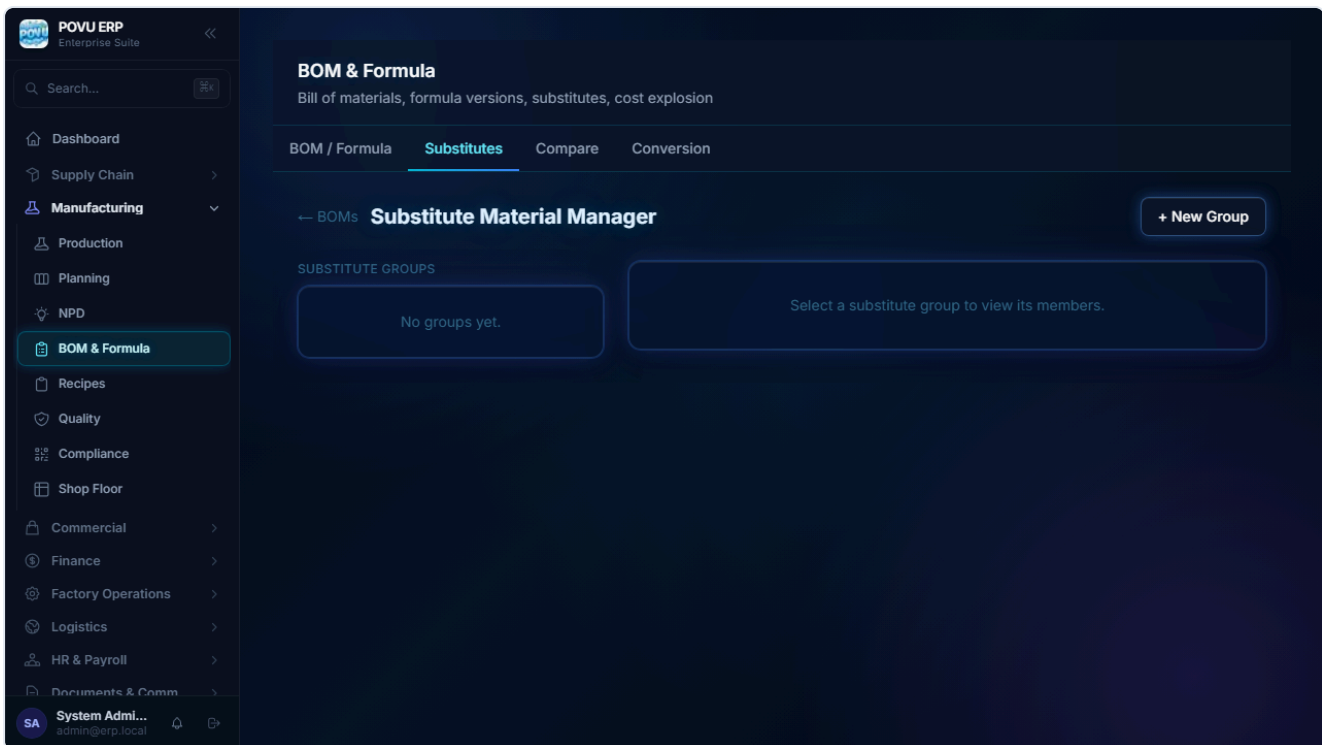
Route: `/dashboard/bom/{bom_id}`

The BOM detail page provides tabs for formula editing, costing, compliance, and cost explosion. Key actions from the Actions column:

- **View** — navigates to `/dashboard/bom/{id}` (main detail)
- **Formula** — navigates to `/dashboard/bom/{id}?tab=formula` (direct tab link)

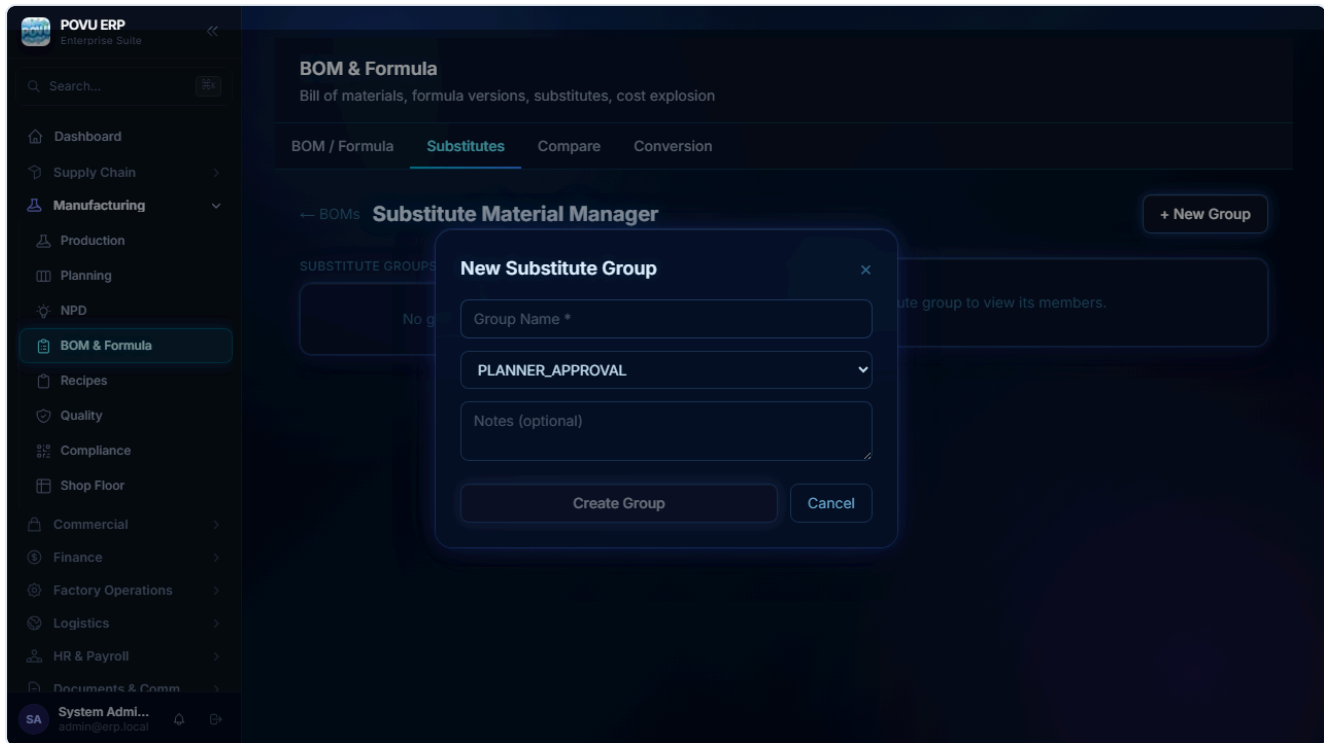
Substitutes Tab

Route: `/dashboard/bom?tab=substitutes`



Substitutes tab listing approved substitute groups and alternative materials.

Button: `+ New Substitute Group`

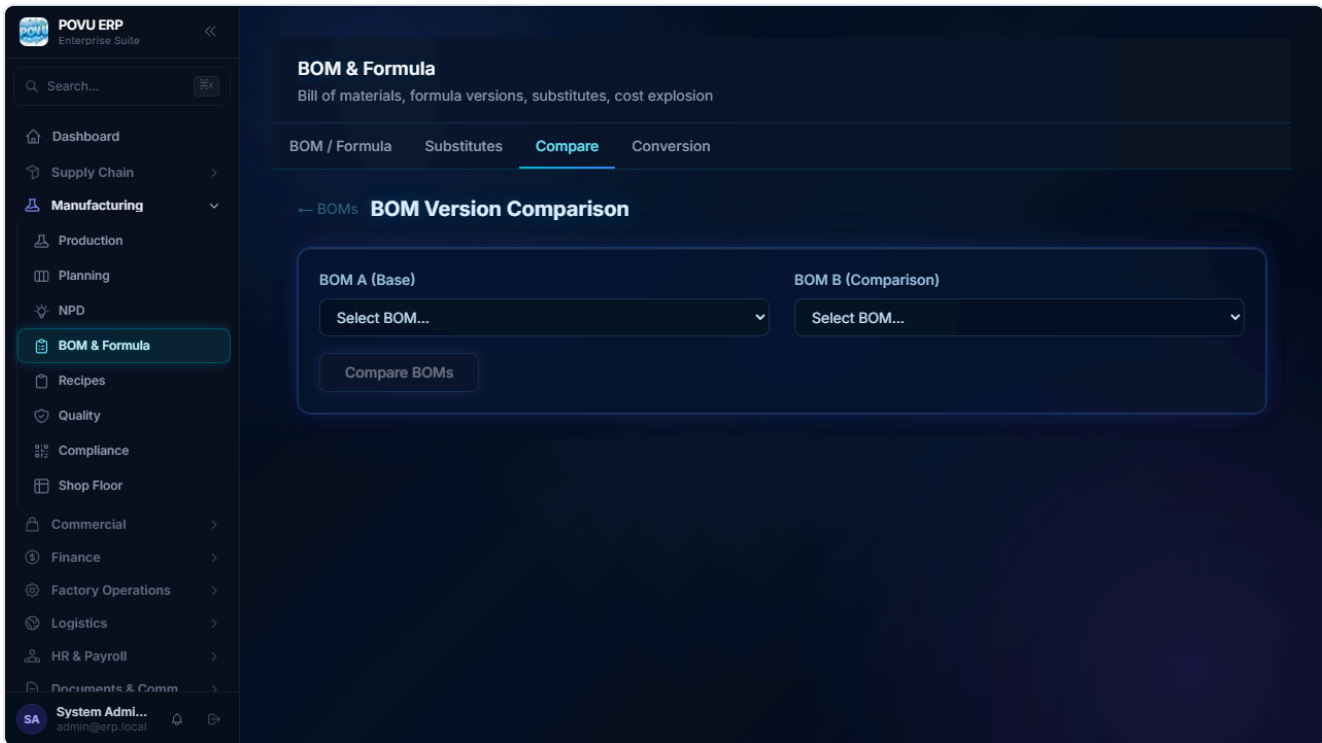


New Substitute Group modal — group code and primary material fields.

Manages approved material substitutions. When a BOM item's primary material is unavailable, an approved substitute with the same `alternative_group` can be used.

Compare Tab

Route: `/dashboard/bom?tab=compare`

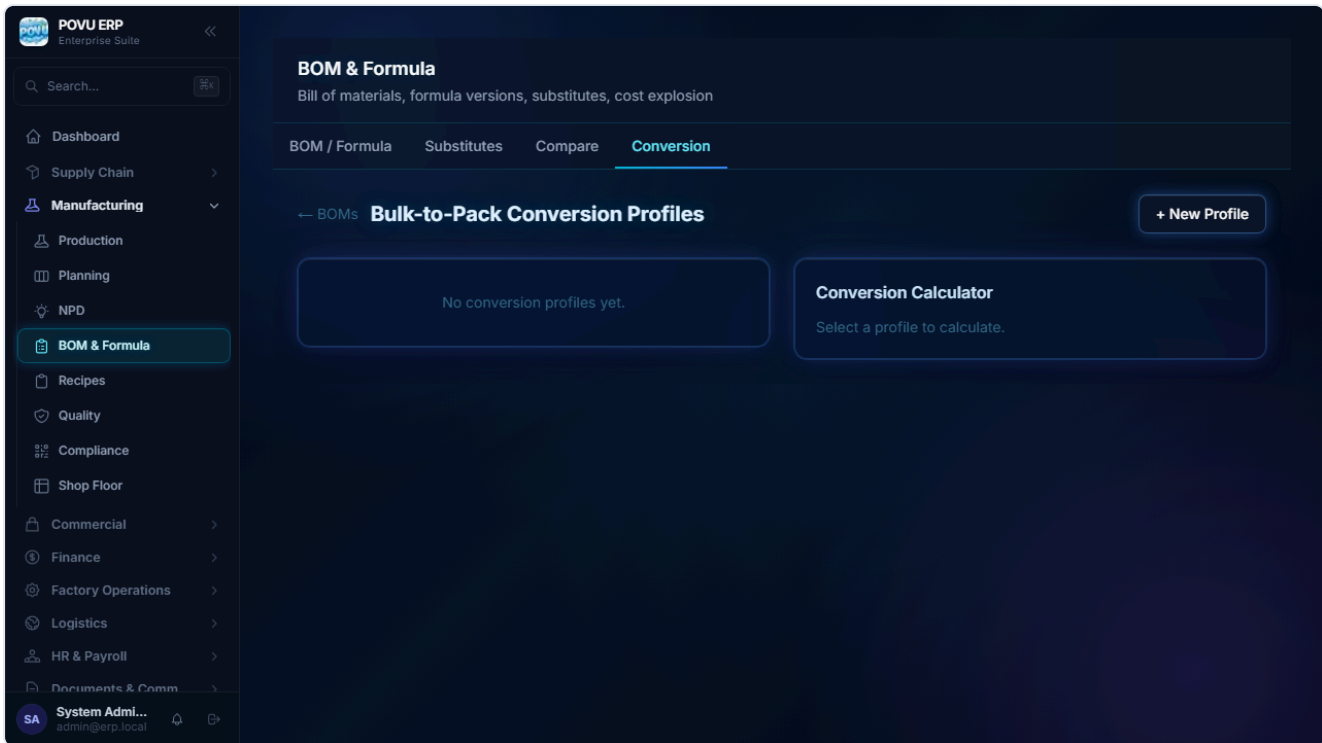


Compare tab — side-by-side view of two BOM versions showing ingredient and cost differences.

Side-by-side comparison of two BOM versions for the same product. Used during reformulation reviews to see cost and ingredient differences.

Conversion Tab

Route: `/dashboard/bom?tab=conversion`



Conversion tab showing UoM conversion factors for bulk-to-packaged unit equivalents.

Manages unit of measure conversions specific to this BOM's product line (e.g. converting bulk KG quantities to packaged unit equivalents).

BOM vs Recipe

Aspect	Recipe	BOM
Purpose	Formulation R&D; process parameters	Production release; cost explosion
Status cycle	DRAFT → APPROVED → OBSOLETE	DRAFT → UNDER_REVIEW → APPROVED → RELEASED → SUPERSEDED → ARCHIVED
Process steps	Yes (temperature, pH, RPM)	No
Cost fields	No	Yes (<code>standard_batch_cost</code>)
Used by work orders	No	Yes (RELEASED BOMs only)
<code>is_default</code> flag	No	Yes

Production Plans

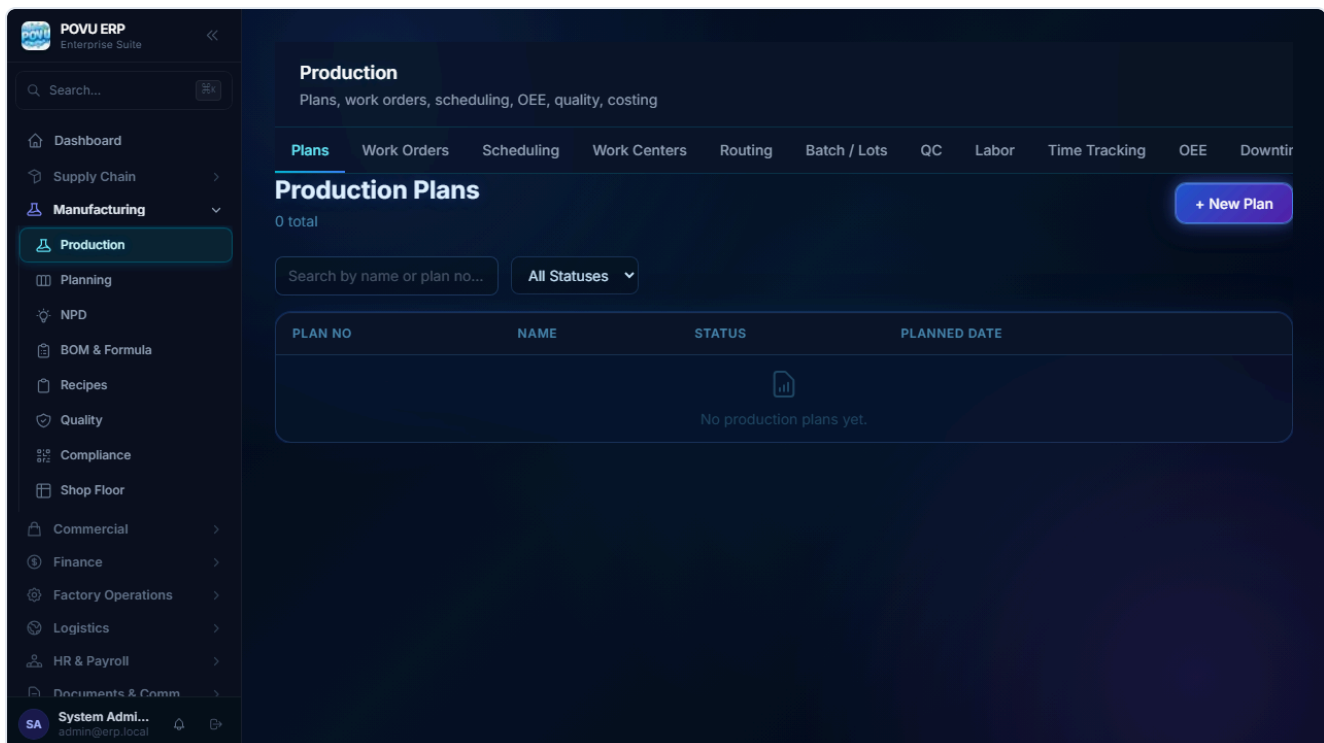
Route: `/dashboard/production` (default tab: Plans)

Permission required: `production.view`

Tab key: `plans`

What It Does

Production Plans group related work orders under a single planning event. A plan defines a target date, a reference number, and a name. Work orders are then raised against a confirmed plan. Plans must be **Confirmed** before work orders can be executed.



Production workspace — Plans tab showing production plan list with status badges and actions.

Plans Tab

Search

Free-text search filters by plan name or plan number (`plan_no`).

Table Columns

Column	Field	Notes
Plan No	<code>plan_no</code>	Monospace; clickable link to plan detail at <code>/dashboard/production/plans/{id}</code>
Name	<code>name</code>	Plan description name
Status	<code>status</code>	Status badge (see Status Values below)
Planned Date	<code>planned_date</code>	Formatted as locale date string
Actions	—	Confirm button (DRAFT only) + View button

Status Values

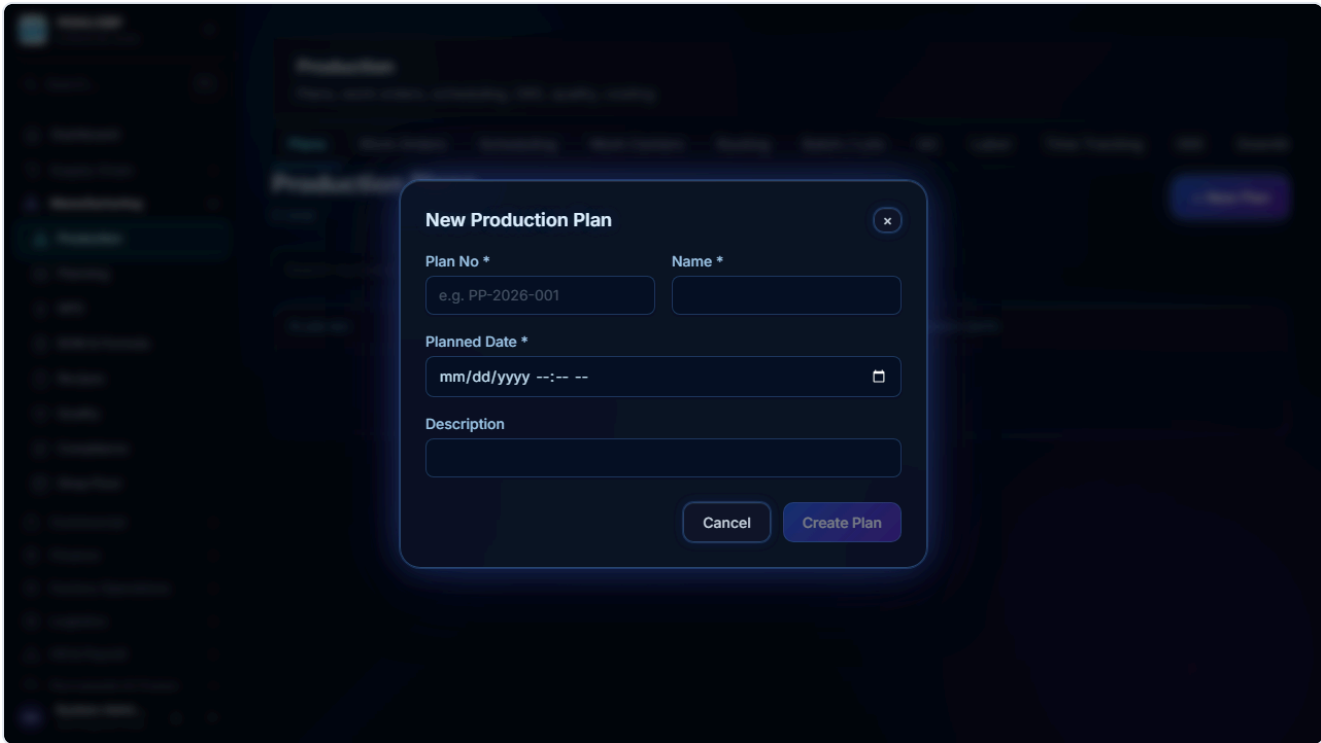
Status	Badge Colour	Meaning
DRAFT	Blue	Plan created; not yet confirmed; work orders cannot be started
CONFIRMED	Green	Plan approved; work orders can proceed
IN_PROGRESS	Green	At least one work order has started
COMPLETED	Green	All work orders completed
CANCELLED	Red	Plan cancelled; no further operations allowed

Confirm Action

The **Confirm** button appears on any row where `status === "DRAFT"`. Clicking it transitions the plan to `CONFIRMED` immediately. There is no additional confirmation dialog.

Creating a Production Plan

Button: `+ New Plan ( data-testid="production-create-plan-button" )`



New Production Plan modal — Plan No, Name, Planned Date and Description fields.

New Production Plan Modal Fields

Field	Label	Required	Default	Notes
plan_no	Plan No	Yes	—	User-assigned reference; e.g. PP-2026-001
name	Name	Yes	—	Descriptive name for this plan
planned_date	Planned Date	Yes	—	datetime-local input — includes time
description	Description	No	—	Optional notes

Submit is disabled until both `plan_no` and `name` are filled. The `planned_date` is required by the form but the submit button enables as soon as `plan_no` and `name` are present (the `datetime-local` field validation is enforced by HTML5).

New plans are always created in `DRAFT` status.

Plan Detail Page

Route: `/dashboard/production/plans/{id}`

The plan detail page shows all work orders assigned to this plan, progress tracking, and links to individual work order detail pages. From here you can also confirm the plan (same action as the list

confirm button).

Production Workspace Tabs

The `/dashboard/production` page contains 20 tabs covering the full production lifecycle:

Tab Key	Label	Content
<code>plans</code>	Plans	Production plan list and creation (this chapter)
<code>orders</code>	Work Orders	Work order management
<code>scheduling</code>	Scheduling	Gantt and scheduling board
<code>work-centers</code>	Work Centers	Work center capacity and configuration
<code>routing</code>	Routing	Operation routing for products
<code>batch-lots</code>	Batch / Lots	Batch and lot tracking
<code>quality-control</code>	QC	In-process quality control
<code>labor</code>	Labor	Labour assignment and tracking
<code>time-tracking</code>	Time Tracking	Operator time recording
<code>oee</code>	OEE	Overall Equipment Effectiveness
<code>downtime</code>	Downtime	Downtime events and root causes
<code>waste-yield</code>	Waste & Yield	Waste recording and yield calculations
<code>wip</code>	WIP	Work-in-progress inventory
<code>costing</code>	Costing	Production cost actuals
<code>variance</code>	Variance	Cost variance analysis
<code>reports</code>	Reports	Production reports
<code>execution</code>	Execution	Production execution dashboard
<code>machine-ops</code>	Machine Ops	Machine operations and maintenance
<code>material-flow</code>	Material Flow	Material movement during production
<code>projects</code>	Projects	Production project management

All tabs require `production.view` permission.

Work Orders & Scheduling

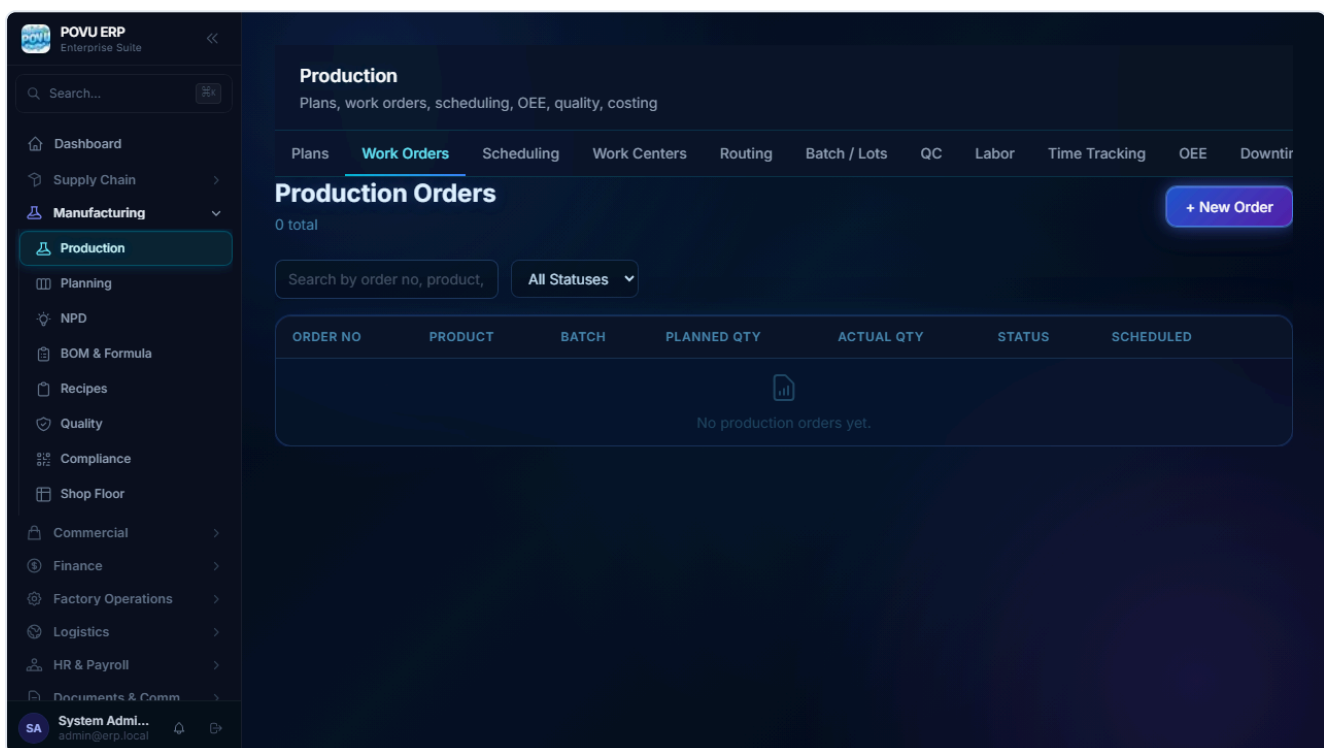
Route: `/dashboard/production?tab=orders` (and related tabs)

Permission required: `production.view`

Work Orders Tab

Tab key: `orders`

Route: `/dashboard/production/orders/page`



Production Work Orders tab listing work orders with status, plan reference, and target quantities.

Creating a Work Order

Button: `+ New Order` (opens the New Work Order modal)

New Production Order

Order No * Batch No

Product *

Recipe (Approved) *

Planned Quantity * Unit

FG Target Warehouse * Material Source Warehouse

Scheduled Start * Scheduled End *

Notes

New Work Order modal — Order No, Product, Recipe, Planned Quantity, UOM, Target Warehouse, Scheduled Start and End fields.

Work orders are the execution units of manufacturing. Each work order references a production plan, a product (or BOM), a target quantity, and a work center. Work orders progress through status stages from DRAFT to COMPLETED.

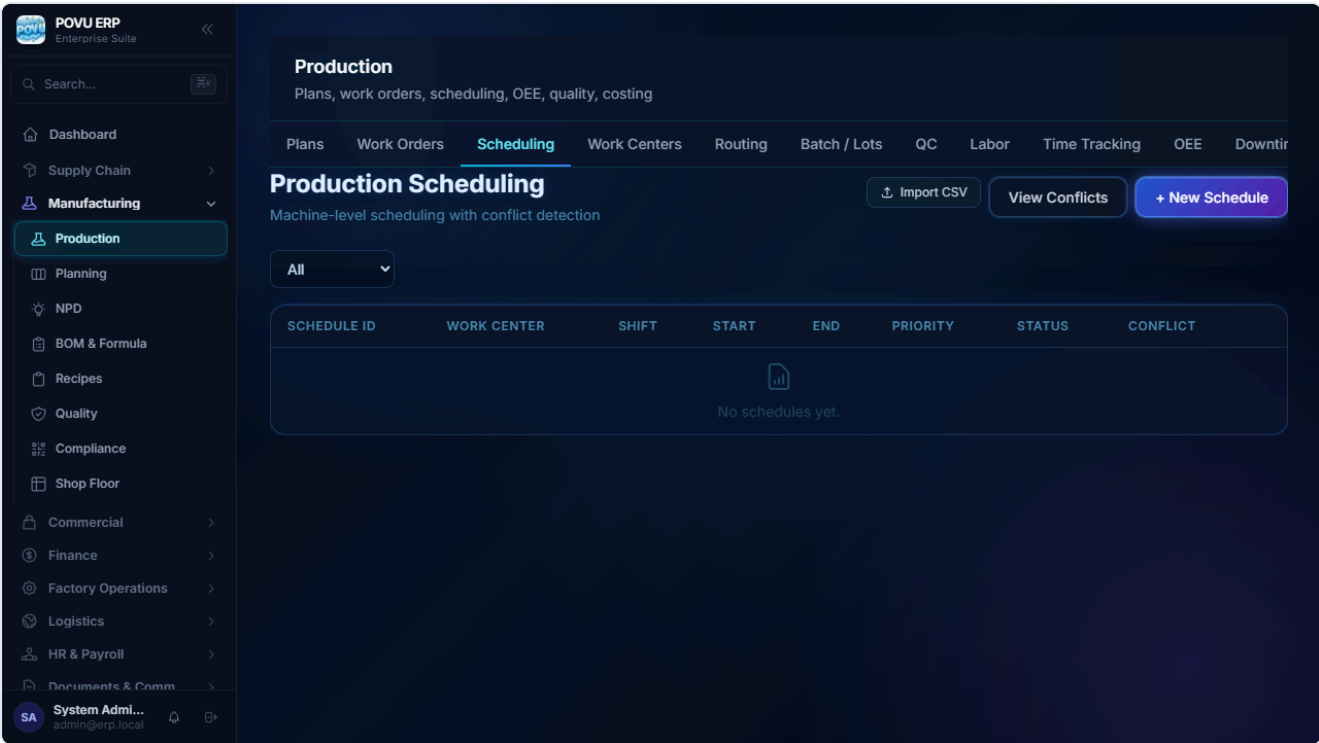
Key Concepts

- A work order is always linked to a **Production Plan**. The plan must be **CONFIRMED** before work orders can start.
- Work orders reference a **BOM** (the RELEASED formulation) and a **Routing** (the sequence of operations across work centers).
- Batch and lot numbers are assigned when a work order is released.
- Labour, time tracking, and OEE data are linked at the work order level.

Scheduling Tab

Tab key: `scheduling`

Route: `/dashboard/production/scheduling/page`



Production Scheduling tab — Gantt-style board for sequencing work orders across work centers and shifts.

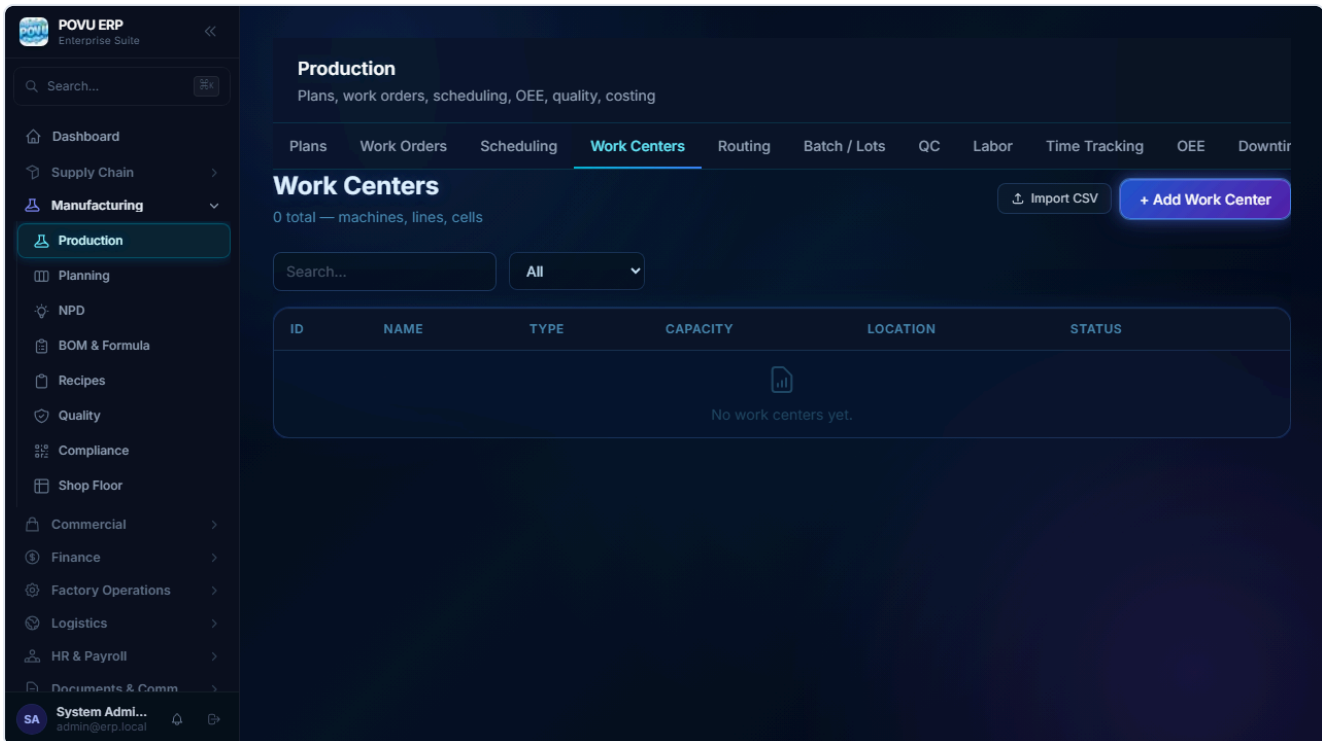
The scheduling tab provides a visual Gantt-style board for sequencing work orders across work centers and shifts. Use this when you need to manually adjust planned start/end times or resolve conflicts.

For advanced finite-capacity scheduling with AI optimisation, use **Advanced Planning** (`/dashboard/planning`).

Work Centers Tab

Tab key: `work-centers`

Route: `/dashboard/production/work-centers/page`



Work Centers tab showing capacity, shifts, and cost rates for each production station.

Work centers represent physical production stations (e.g. Blending Line 1, Filling Line 3, Pasteuriser). Each work center has:

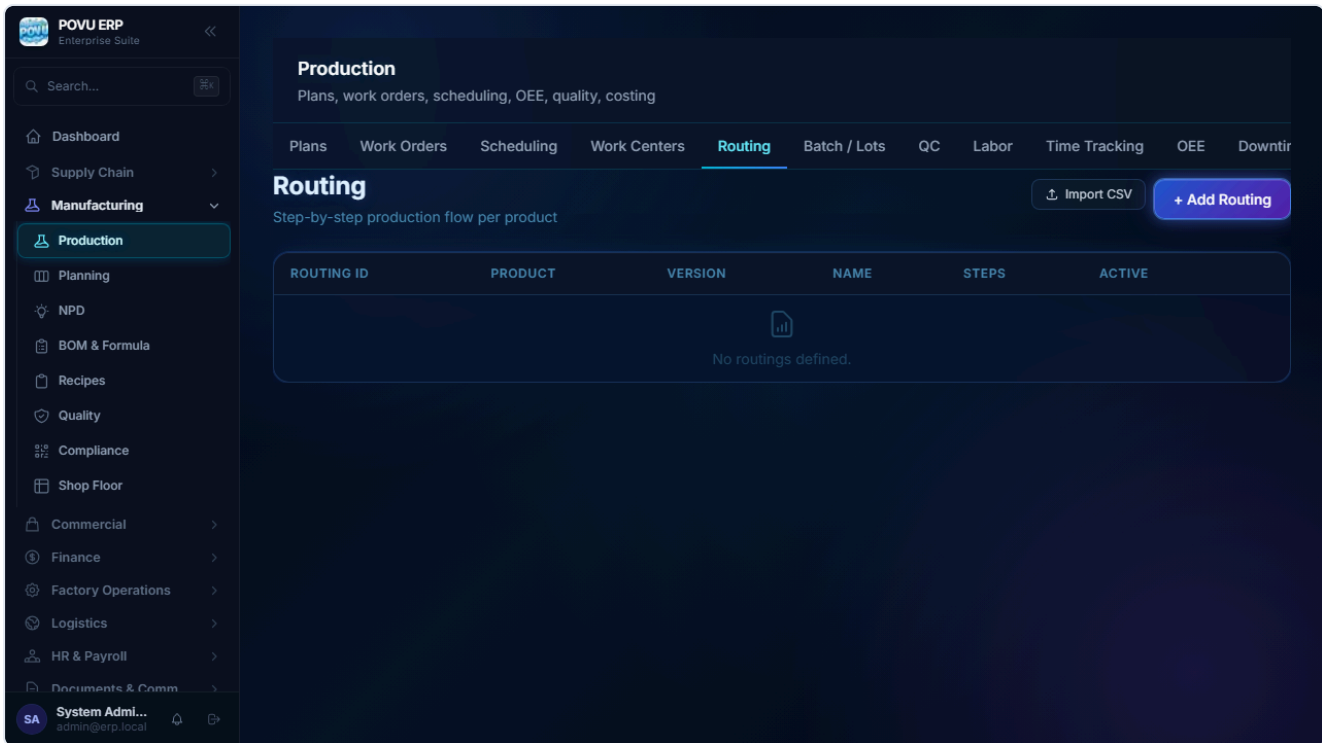
- A capacity (units per hour or KG per shift)
- Operating shifts
- A cost rate (used for costing work orders)
- Downtime tracking linkage

Work centers are also managed from the Shop Floor module for real-time status.

Routing Tab

Tab key: routing

Route: /dashboard/production/routing/page



Routing tab defining operation sequences and work center assignments per product.

A routing defines the sequence of operations (and the work center for each) required to produce a product. Example routing for a juice product:

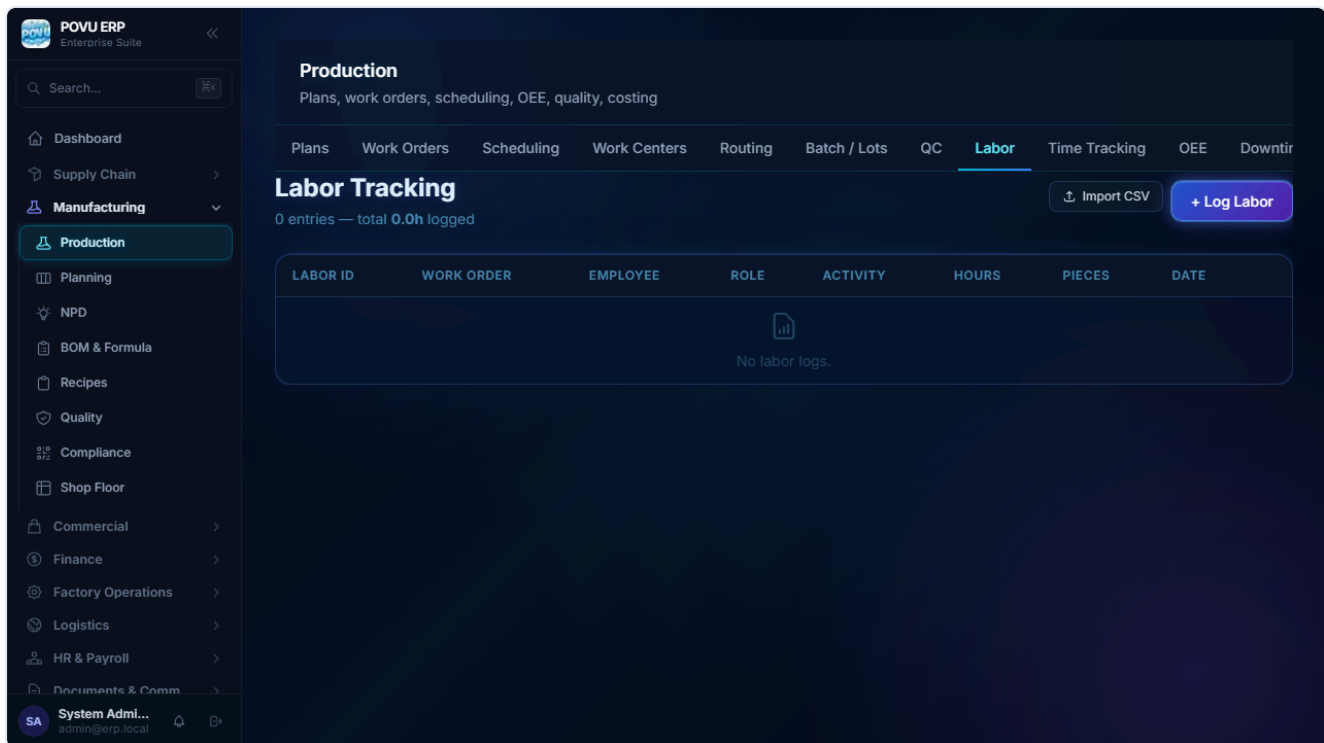
Step	Operation	Work Center
10	Pre-mixing	Blending Line 1
20	Pasteurisation	Pasteuriser A
30	Filling	Filling Line 2
40	Labelling	Labelling Machine 1

Routings are linked to products (or BOMs). When a work order is created, the routing determines which work centers are scheduled and in what sequence.

Labor Tab

Tab key: labor

Route: /dashboard/production/labor/page



Labor tab showing operator assignments per work order and shift.

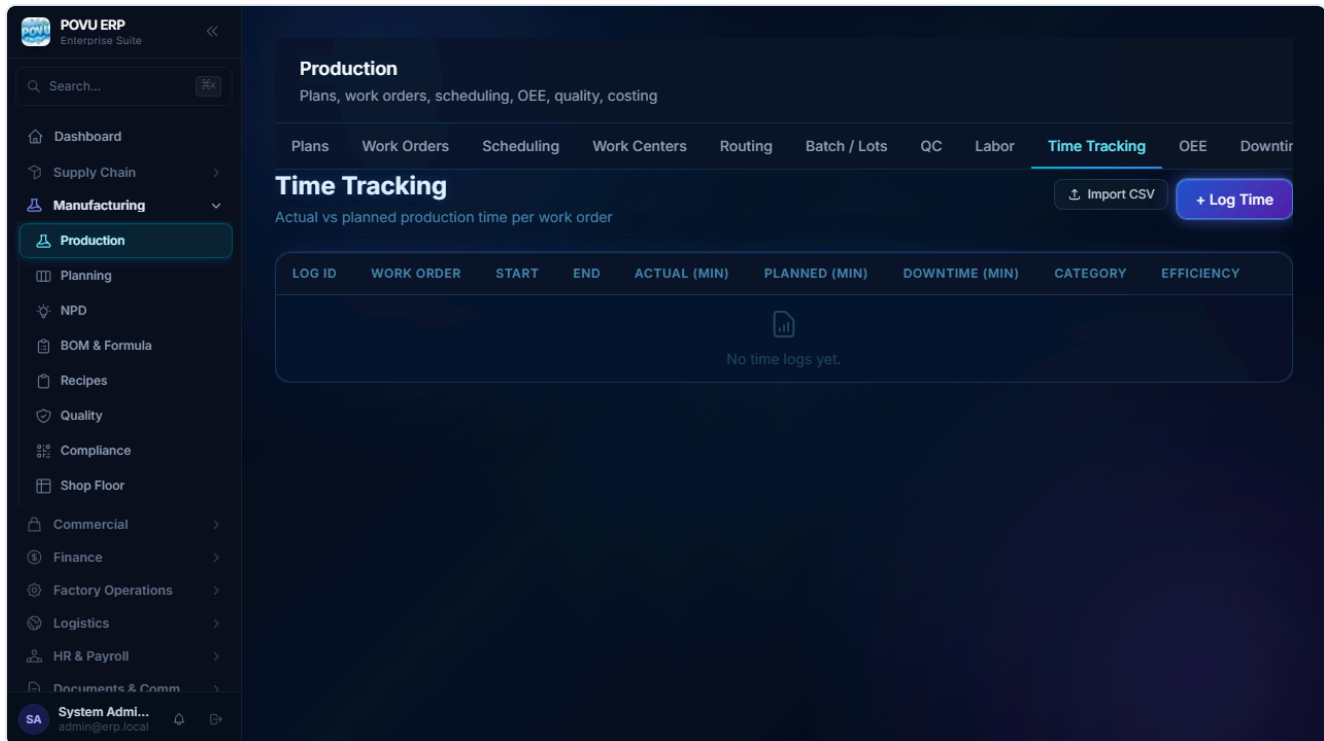
Labour assignments link operators to specific work orders and shifts. This feeds into:

- Labour costing per work order
- Time and attendance reconciliation
- OEE workforce efficiency metrics

Time Tracking Tab

Tab key: `time-tracking`

Route: `/dashboard/production/time-tracking/page`



Time Tracking tab with operator time entries against work orders.

Operators log their productive time against work orders. Time tracking data feeds into:

- Labour cost actuals
- Work order completion progress
- Efficiency benchmarks vs. standard routing times

WIP Tab

Tab key: `wip`

Route: `/dashboard/production/wip/page`

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admin@erp.local

Production
Plans, work orders, scheduling, OEE, quality, costing

Plans Work Orders Scheduling Work Centers Routing Batch / Lots QC Labor Time Tracking OEE Downtir

WIP Valuation
Work-in-progress financial value for all active production orders

Total WIP Value
Ksh 0.00

Material Cost
Ksh 0.00

Labor Cost
Ksh 0.00

Machine Cost
Ksh 0.00

In-Progress Orders — WIP Detail
0 active orders - refreshes every 60s

ORDER	PRODUCT	PLANNED QTY	ACTUAL QTY	PROGRESS	MATERIAL	LABOR	MACHINE	WIP TOTAL	DUE
No in-progress production orders									

WIP tab showing materials issued to production and unconverted quantities.

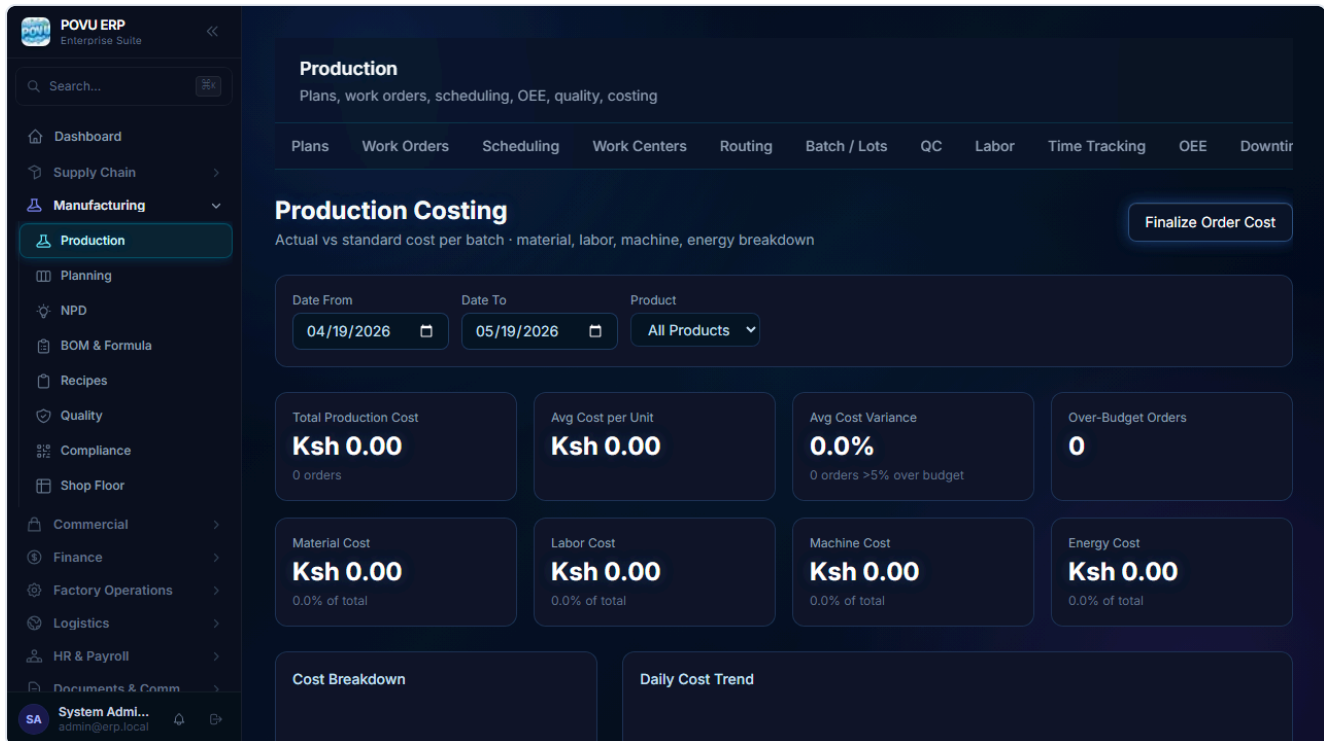
Work-in-Progress (WIP) tracks materials that have been issued to production but not yet converted into finished goods. WIP values are used in:

- Balance sheet inventory valuation
- Production cost control
- Period-end closing

Costing Tab

Tab key: costing

Route: /dashboard/production/costing/page



Costing tab showing standard vs. actual cost breakdown per work order.

Production costing aggregates:

- Material costs (actual quantities issued × material cost)
- Labour costs (logged hours × labour rate)
- Overhead absorption (machine hours × overhead rate)

Costing is performed per work order and rolled up to the production plan level.

Variance Tab

Tab key: variance

Route: /dashboard/production/variance/page

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Production
Plans, work orders, scheduling, OEE, quality, costing

Plans Work Orders Scheduling Work Centers Routing Batch / Lots QC Labor Time Tracking OEE Downtir

Production Variance & Capacity
Standard vs actual cost by component · work center utilization

From 04/19/2026 To 05/19/2026

Variance Detail Work Center Utilization

Material Variance
Ksh 0.00

Labor Variance
Ksh 0.00

Machine Variance
Ksh 0.00

Total Variance
Ksh 0.00

Order-Level Variance (0 orders)

ORDER	PRODUCT	STATUS	STD MAT	ACT MAT	MAT VAR	STD LABOR	ACT LABOR	LABOR VAR	TOTAL VAR	VAR %
No variance data for selected period										

SA System Admin...
admin@erp.local

Variance tab comparing standard vs. actual cost with material, labour, and overhead breakdown.

Variance analysis compares:

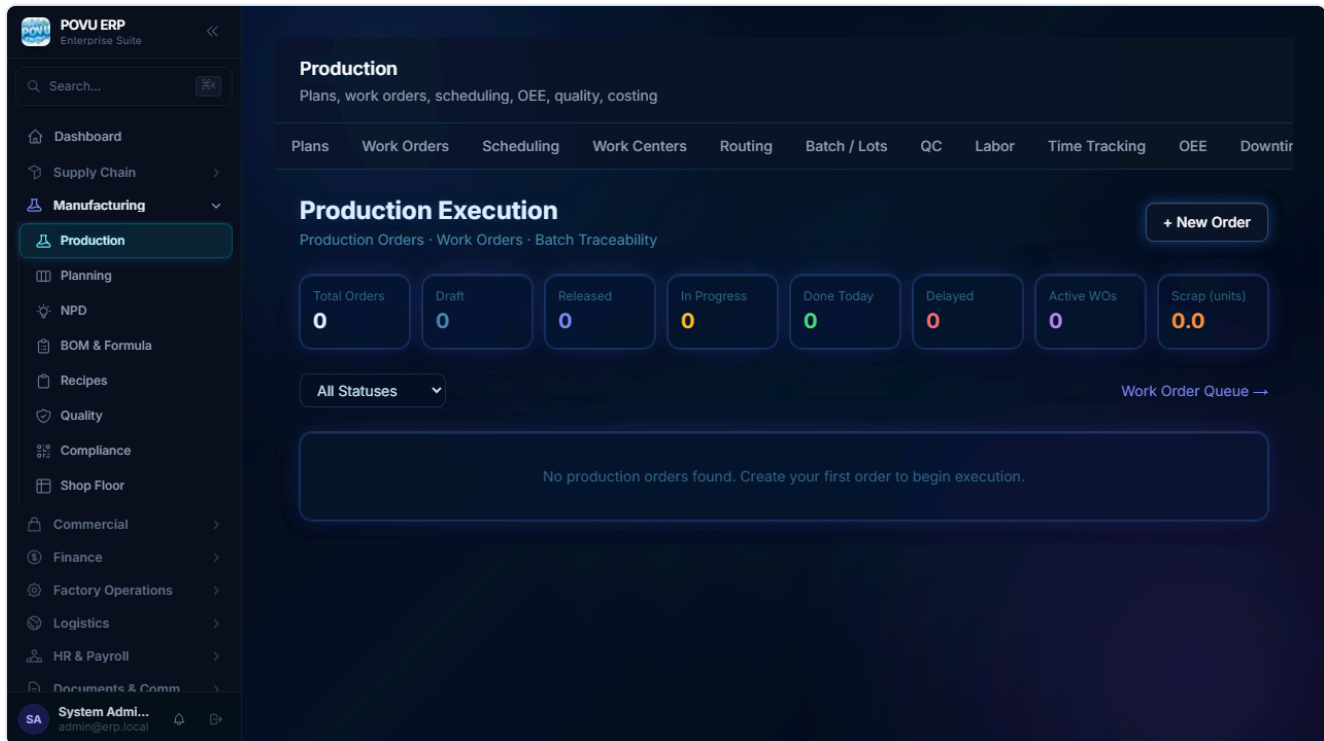
- Standard cost (from BOM + Routing) vs. Actual cost (from work order)
- Material usage variance
- Labour efficiency variance
- Overhead absorption variance

Significant variances are flagged for investigation.

Execution Tab

Tab key: `execution`

Route: `/dashboard/production-execution/page`



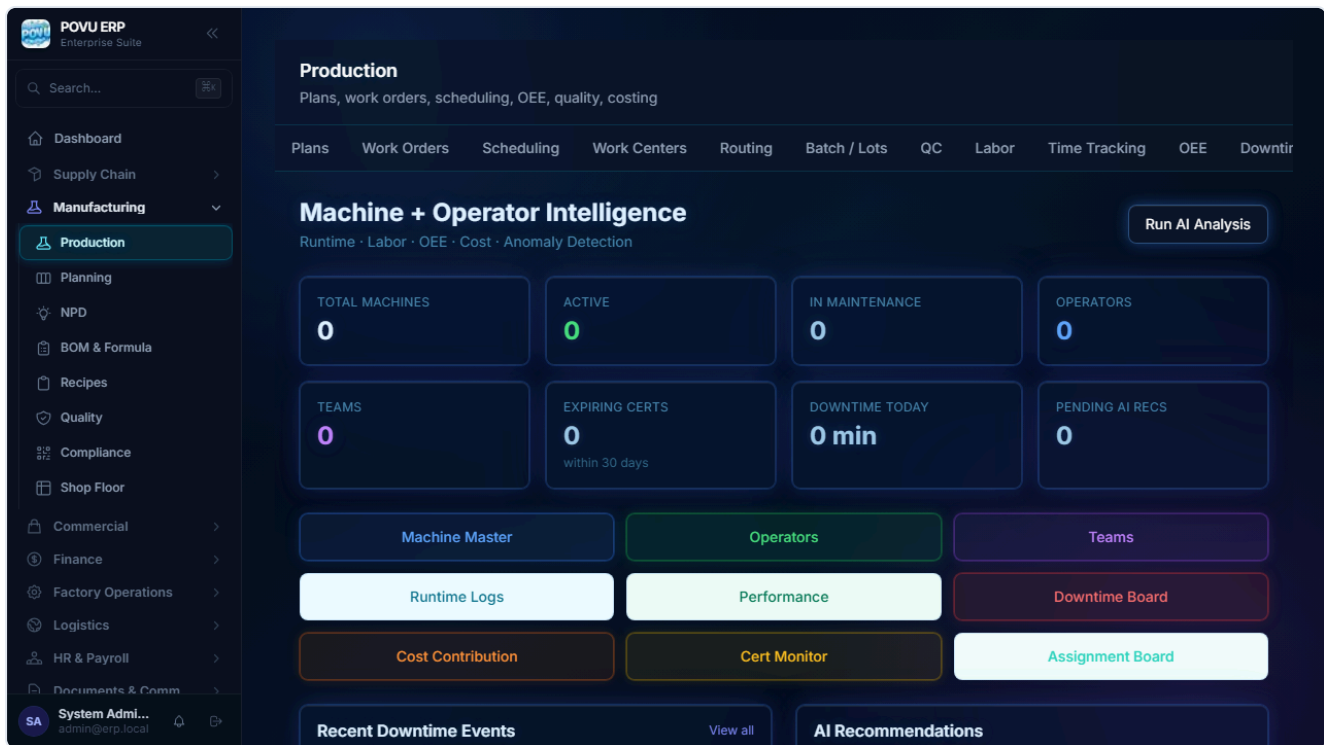
Execution tab showing real-time work order progress, percentages, and exception alerts.

Production execution provides a manager-level real-time view of all active work orders, their progress percentages, and exception alerts. It complements the Shop Floor Supervisor console.

Machine Ops Tab

Tab key: `machine-ops`

Route: `/dashboard/machine-ops/page`



Machine Ops tab with maintenance schedules, checklists, and breakdown reports.

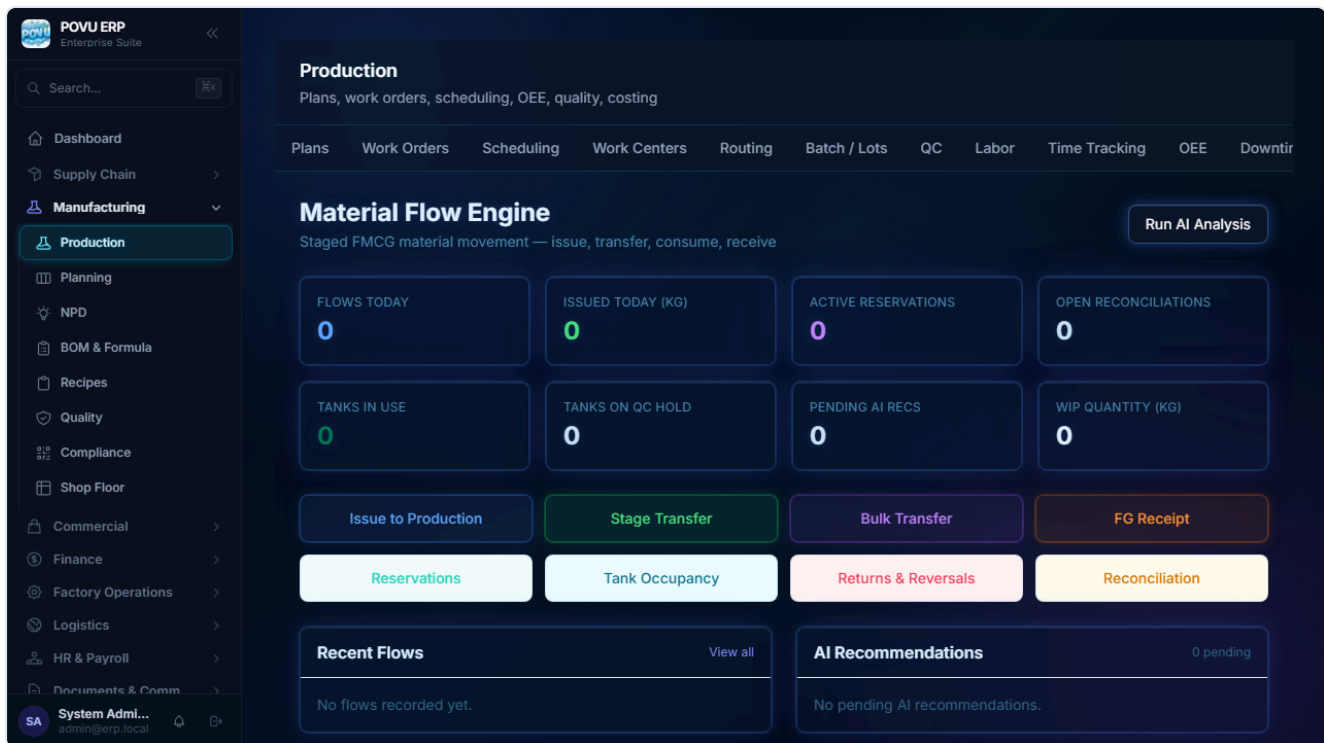
Machine operations management covers:

- Machine maintenance schedules
- Preventive maintenance checklists
- Breakdown reporting
- Integration with downtime events in OEE

Material Flow Tab

Tab key: `material-flow`

Route: `/dashboard/material-flow/page`



Material Flow tab tracking goods issue from warehouse to production and goods receipt back to stock.

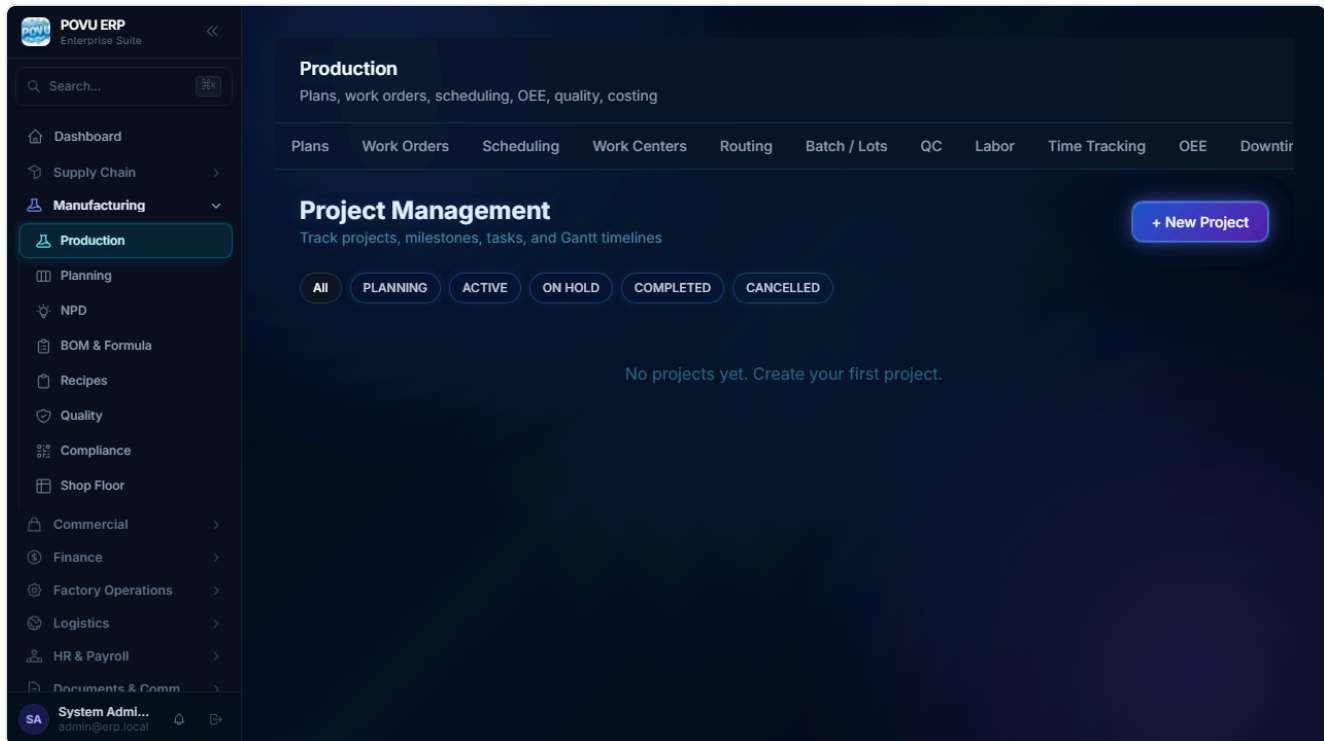
Material flow tracks the movement of materials through the production process:

- Goods Issue (materials from warehouse to production)
- Goods Receipt (finished goods from production to warehouse)
- Transfer orders between production stages

Projects Tab

Tab key: `projects`

Route: `/dashboard/projects/page`



Projects tab showing production project campaigns with timelines and work order counts.

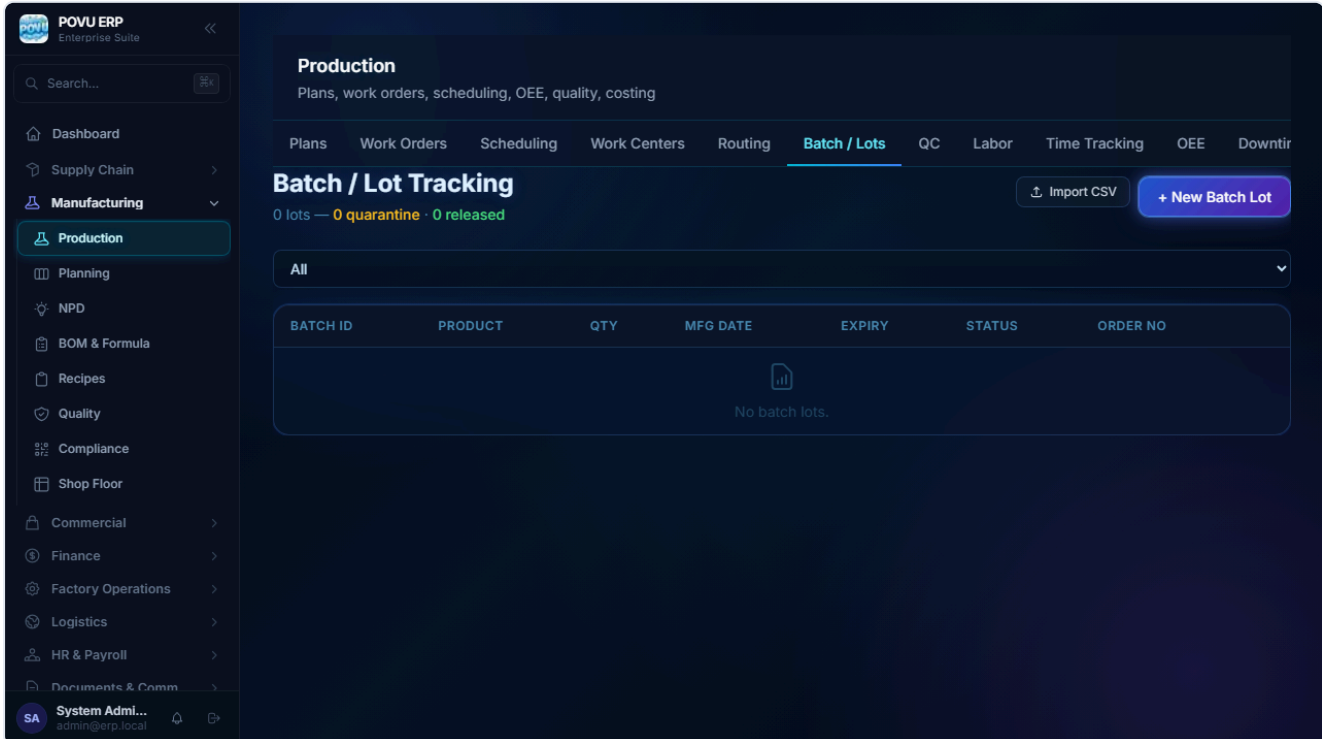
Production projects manage multi-work-order production campaigns (e.g. a seasonal promotion requiring co-ordinated runs across multiple lines). Projects group plans and work orders with shared timelines and budgets.

Batch & Lots

Route: /dashboard/production?tab=batch-lots

Tab key: batch-lots

Permission required: production.view



Batch / Lots tab showing batch and lot numbers assigned during production runs.

What It Does

The Batch & Lots tab tracks the batch numbers and lot numbers assigned during production. Batch/lot traceability is required for:

- Food safety recalls (trace finished goods back to raw material lots)
- Shelf-life management (FIFO/FEFO rotation)
- Quality holds (quarantine a specific batch without affecting stock of other batches)
- Customer complaints (identify which production run a product came from)

Key Concepts

Batch vs. Lot

Term	Meaning in this ERP
Batch	A single production run — all output from one work order or a portion of it
Lot	A traceability unit used for materials received from suppliers

These terms are sometimes used interchangeably in the UI. In reports, `batch_no` refers to production output; `lot_number` refers to incoming material.

Batch Number Format

Batch numbers are either:

- System-generated (auto-assigned when a work order is released)
- User-assigned (manually entered during work order creation)

Typical format: `BTH-{YYYYMMDD}-{sequence}` (e.g. `BTH-20260519-001`).

Lot Number Format

Lot numbers come from the supplier delivery. They are recorded during Goods Receipt in the Procurement/Inventory module and carried through to production when materials are issued.

Batch Record Fields

A batch record captures:

Field	Description
Batch No	Unique identifier for this production batch
Work Order	The work order that produced this batch
Product	Finished good produced
Quantity Produced	Actual good quantity
Quantity Scrapped	Quantity rejected / written off
Yield %	$\text{quantity_produced} / (\text{quantity_produced} + \text{quantity_scrapped}) \times 100$
Production Date	Date the batch was manufactured
Expiry Date	Calculated from production date + product shelf life
Status	PENDING / RELEASED / QUARANTINED / REJECTED / CONSUMED
QC Result	Link to the QC inspection for this batch

Field	Description
Warehouse Location	Where the finished batch was stored

Traceability

Forward Trace (Batch → Customer)

Starting from a batch number, trace:

1. Which work order produced it
2. Which materials (with lot numbers) were consumed
3. Which delivery notes / invoices it appeared on
4. Which customers received it

Backward Trace (Material Lot → Batch)

Starting from a supplier lot number, trace:

1. Which purchase orders received this lot
2. Which production work orders consumed it
3. Which product batches it contributed to
4. Which customers may have received those batches

QC Integration

When a production batch is created, it appears in the **QC Inspections** queue for a **Finished Goods** inspection. The QC result (PASSED / FAILED / CONDITIONAL_RELEASE) is linked back to the batch record and determines whether the batch can be released to the warehouse.

- `PASSED` → batch status moves to RELEASED, stock credited to finished goods
- `FAILED` → batch status moves to QUARANTINED or REJECTED
- `CONDITIONAL_RELEASE` → batch released with restrictions; notes recorded

Shelf Life & Expiry

Each product has a configured shelf life (in days). When a batch is created:

- `expiry_date = production_date + shelf_life_days`

Batches approaching expiry are flagged in inventory reports. FEFO (First Expiry First Out) picking enforces expiry order during sales order fulfilment.

Batch Quarantine

A batch can be quarantined at any point by a Quality officer. While quarantined:

- The batch cannot be picked for sales orders
- The stock is moved to a quarantine location in the WMS
- All movements require QC manager authorisation

To release from quarantine: navigate to the QC inspection for this batch, record the decision, and change batch status to RELEASED.

QC Inspections

Route: `/dashboard/quality` (default tab: Inspections)

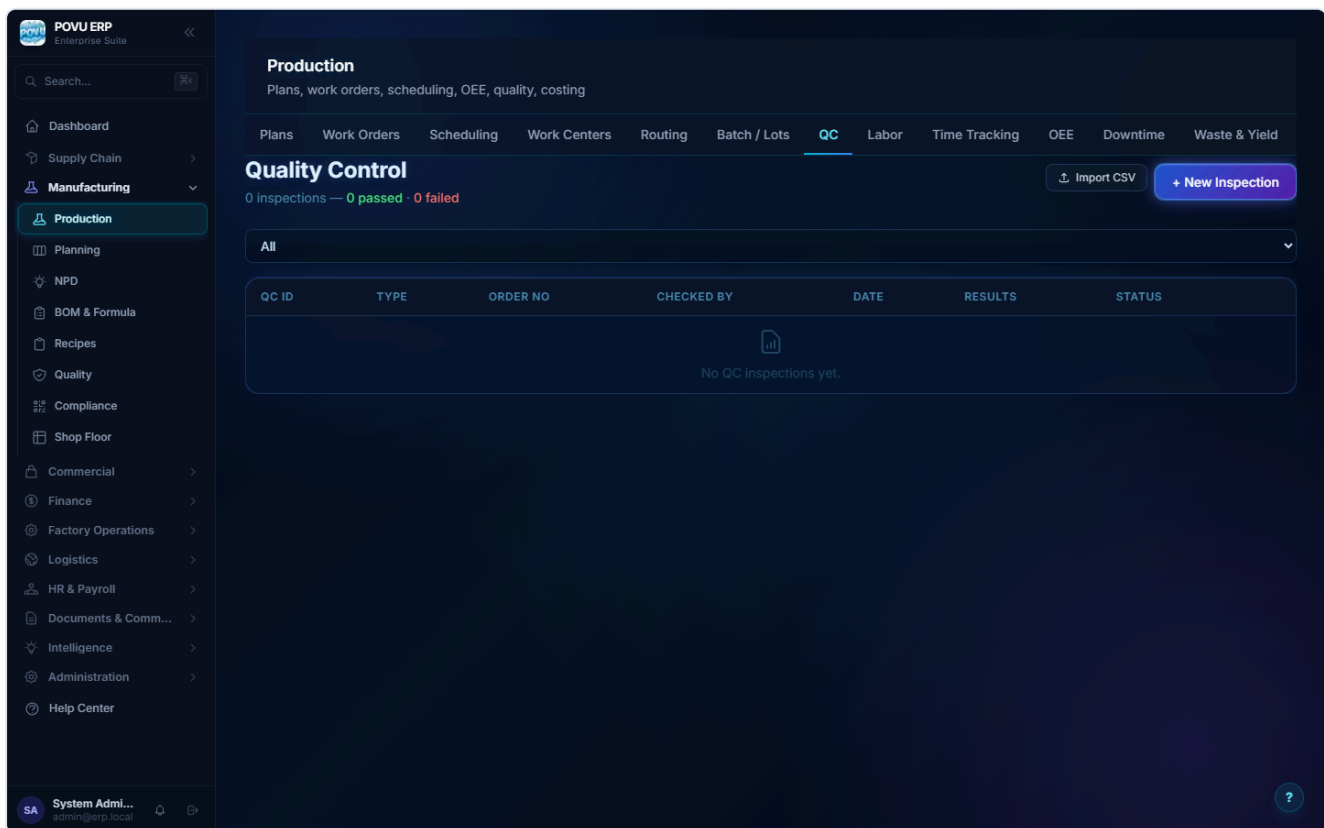
Permission required: `quality.view`

Tab key: `inspections`

Workspace tabs: Inspections, Certificates, Parameters, Consumer Complaints, Reports, QMS, Allergen, Brand Assets

What It Does

The Quality module manages all QC inspections across three inspection contexts: incoming raw materials, in-process production checks, and finished goods release. Each inspection records test results, a pass/fail decision, and optional quarantine actions.



Quality workspace showing the Inspections tab with seeded QC inspections, dashboard counters, and inspection list.

Inspection List

Dashboard Counters

Counter	Description
Pending / Active	Inspections with status PENDING or IN_PROGRESS
Passed	Inspections with status PASSED
Failed	Inspections with status FAILED
Critical Failures	Inspections where <code>critical_fail = true</code>

Critical failures are highlighted in bold red. These indicate a safety or regulatory failure requiring immediate supervisor action.

Filters

Filter	Options
Type	All Types / Incoming QC / In-Process QC / Finished Goods QC
Status	All Statuses / Pending / In Progress / Passed / Failed / Conditional Release / Cancelled

Table Columns

Column	Field	Notes
Inspection No	<code>inspection_no</code>	Monospace; clickable link to detail at <code>/dashboard/quality/{id}</code>
Type	<code>qc_type</code>	Coloured badge: INCOMING (blue), IN_PROCESS (blue), FINISHED_GOODS (green)
Item	<code>material_code</code> + <code>material_name</code> or <code>product_name</code>	Material code shown in monospace prefix
Supplier	<code>supplier_name</code>	Blank for non-incoming inspections
Lot / Batch	<code>lot_number</code> or <code>batch_no</code>	Whichever is present
Date	<code>inspection_date</code>	Locale date format
Tests	<code>pass_count</code> ✓ <code>fail_count</code> X	Green pass count, red fail count; "CRIT" badge if <code>critical_fail</code>
Status	<code>status</code> + <code>quarantine_applied</code>	Status badge; QUARANTINED badge shown separately if quarantine applied

Creating a QC Inspection

Button: `+ New Inspection` (`data-testid="quality-create-inspection-button"`)

New QC Inspection form. The Supplier and Material fields only appear when qc_type is INCOMING.

New QC Inspection Modal Fields

Field	Label	Required	Type	Notes
<code>inspection_no</code>	Inspection No	Yes	string	User-assigned; e.g. <code>QC-2026-001</code>
<code>qc_type</code>	QC Type	Yes	enum	INCOMING / IN_PROCESS / FINISHED_GOODS
<code>inspection_date</code>	Inspection Date	Yes	date	Date picker
<code>lot_number</code>	Lot Number	No	string	Supplier lot for incoming
<code>batch_no</code>	Batch No	No	string	Production batch for in-process or FG
<code>supplier_id</code>	Supplier	No (shown for INCOMING)	select	Appears only when <code>qc_type = INCOMING</code>
<code>material_id</code>	Material	No (shown for INCOMING)	select	Appears only when <code>qc_type = INCOMING</code>

Field	Label	Required	Type	Notes
warehouse_id	Warehouse	No	select	Which warehouse this inspection is associated with
sample_size	Sample Size	No	decimal	Number (e.g. 2.5)
sample_unit	Sample Unit	No	string	Default "KG" ; free text
notes	Notes	No	string	Free text

Conditional display: The `supplier_id` and `material_id` fields are shown only when `qc_type` is set to `INCOMING`. They are hidden for `IN_PROCESS` and `FINISHED_GOODS` types.

Submit is disabled until `inspection_no` and `inspection_date` are filled.

After creation, the user is immediately navigated to the inspection detail page at

`/dashboard/quality/{id}`.

QC Inspection Types

Type	When Used	Typical Item
INCOMING	On receipt of raw materials from supplier	Raw material + lot number
IN_PROCESS	During active production run	Intermediate product + batch number
FINISHED_GOODS	On completion of a production batch	Finished product + batch number

Inspection Status Values

Status	Meaning
PENDING	Created but no tests recorded yet
IN_PROGRESS	Tests being entered; not yet decided
PASSED	All tests passed; item released
FAILED	One or more tests failed; item rejected or reworked
CONDITIONAL_RELEASE	Released despite minor non-conformance; conditions noted
CANCELLED	Inspection voided; no action taken

QC Decision Values

Decision	Badge Colour	Meaning
ACCEPT	Green	Item fully accepted
REJECT	Red	Item rejected; return to supplier or dispose
REWORK	Red	Item to be reworked before re-inspection
CONDITIONAL_RELEASE	Blue	Accepted with conditions and documentation

Critical Fail Flag

`critical_fail = true` is set when at least one test result breaches a critical specification limit (e.g. microbiological contamination, allergen mislabelling, regulatory parameter breach). Critical failures:

- Are displayed as bold "CRIT" in the test column
- Increment the Critical Failures dashboard counter
- Typically trigger automatic quarantine of the affected batch/lot

Quarantine

`quarantine_applied = true` indicates that the warehouse stock associated with this inspection has been moved to a quarantine location. The QUARANTINED badge is shown alongside the status badge in the list.

To release from quarantine, record the ACCEPT or CONDITIONAL_RELEASE decision on the inspection detail page.

Inspection Detail Page

Route: `/dashboard/quality/{inspection_id}`

The detail page provides:

- Full header information
- Test result entry (each parameter with target, actual, and pass/fail)
- Decision recording (ACCEPT / REJECT / REWORK / CONDITIONAL_RELEASE)
- Quarantine management
- Attached certificates (certificates linked from the Certificates tab)
- History log of status changes

Other Quality Tabs

The screenshot displays the 'Quality' section of the POVU ERP interface. The left sidebar shows the 'Quality' tab selected. The main content area is titled 'Quality' and includes a sub-header 'QC inspections, certificates, parameters, allergen, brand assets'. Below this, there are tabs for 'Inspections', 'Certificates', 'Parameters', 'Consumer Complaints', 'Reports', 'QMS', 'Allergen', and 'Brand Assets'. The 'Certificates' tab is active, showing the 'Regulatory Certificate Register'. A red error message is visible: 'SyntaxError: Unexpected token '<', '<!DOCTYPE "... is not valid JSON'. Below the error, there are filters for 'Status' (All) and 'Entity Type' (All), with a checkbox for 'Expiring / Expired only' and a 'Clear' button. The main area shows 'Certificates (0)' and a message 'No certificates found.'

Certificates tab listing regulatory and quality certificates with status and expiry dates.

The screenshot displays the 'Quality' section of the POVU ERP interface, specifically the 'Parameters' tab. The left sidebar shows the 'Quality' tab selected. The main content area is titled 'Quality' and includes a sub-header 'QC inspections, certificates, parameters, allergen, brand assets'. Below this, there are tabs for 'Inspections', 'Certificates', 'Parameters', 'Consumer Complaints', 'Reports', 'QMS', 'Allergen', and 'Brand Assets'. The 'Parameters' tab is active, showing the 'QC Parameters' section. A red error message is visible: 'SyntaxError: Unexpected token '<', '<!DOCTYPE "... is not valid JSON'. Below the error, there are buttons for 'Import CSV' and '+ New Parameter'. The main area shows a table with columns: 'NAME', 'TYPE', 'UNIT', 'SPEC RANGE', 'CRITICAL', 'APPLICABLE TO', and 'STATUS'. The table is empty, and a message says 'No parameters. Add templates to speed up QC data entry.'

Parameters tab with specification limits per product and material.

The screenshot shows the POVI ERP Enterprise Suite interface. The left sidebar contains a navigation menu with categories like Dashboard, Supply Chain, Manufacturing, Production, Planning, NPD, BOM & Formula, Recipes, Quality (highlighted), Compliance, Shop Floor, Commercial, Finance, Factory Operations, Logistics, HR & Payroll, and Documents & Comm. The main content area is titled 'Quality' and includes a sub-header 'QC inspections, certificates, parameters, allergen, brand assets'. Below this is a tabbed interface with 'Inspections', 'Certificates', 'Parameters', 'Consumer Complaints' (active), 'Reports', 'QMS', 'Allergen', and 'Brand Assets'. The 'Consumer Complaint Management' section features a '+ Log Complaint' button and a filter bar with 'Severity' (All), 'Status' (All), and 'Lot Number' (e.g. LOT-2026-001). A table titled 'Complaints (0)' shows 'No complaints found.'

Consumer Complaints tab tracking customer-reported product issues linked to batches.

The screenshot shows the POVI ERP Enterprise Suite interface, specifically the 'Reports' tab under the 'Quality' module. The left sidebar is identical to the previous screenshot. The main content area is titled 'QC Reports' with a sub-header 'Inspection outcomes, rejection analysis, and compliance overview'. Below this is a tabbed interface with 'Overview' (active), 'Rejection Summary', 'Incoming QC', 'In-Process', and 'Finished Goods'. The 'Overview' tab displays a grid of KPI cards: 'Total Inspections' (0), 'Passed' (0), 'Failed' (0), 'Conditional' (0), 'Quarantined' (0), 'Rework Candidates' (0), 'Critical Failures' (0), and 'Pass Rate' (a bar chart). At the bottom, there are three summary cards for 'Incoming QC', 'In-Process QC', and 'Finished Goods QC', each showing '0' and '0 passed of 0'.

Quality Reports tab showing KPI trends and generated quality reports.

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Quality
QC inspections, certificates, parameters, allergen, brand assets

Inspections Certificates Parameters Consumer Complaints Reports **QMS** Allergen Brand Assets

Quality Management System
QC checkpoints · HACCP · CCP monitoring · Deviations · Corrective actions · Release control

QC INSPECTIONS

TOTAL 0	PENDING 0	PASSED TODAY 0	FAILED TODAY 0	CONDITIONAL TODAY 0	PASS RATE (7D) 0%
------------	--------------	-------------------	-------------------	------------------------	----------------------

HACCP & CCP

ACTIVE CCPS 0	CCP VIOLATIONS (7D) 0	ALLERGEN VALIDATIONS PENDING 0
------------------	--------------------------	-----------------------------------

DEVIATIONS & CORRECTIVE ACTIONS

OPEN DEVIATIONS 0	CRITICAL DEVIATIONS 0	OPEN CORRECTIVE ACTIONS 0	OVERDUE CAS 0
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LOT QUALITY & RELEASE

QMS tab for non-conformance reports, CAPA, and SOPs.

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Documents & Comm
System Admin...
admin@erp.local

Quality
QC inspections, certificates, parameters, allergen, brand assets

Inspections Certificates Parameters Consumer Complaints Reports QMS **Allergen** Brand Assets

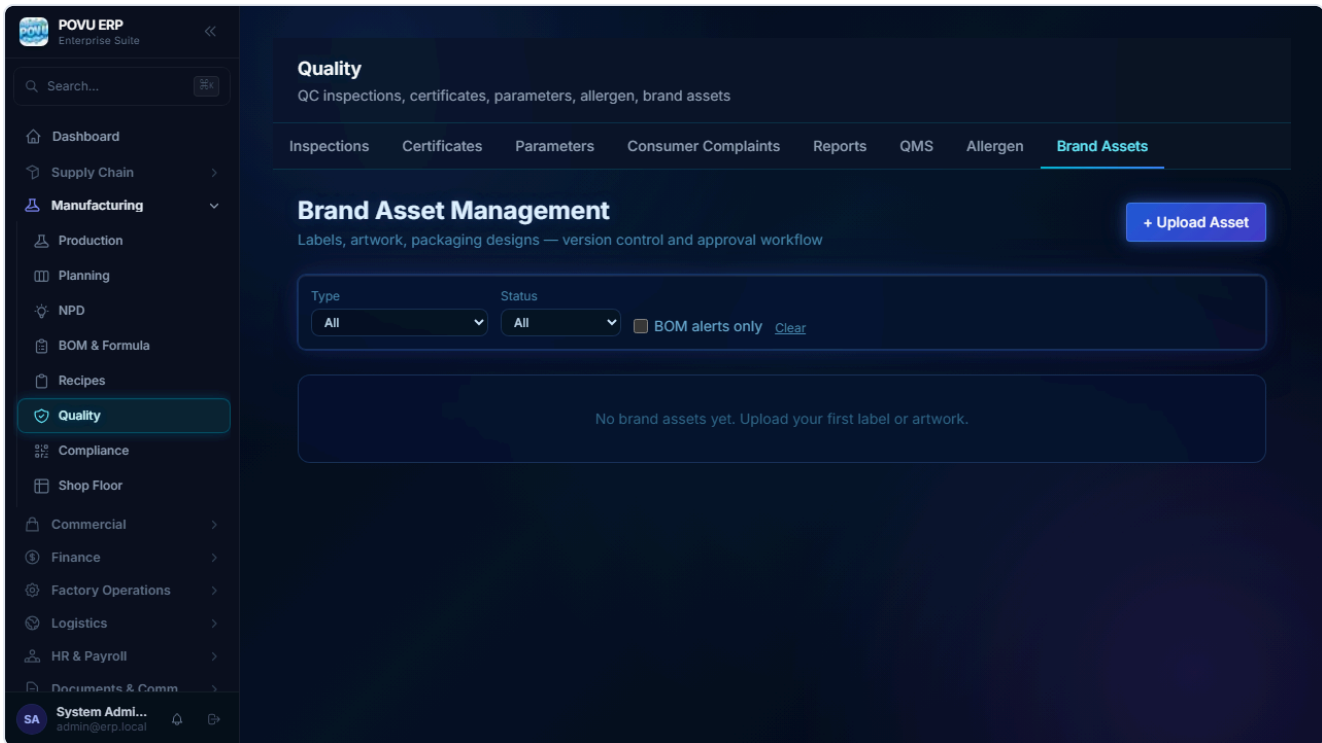
Allergen + Nutrition Management
Allergen master · Roll-up engine · Nutrition panels · Label readiness

Seed Allergens Run All AI Agents

ALLERGENS 0	MATERIALS PROFILED 0	MISSING PROFILES 0	PRODUCT ALLERGEN SUMMARIES 0	PRODUCT NUTRITION SUMMARIES 0	STALE ALLERGEN SUMMARIES 0
STALE NUTRITION SUMMARIES 0	OPEN CHANGE LOGS 0	PENDING LABEL REVIEWS 0	HIGH CROSS-CONTACT RISK 0	AI PENDING 0	
Missing allergen profiles materials without allergen data		Stale summaries need recalculation		Label reviews pending allergen changes to review	

Quick Navigation

Allergen management tab showing allergen declarations and cross-contamination matrix.



Brand Assets tab for brand-related quality standards and packaging specifications.

Tab	Content
Certificates	Quality / regulatory certificates linked to suppliers and products
Parameters	Standard test parameters and specification limits per product/material
Consumer Complaints	Customer-reported product complaints; linked to batches
Reports	Quality reports and KPI trends
QMS	Quality Management System documents and workflows
Allergen	Allergen declarations per product and BOM; cross-contamination risk matrix
Brand Assets	Brand-related quality standards and packaging specifications

Demo Data — QC Inspection Seed (TASK-015.2)

The system ships with seeded QC inspections covering all three inspection types. These use the production master data (work centers, products, materials, batch lots) seeded in TASK-015.

Seeded Inspection Types

QC Type	Scope
INCOMING	Raw material receipt inspections with supplier and material references
IN_PROCESS	In-process checks linked to production work orders and batch numbers
FINISHED_GOODS	Final product release inspections

Seeded Status Distribution

Status	Meaning
PASSED	Inspection completed with all tests within spec
FAILED	One or more tests outside spec
CONDITIONAL_RELEASE	Released with documented conditions
PENDING	Created; awaiting test entry

Seeded Test Parameters

Each seeded inspection has test results with `actual_value`, `target_value`, and `pass_fail` recorded per parameter. Critical fail (`critical_fail = true`) is set on inspections where a critical specification breach is simulated.

Idempotency

The QC seed is idempotent — re-running Docker startup does not duplicate inspections. The `_get_or_create_qc` helper uses `inspection_no` as the unique key.

Shop Floor Operations

Route: `/dashboard/shop-floor`

Permission required: `production.view`

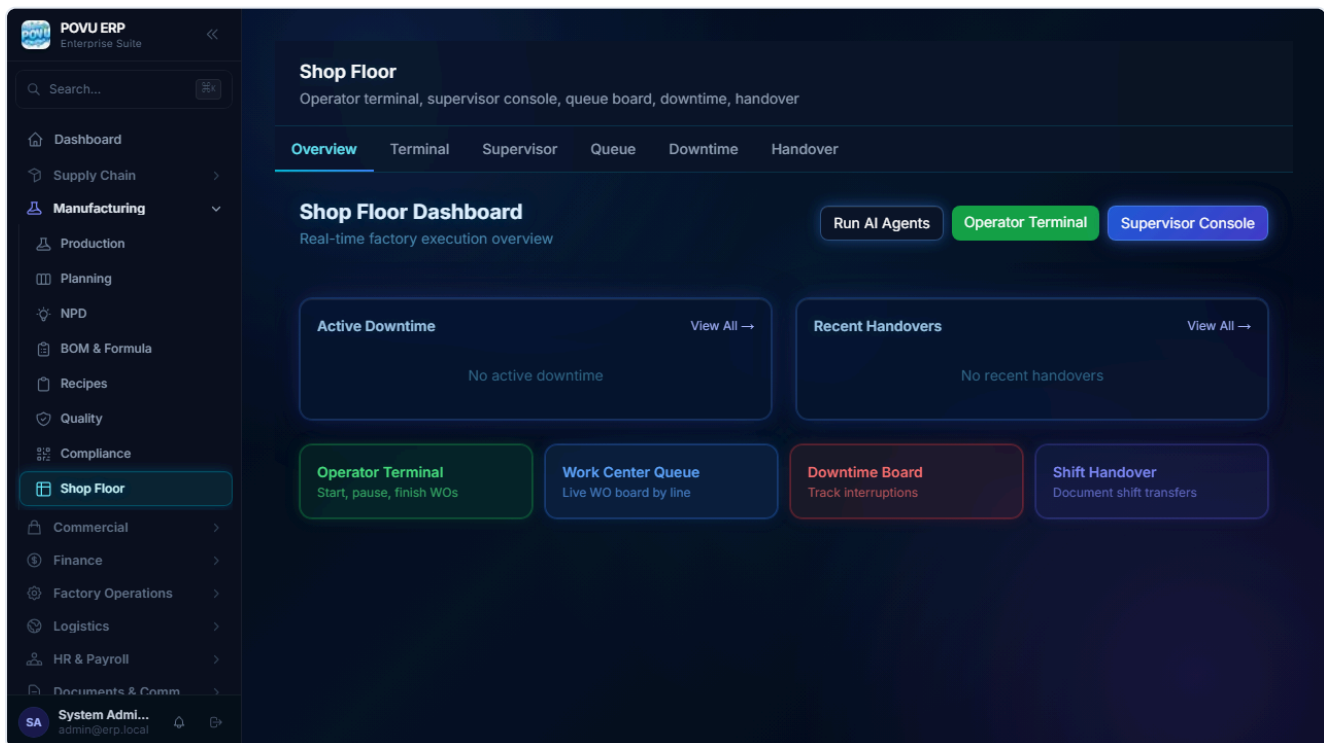
Workspace tabs: Overview, Terminal, Supervisor, Queue, Downtime, Handover

Auto-refresh: Every 30 seconds

What It Does

The Shop Floor module is the real-time factory execution layer. Operators log in at the terminal to start, pause, and finish work orders. Supervisors monitor all lines from the supervisor console. Downtime events are logged and categorised. Shifts are handed over with documented records.

The dashboard auto-refreshes every 30 seconds to reflect live factory status.



Shop Floor dashboard showing live KPI tiles: active sessions, running work orders, good quantity today, and help requests.

Overview Tab

Tab key: `overview`

Dashboard KPIs

KPI	Field	Colour
Active Sessions	<code>active_sessions</code>	Indigo
Running WOs	<code>running_work_orders</code>	Green
Paused WOs	<code>paused_work_orders</code>	Yellow
QC Holds	<code>qc_hold_count</code>	Purple
Active Downtime	<code>active_downtime_count</code>	Red
Help Requests	<code>help_requests_open</code>	Orange
Good Qty Today	<code>total_good_qty_today</code> (rounded to 0 dp)	Teal
Scrap Today	<code>total_scrap_today</code> (rounded to 0 dp)	Red-400
Pending AI Recs	<code>pending_ai_recs</code>	Blue
Overrides Today	<code>supervisor_overrides_today</code>	Grey

Alerts Panel

Active alerts are shown when present. Alerts with severity `HIGH` or `CRITICAL` are displayed on a red background; others on yellow. Each alert shows:

- `type` — alert category (e.g. "QC_HOLD", "DOWNTIME", "HELP_REQUEST")
- `message` — human-readable description

Active Downtime Panel

Shows currently open downtime events with:

- `downtime_category` — displayed as human-readable label
- `line_id` — production line identifier
- `machine_name` — specific machine affected
- Elapsed minutes since `start_time`
- `impact_level` — coloured badge

Recent Handovers Panel

Shows the 4 most recent shift handover records with:

- `shift_from_name` → `shift_to_name`
- `line_id` or "All lines"

- `outgoing_supervisor`
- Approved / Pending status badge

Quick Navigation Cards

Card	Route	Description
Operator Terminal	<code>/dashboard/shop-floor/terminal</code>	Start, pause, finish work orders
Work Center Queue	<code>/dashboard/shop-floor/queue</code>	Live WO board by production line
Downtime Board	<code>/dashboard/shop-floor/downtime</code>	Track and categorise interruptions
Shift Handover	<code>/dashboard/shop-floor/handover</code>	Document shift transfer records

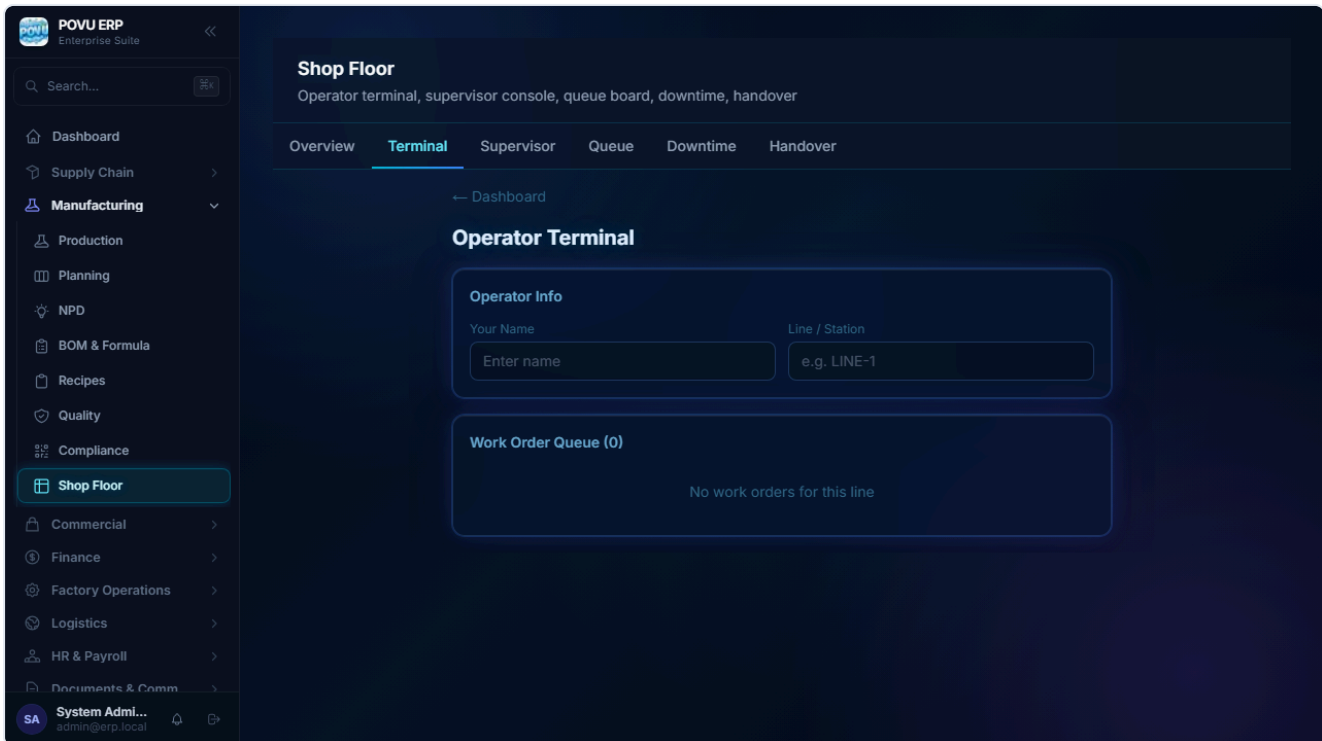
Action Buttons (header)

Button	Action
Run AI Agents	Triggers AI analysis of current shop floor state; invalidates dashboard cache on completion
Operator Terminal	Direct link to <code>/dashboard/shop-floor/terminal</code>
Supervisor Console	Direct link to <code>/dashboard/shop-floor/supervisor</code>

Terminal Tab

Tab key: `terminal`

Route: `/dashboard/shop-floor/terminal/page`



Operator Terminal: full-screen touch interface for starting, pausing, and completing work orders.

The Operator Terminal is a full-screen, simplified interface designed for use on touch screens and shared factory tablets.

Key functions:

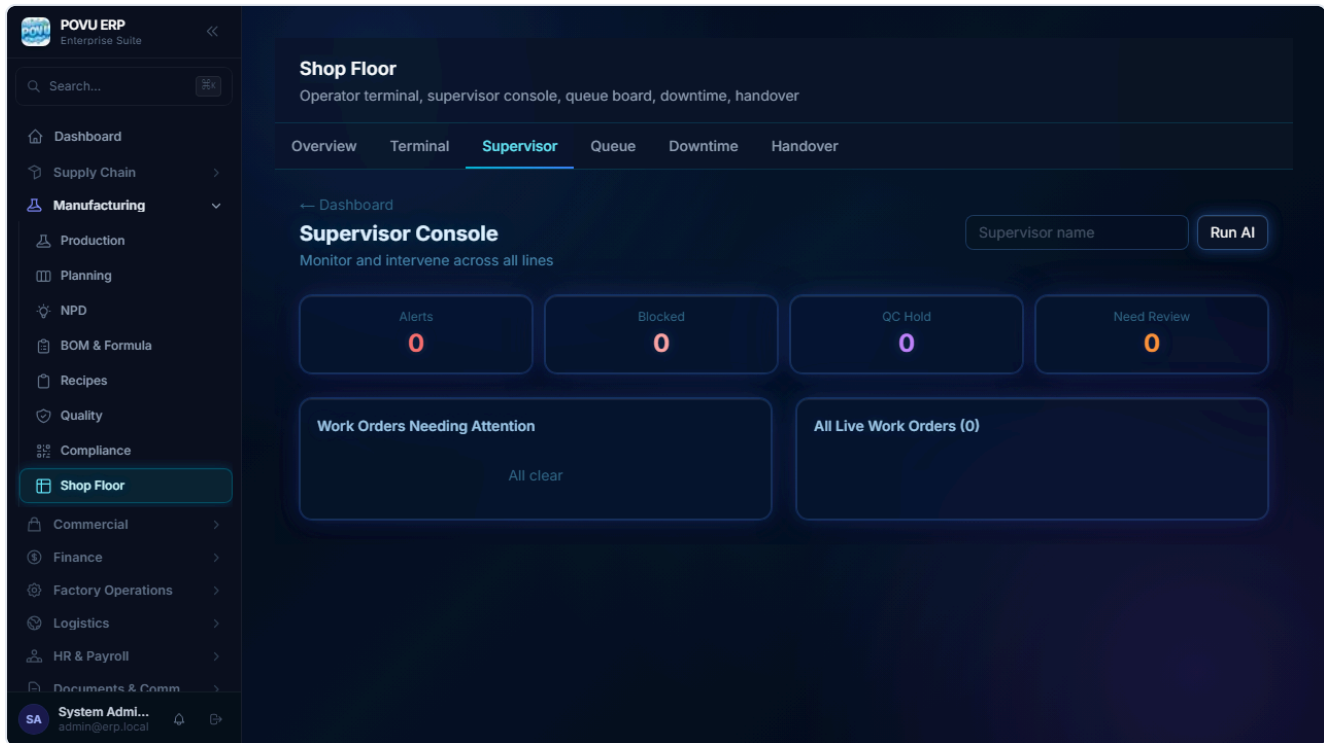
- **Start** a work order — operator selects their work center and the work order to begin; records start time and operator ID
- **Pause** a work order — records pause time and reason (downtime category)
- **Resume** a paused work order
- **Finish** a work order — records completion time, good quantity produced, and scrap quantity

The terminal is accessible directly at `/dashboard/shop-floor/terminal` (standalone page, no sidebar).

Supervisor Tab

Tab key: `supervisor`

Route: `/dashboard/shop-floor/supervisor/page`



Supervisor Console showing all lines, active work orders, and override controls.

The Supervisor Console provides a manager-level view of all active work orders across all lines. Key capabilities:

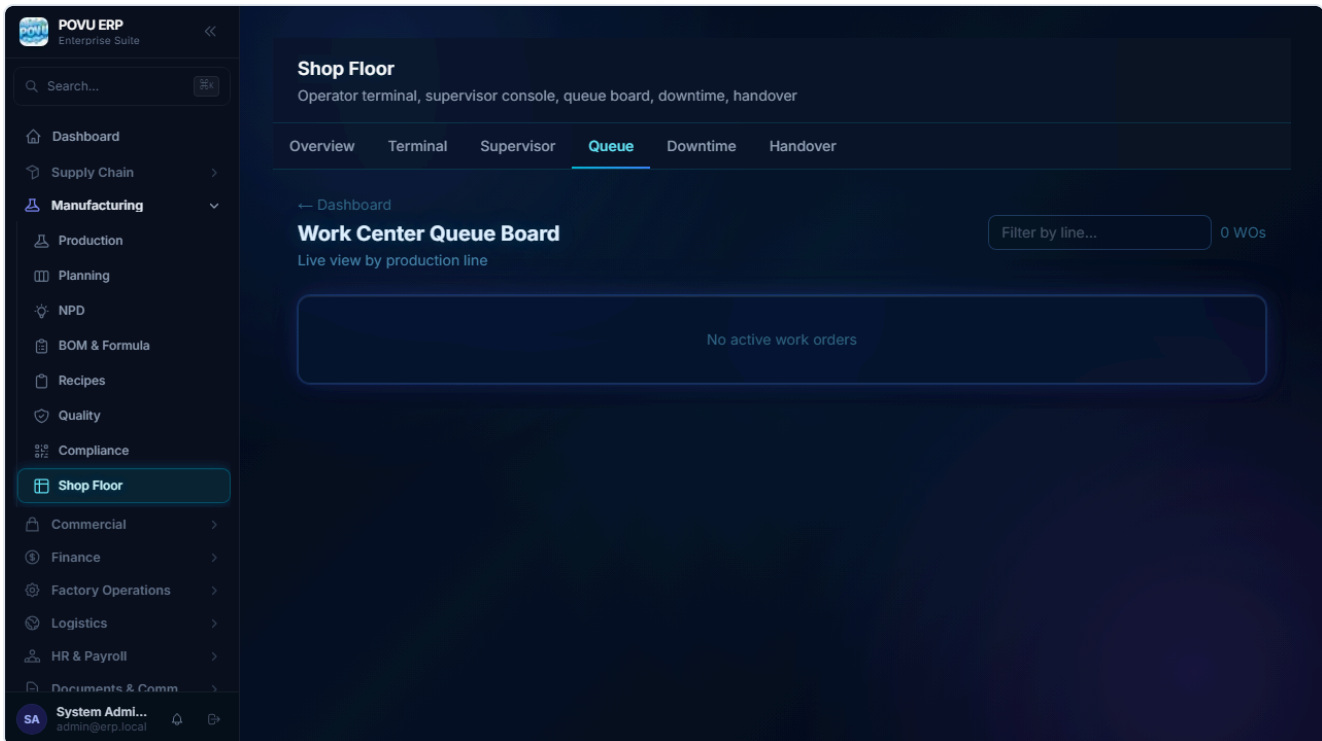
- Override AI recommendations (recorded as `supervisor_overrides`)
- Approve or reject help requests from operators
- Trigger QC holds on specific batches or lines
- View real-time OEE per work center

The supervisor console is accessible directly at `/dashboard/shop-floor/supervisor` (standalone page).

Queue Tab

Tab key: `queue`

Route: `/dashboard/shop-floor/queue/page`



Work Center Queue — Kanban board showing work orders organised by production line.

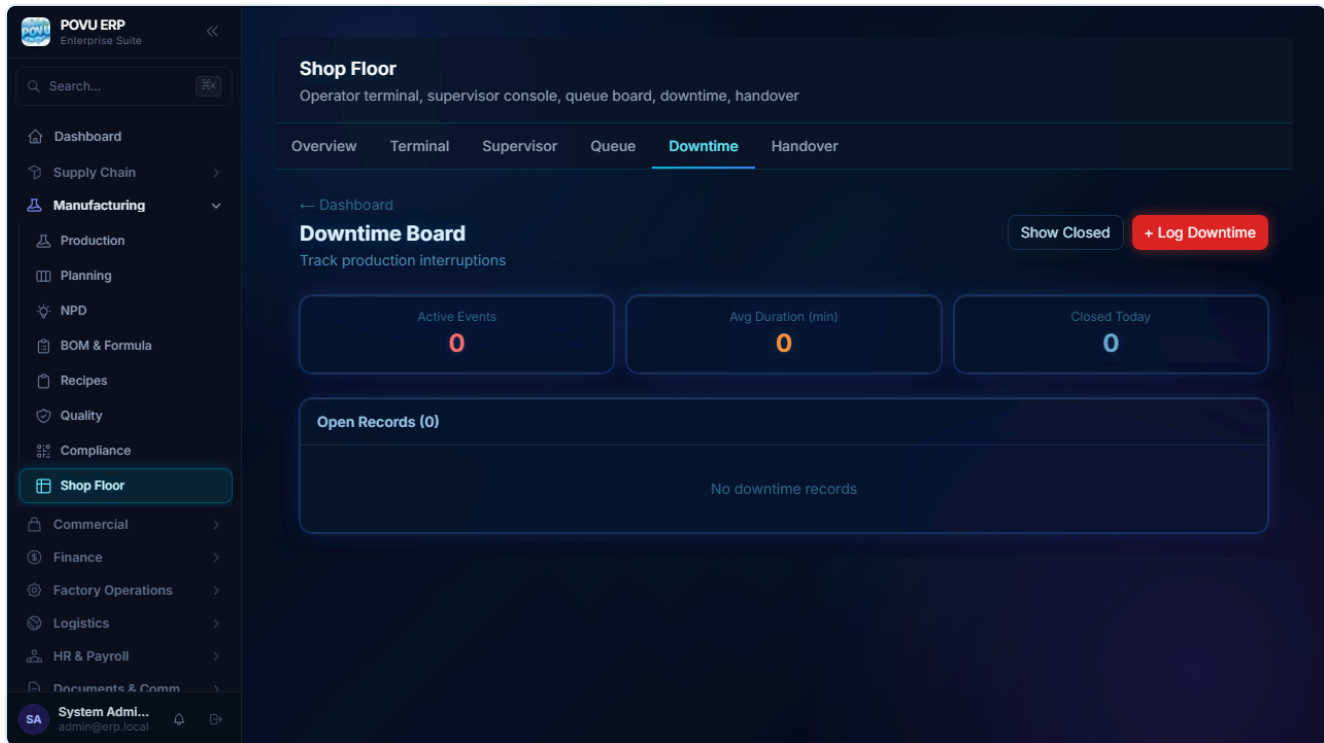
The Work Center Queue is a Kanban-style board showing all work orders organised by production line. Columns represent production lines; cards represent work orders with their current status, target quantity, and operator.

Supervisors can drag work orders between time slots or reassign them to different work centers from this view.

Downtime Tab

Tab key: `downtime`

Route: `/dashboard/shop-floor/downtime/page`



Downtime tab showing open and resolved downtime events with category and impact level.

Downtime Categories

Downtime events are categorised to enable root-cause analysis and OEE calculation. The `DOWNTIME_CAT_LABEL` map translates category codes to human-readable labels. Typical categories:

Category Code	Display Label
<code>BREAKDOWN</code>	Machine Breakdown
<code>PLANNED_MAINTENANCE</code>	Planned Maintenance
<code>CHANGEOVER</code>	Product Changeover
<code>MATERIAL_SHORTAGE</code>	Material Shortage
<code>QUALITY_HOLD</code>	Quality Hold
<code>UTILITIES</code>	Utilities Failure
<code>OPERATOR_ABSENCE</code>	Operator Absence
<code>OTHER</code>	Other

Impact Level

Level	Badge Colour	Meaning
LOW	Grey	Minor delay; line continues
MEDIUM	Yellow	Significant delay; partial output
HIGH	Orange	Line stopped; production halted
CRITICAL	Red	Multi-line or extended stoppage

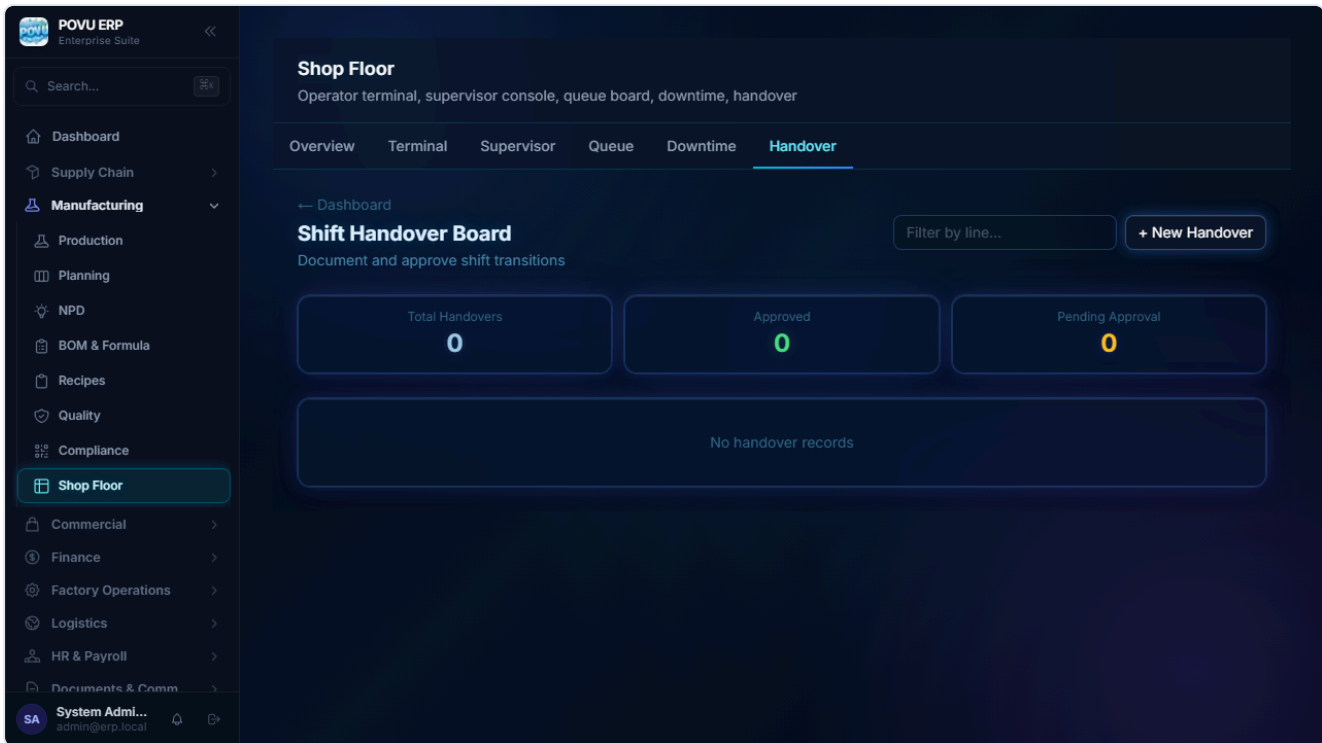
Downtime Record Fields

Field	Description
line_id	Production line affected
machine_name	Specific machine (optional)
downtime_category	Root cause category
impact_level	LOW / MEDIUM / HIGH / CRITICAL
start_time	When downtime started
end_time	When downtime ended (null if still active)

Handover Tab

Tab key: handover

Route: /dashboard/shop-floor/handover/page



Handover tab with shift transfer records, approval status, and open items being handed over.

Shift handover documents the formal transfer of production responsibility between shifts.

Handover Record Fields

Field	Description
<code>shift_from_name</code>	Outgoing shift name (e.g. "Morning Shift")
<code>shift_to_name</code>	Incoming shift name (e.g. "Afternoon Shift")
<code>line_id</code>	Production line(s) covered; "All lines" if facility-wide
<code>outgoing_supervisor</code>	Name of outgoing supervisor
<code>is_approved</code>	Whether incoming supervisor has approved the handover

A handover is not complete until the incoming supervisor marks `is_approved = true`. Pending handovers (not yet approved) are shown with a yellow badge.

What a Handover Records

- Active work orders at shift end (status, quantity produced so far)
- Open downtime events being passed over
- Quality holds in effect
- Help requests unresolved

- Machine status notes
- Any safety incidents during the shift

Advanced Planning & MRP

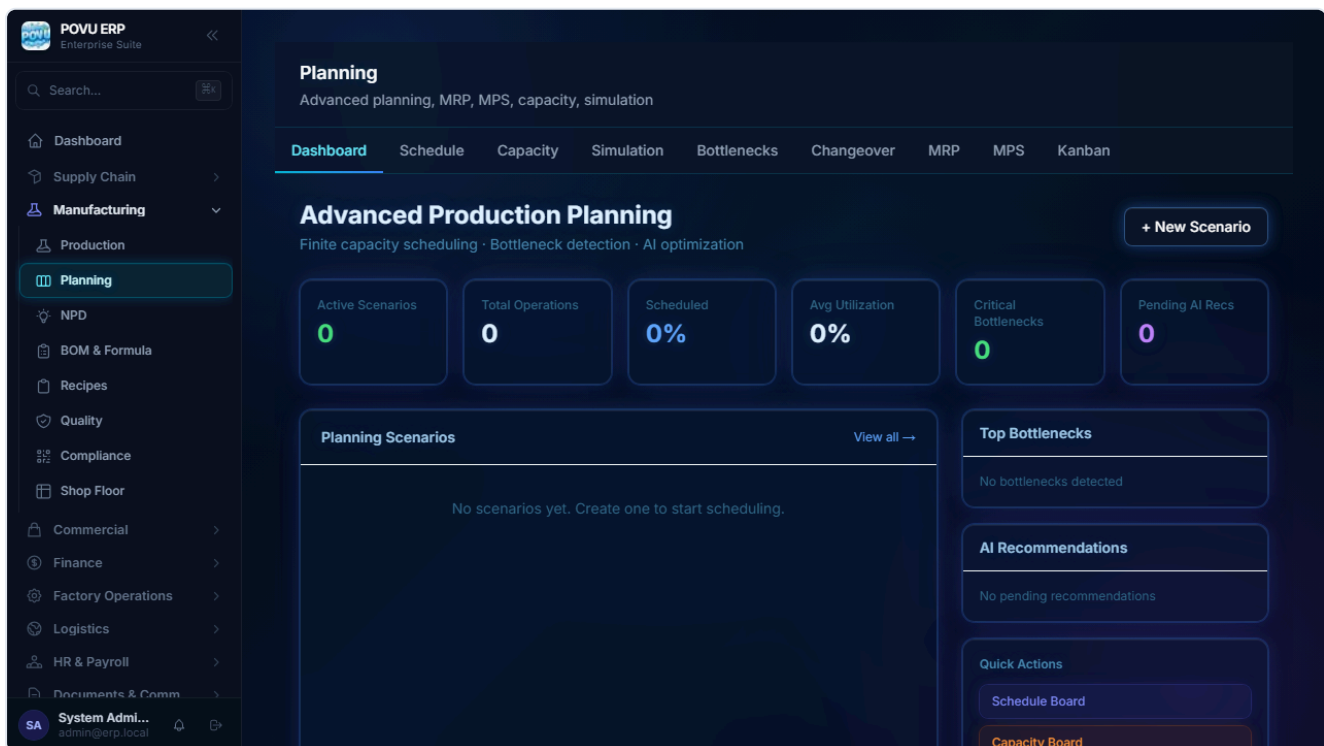
Route: /dashboard/planning

Permission required: planning.view

Workspace tabs: Dashboard, Schedule, Capacity, Simulation, Bottlenecks, Changeover, MRP, MPS, Kanban

What It Does

The Advanced Planning module provides finite and infinite capacity scheduling, material requirements planning (MRP), master production scheduling (MPS), bottleneck detection, and AI-driven optimisation. It is built for production planners who need to schedule across multiple work centers, manage capacity constraints, and align production with demand.



Advanced Planning workspace showing the dashboard with scenario KPIs, scenario list, and navigation tabs.

Planning Dashboard Tab

Tab key: advanced

Dashboard KPIs

KPI	Field	Description
Active Scenarios	<code>active_scenarios</code>	Planning scenarios currently in use
Total Ops Today	<code>total_ops_today</code>	Total scheduled operations for today
Scheduled %	<code>scheduled_pct</code>	Percentage of operations scheduled vs. total required
Avg Utilisation	<code>avg_utilization_pct</code>	Average capacity utilisation across all work centers
Critical Bottlenecks	<code>critical_bottlenecks</code>	Count of work centers at critical overload
Pending AI Recs	<code>pending_ai_rec</code>	AI recommendations awaiting review

Scenario List

Each scenario row shows:

- `scenario_no` — system reference number
- `scenario_name` — user-assigned name (clickable link to Schedule Board)
- `status` — scenario state (coloured badge)
- `mode` — FINITE or INFINITE capacity mode
- `horizon_start` → `horizon_end` — planning horizon date range
- `scheduled_ops` / `total_ops` — progress fraction
- `bottleneck_count` — count of detected bottlenecks
- `calculated_at` — date of last calculation

Status badge colours depend on the scenario status (DRAFT, ACTIVE, ARCHIVED, etc.).

Top Bottlenecks Panel

Shows the most constrained work centers with:

- `work_center_name` — name of the bottleneck work center
- `severity` — bottleneck severity badge
- `peak_utilization_pct` — highest utilisation percentage in the horizon
- `overloaded_days` — count of days with utilisation > 100%

AI Recommendations Panel

Pending AI recommendations from the planning agents:

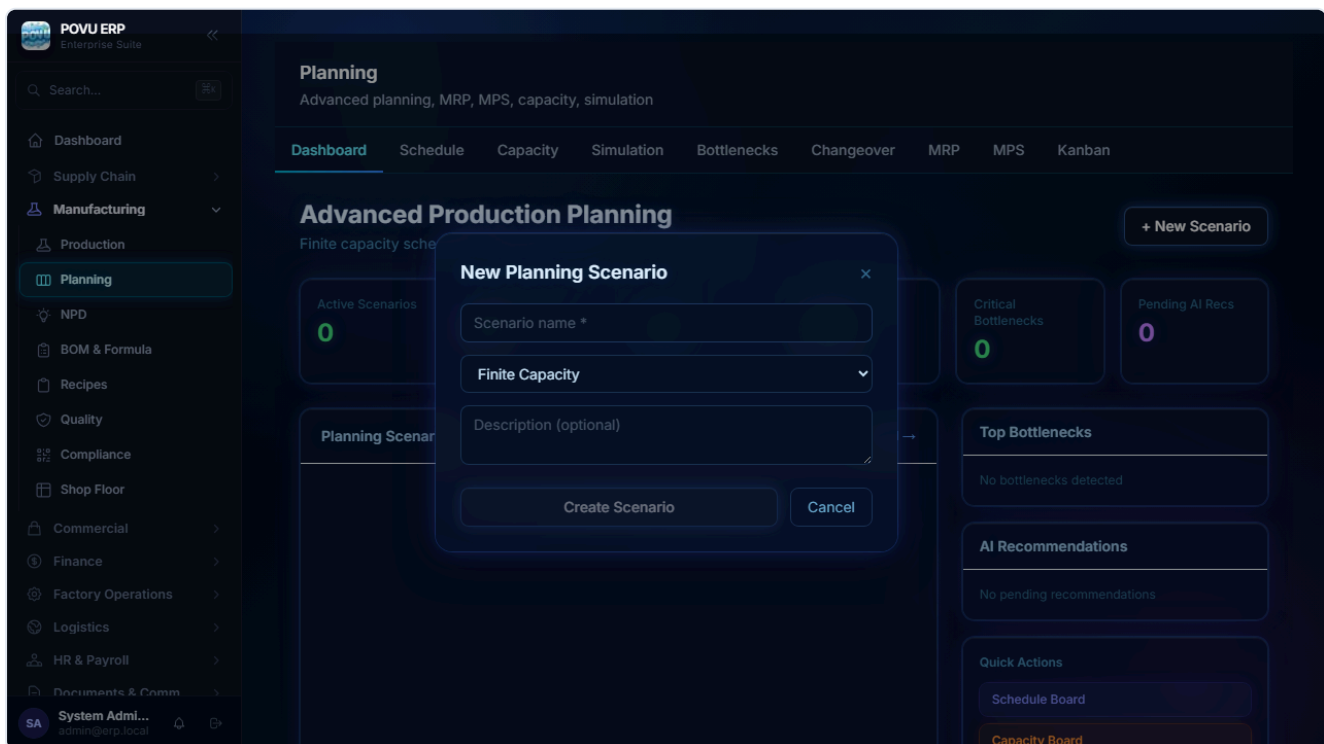
- `agent_type` — which AI agent generated it (coloured badge)
- `title` — recommendation summary

Quick Navigation

Action	Route
Schedule Board	<code>/dashboard/planning/schedule</code>
Capacity Board	<code>/dashboard/planning/capacity</code>
Bottleneck Explorer	<code>/dashboard/planning/bottlenecks</code>
Simulation Sandbox	<code>/dashboard/planning/simulation</code>
Changeover Matrix	<code>/dashboard/planning/changeover</code>

Creating a Planning Scenario

Button: `+ New Scenario`



New Planning Scenario modal — Scenario Name, Capacity Mode, and Description fields.

New Planning Scenario Fields

Field	Label	Required	Default	Notes
<code>scenario_name</code>	Scenario name	Yes	—	Descriptive name; e.g. "Q2 2026 Baseline"

Field	Label	Required	Default	Notes
mode	Capacity mode	Yes	FINITE	FINITE (respects work center limits) / INFINITE (no capacity constraints)
description	Description	No	—	Optional notes

Submit disabled until `scenario_name` is filled.

Capacity Modes

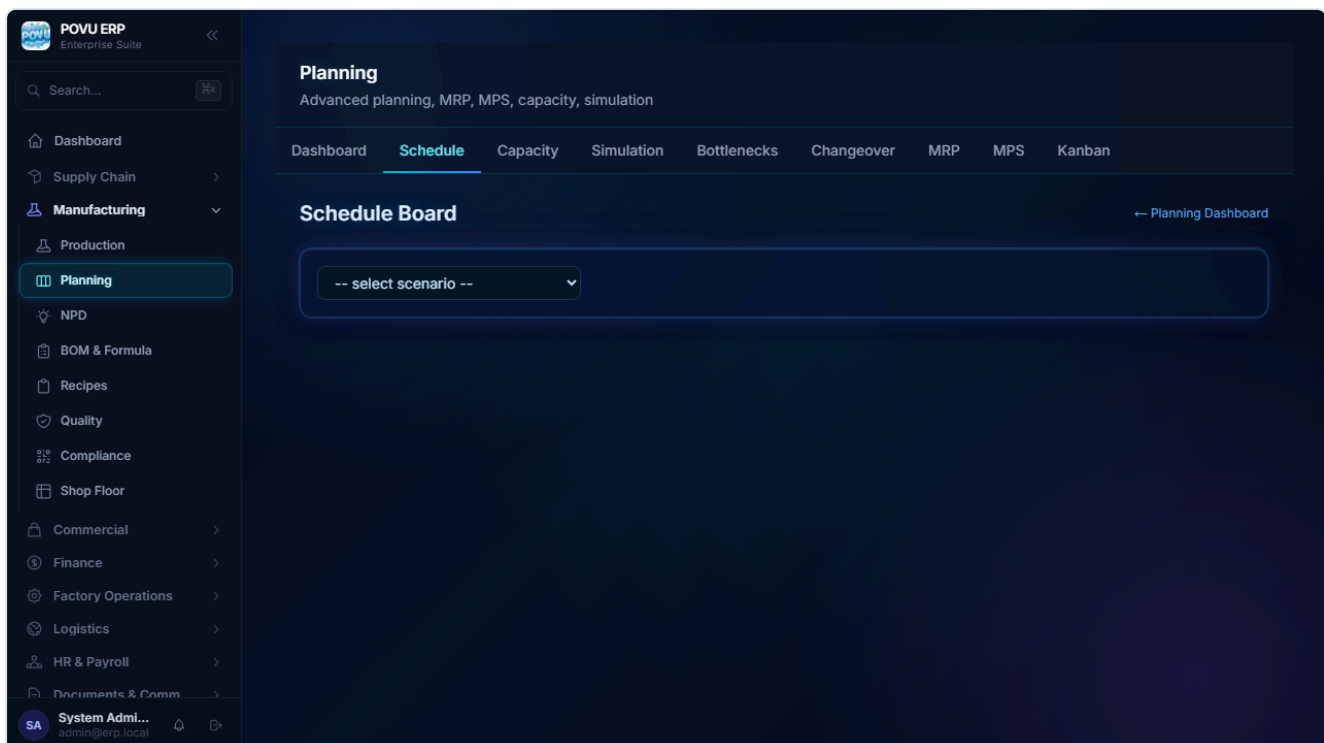
Mode	When to Use
FINITE	Standard scheduling — work orders queue when capacity is full; bottlenecks are identified
INFINITE	"What-if" analysis — schedule without capacity limits; shows theoretical throughput

Start with a FINITE scenario for operational planning. Use INFINITE for demand-side what-if analysis.

Schedule Tab

Tab key: `schedule`

Route: `/dashboard/planning/schedule/page`



Schedule tab — interactive Gantt board showing operations per work center per day.

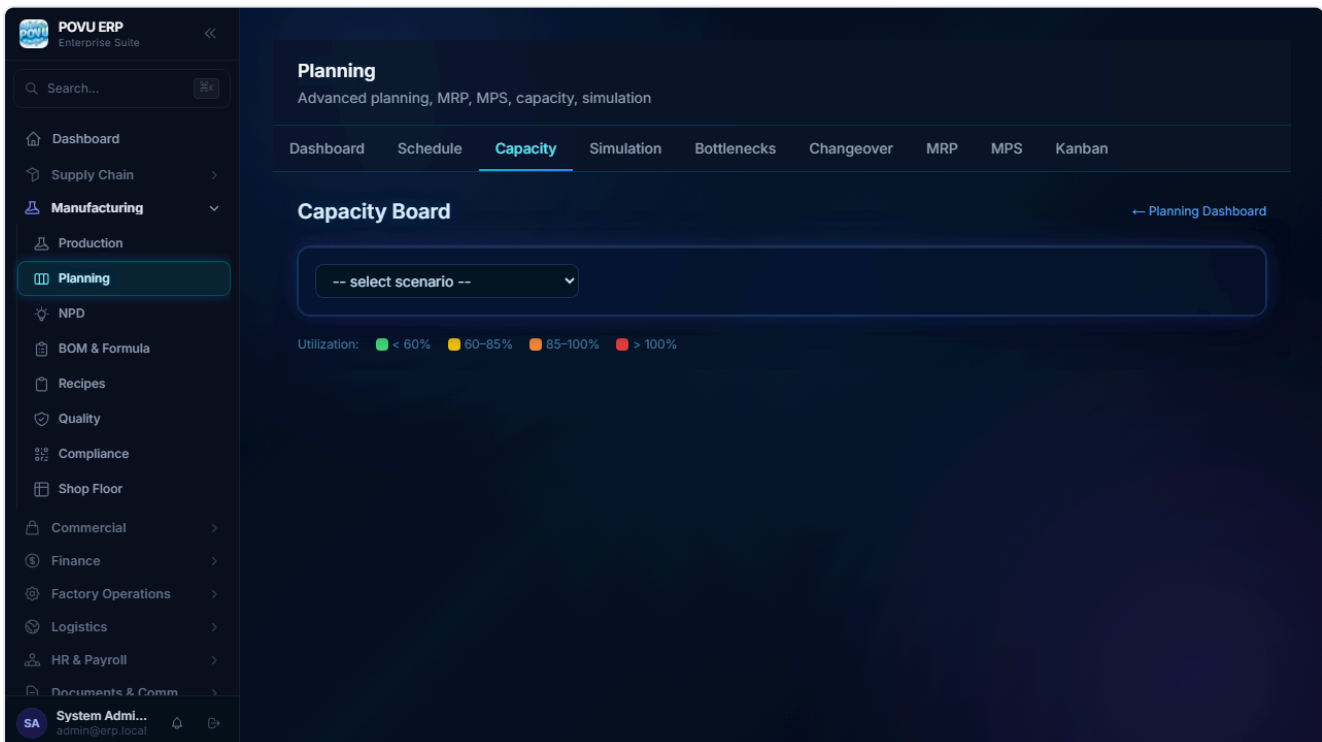
Interactive Gantt board showing scheduled operations per work center per day/shift. Drag-and-drop to reschedule. Conflicts highlighted in red.

Navigate to a specific scenario via URL: `/dashboard/planning/schedule?scenario={id}`

Capacity Tab

Tab key: `capacity`

Route: `/dashboard/planning/capacity/page`



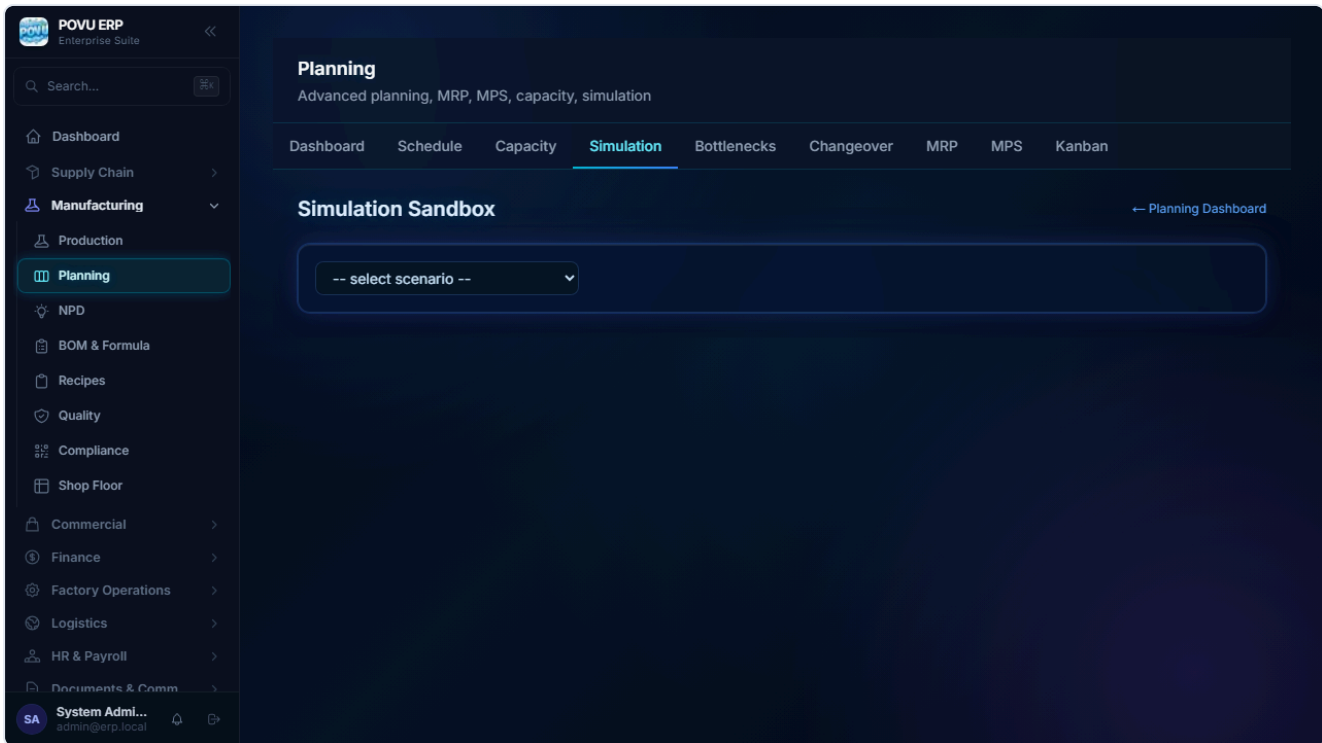
Capacity tab showing work center load vs. available capacity for the planning horizon.

Visual capacity loading chart per work center for the planning horizon. Shows available capacity, scheduled load, and overloaded periods. Used to identify where to offload or reschedule.

Simulation Tab

Tab key: `simulation`

Route: `/dashboard/planning/simulation/page`



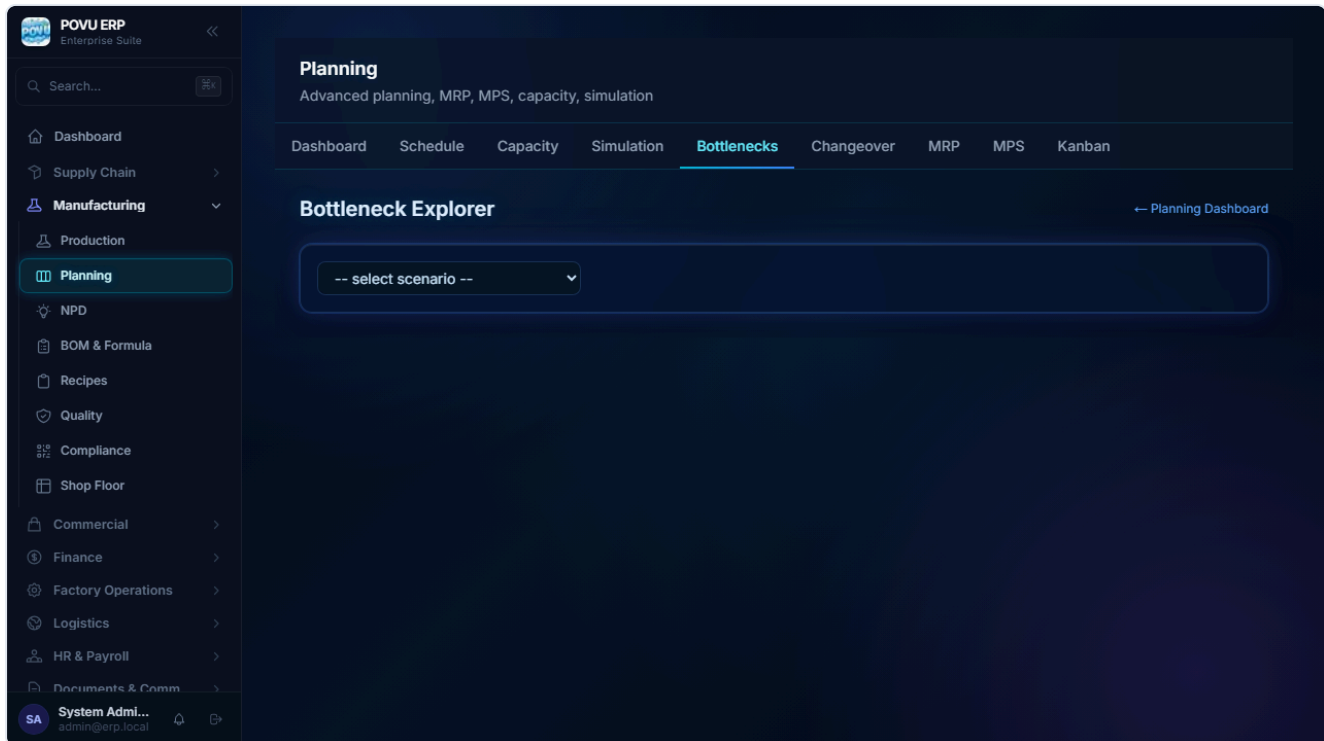
Simulation tab — sandbox environment for testing alternative schedules without affecting operational plans.

Sandbox environment to test alternative schedules without affecting operational plans. Create a simulation scenario, apply changes (shift priorities, add/remove work orders, change capacity), and compare results with the baseline.

Bottlenecks Tab

Tab key: `bottlenecks`

Route: `/dashboard/planning/bottlenecks/page`



Bottlenecks tab showing constrained work centers with overload severity and AI recommendations.

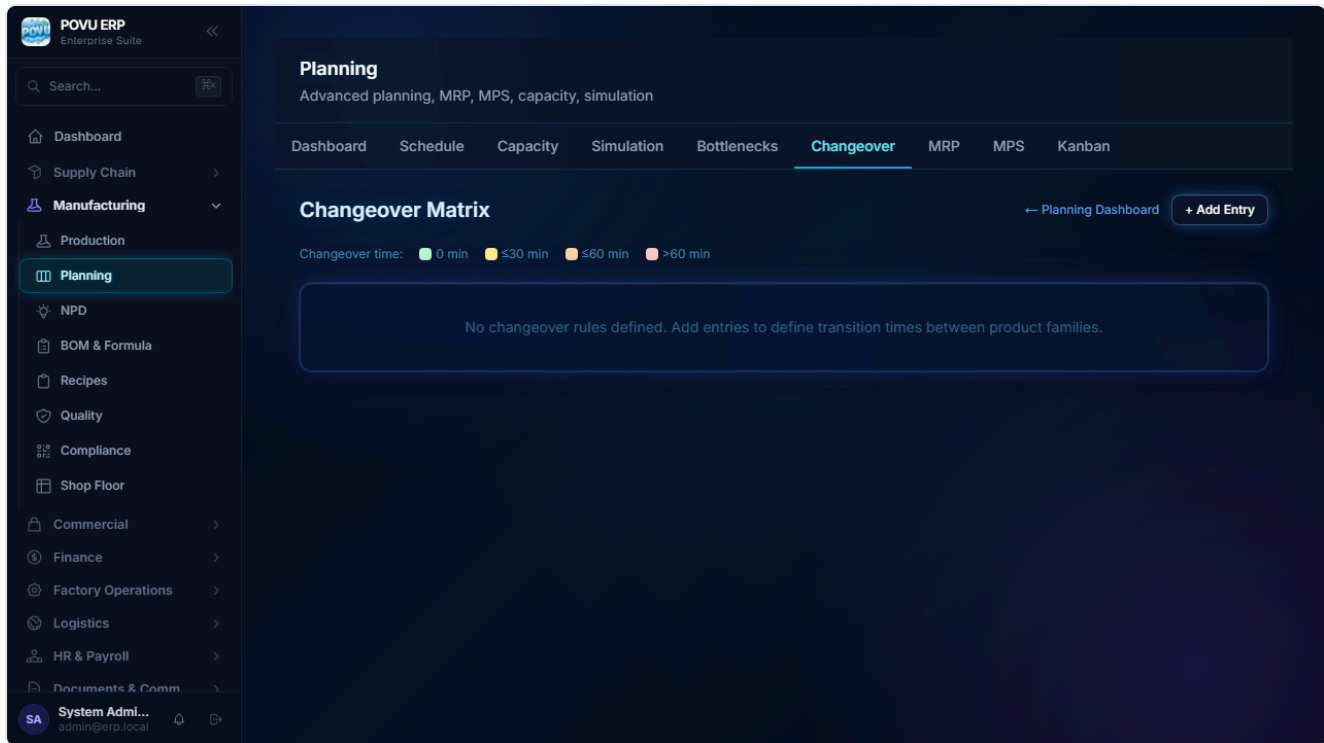
Detailed bottleneck analysis explorer. Shows each constrained work center with:

- Overload severity
- Number of overloaded days in the horizon
- Peak utilisation date
- Work orders contributing to overload
- AI recommendations for resolution

Changeover Tab

Tab key: `changeover`

Route: `/dashboard/planning/changeover/page`



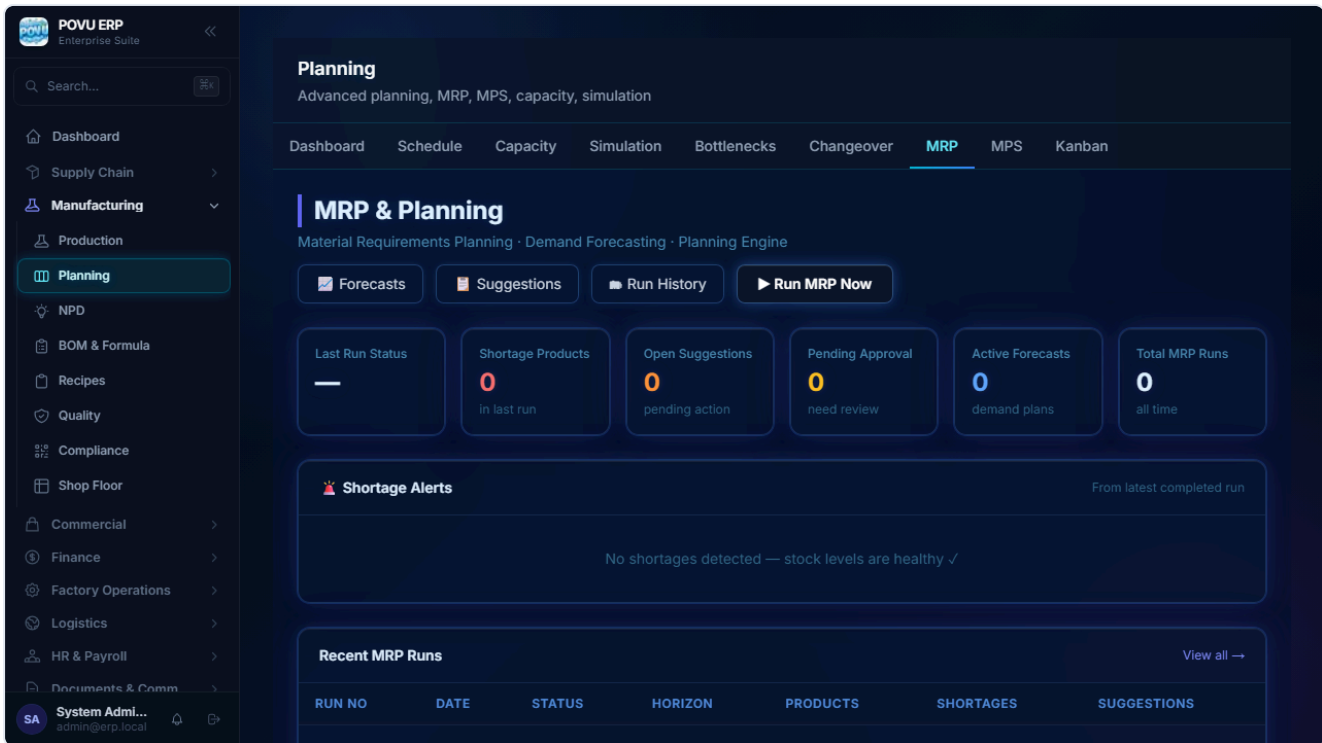
Changeover Matrix tab defining product-to-product setup times per work center.

Changeover matrix management. Defines the setup time required when switching between products on a given work center (e.g. from product A to product B on Line 1 = 45 minutes). The scheduler uses these times when sequencing work orders to minimise total changeover duration.

MRP Tab

Tab key: `mrp`

Route: `/dashboard/mrp/page`



MRP tab showing material requirements plan output with planned orders and suggested actions.

Material Requirements Planning calculates what materials need to be ordered or produced, by when, to fulfil the demand plan.

MRP run inputs:

- Demand (from MPS or sales orders)
- Current stock on hand
- Open purchase orders
- BOM explosions
- Planned production orders

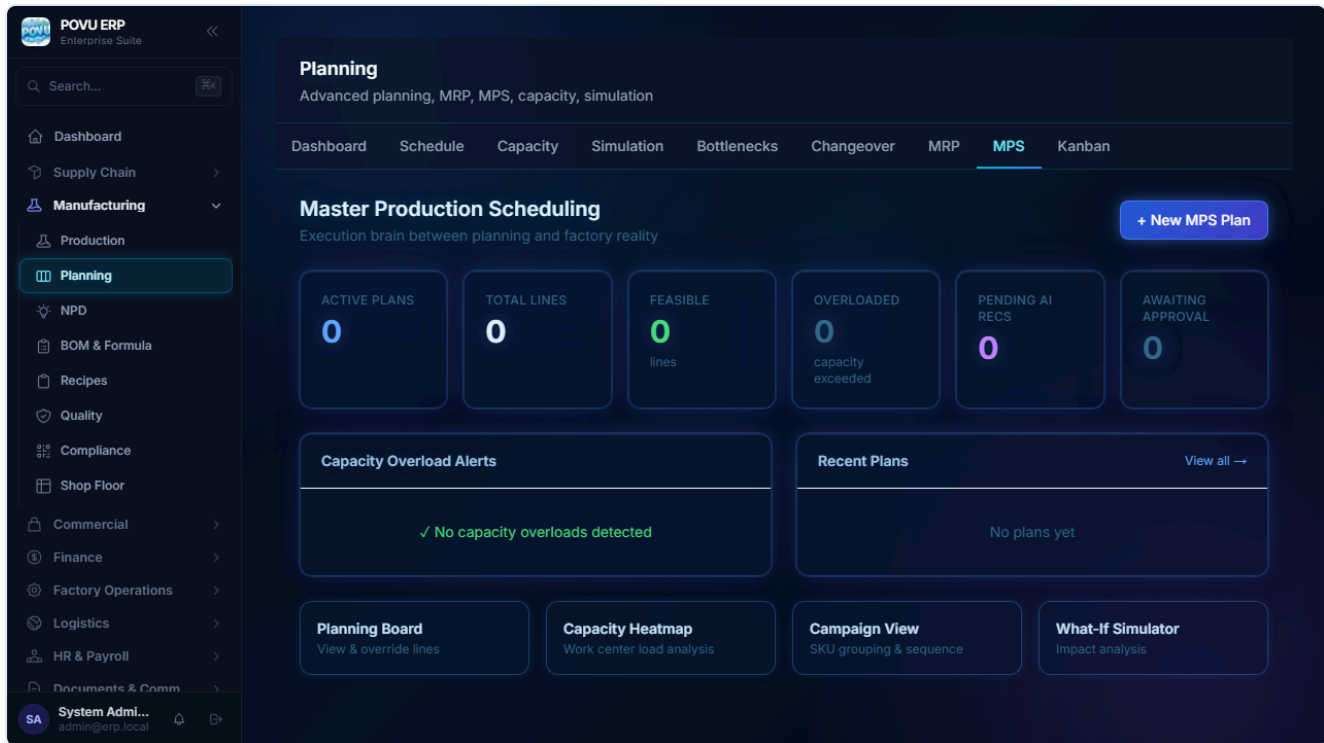
MRP run outputs:

- Planned purchase orders (with suggested due dates)
- Planned production orders
- Action messages (expedite, defer, cancel)

MPS Tab

Tab key: mps

Route: /dashboard/mps/page



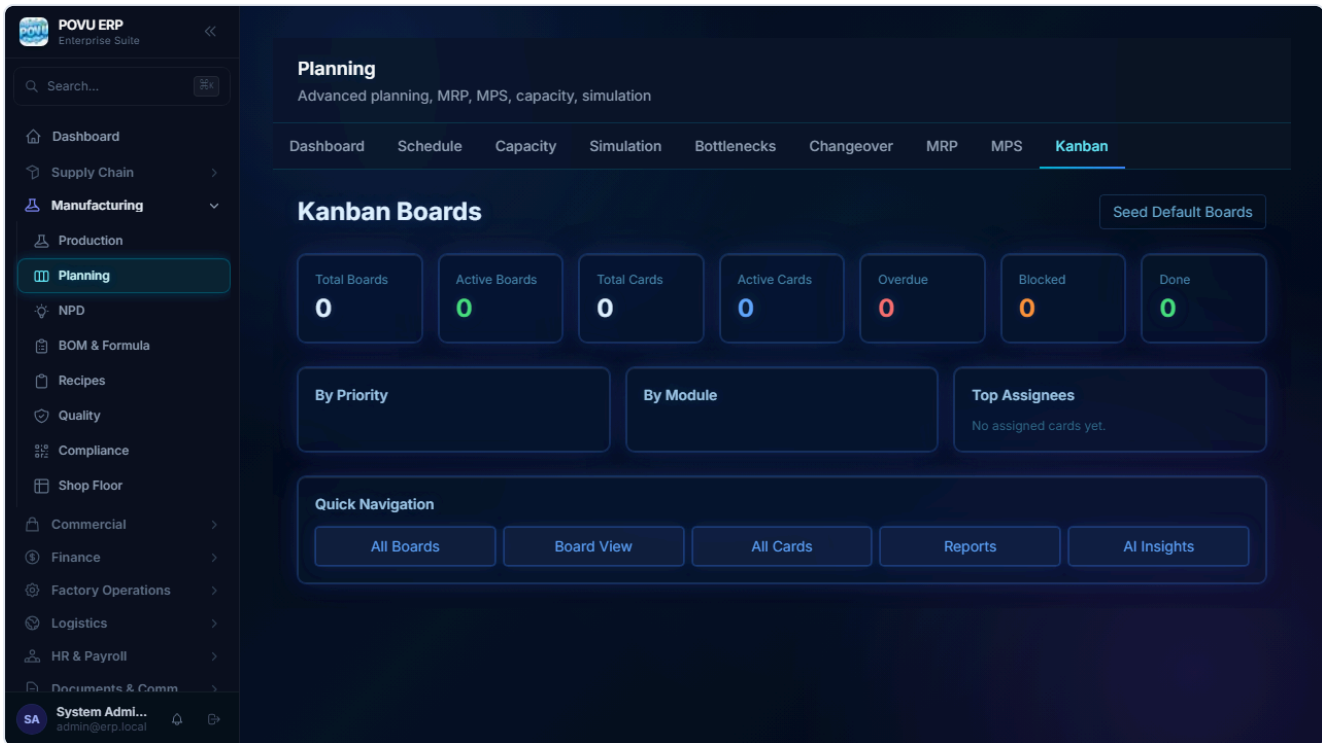
Master Production Schedule tab showing the time-phased production schedule by product.

Master Production Schedule defines the production plan at the finished goods level across the planning horizon. MPS drives MRP. Planners manage MPS quantities by product and week/day.

Kanban Tab

Tab key: kanban

Route: /dashboard/kanban/page



Kanban tab for pull-based replenishment — card quantities, bin sizes, and replenishment triggers.

Kanban-based replenishment for high-velocity materials or work centers that operate on pull scheduling rather than MRP push scheduling. Manages Kanban card quantities, bin sizes, and replenishment triggers.

New Product Development

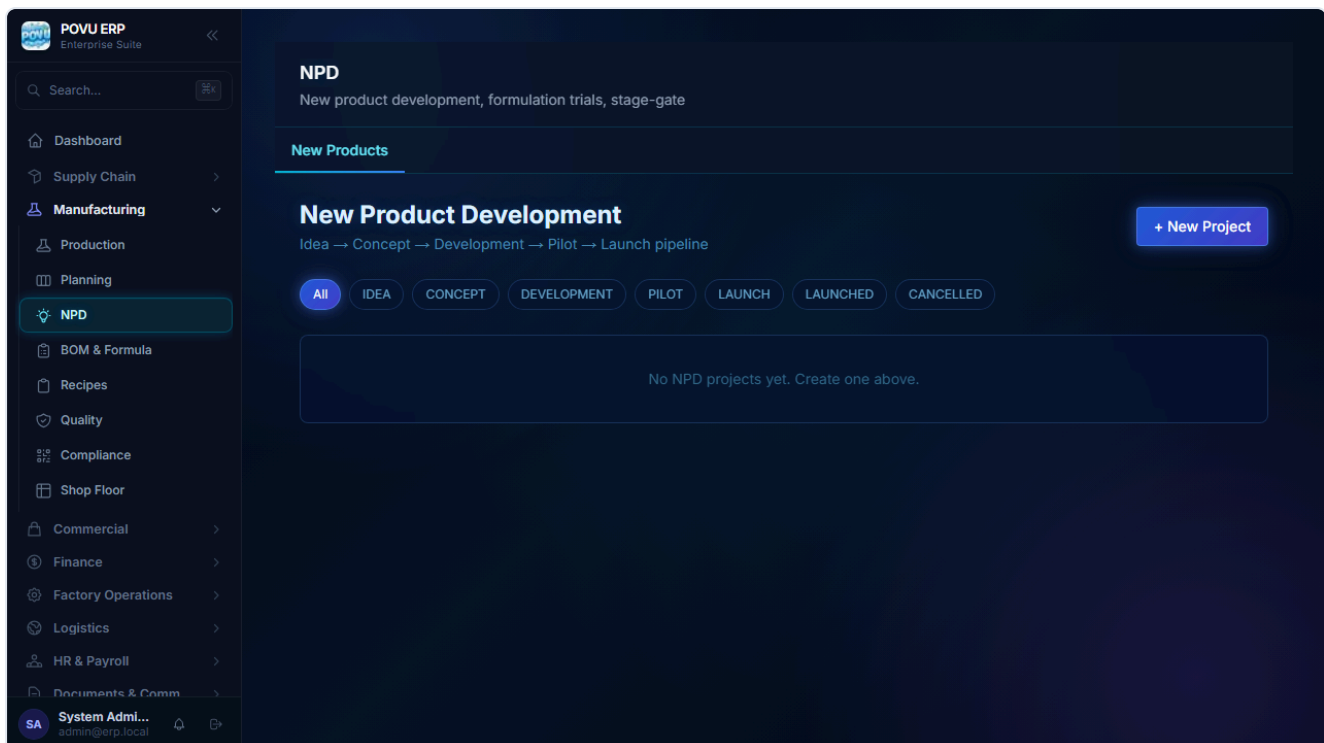
Route: `/dashboard/npd`

Permission required: `npd.view`

Workspace tabs: New Products (single tab)

What It Does

The NPD module manages the stage-gate pipeline for new product development — from an initial idea through concept, development, pilot trials, and commercial launch. Each project tracks the target market, estimated cost of goods, brand, and target launch date.



NPD workspace showing the stage-gate pipeline with project cards grouped by stage.

NPD Dashboard

KPI Cards

Card	Description
Total Active	Count of all projects not in CANCELLED or LAUNCHED status

Card	Description
PILOT	Projects currently in the PILOT stage
LAUNCH	Projects in the LAUNCH stage (about to go live)
LAUNCHED	Projects that have reached commercial launch

Stage Pipeline Bar

Clickable filter chips for each stage:

- **All** — show all projects
- **IDEA** — initial idea; not yet validated
- **CONCEPT** — concept approved; preliminary costings started
- **DEVELOPMENT** — active R&D; formula trials underway
- **PILOT** — small-scale trial production run
- **LAUNCH** — final commercial approval; supply chain being set up
- **LAUNCHED** — live in the market
- **CANCELLED** — discontinued

Clicking a stage chip filters the project grid. Clicking the active chip again clears the filter.

Project Card Grid

Projects are displayed as cards (not a table). Each card shows:

- `name` — project/product name
- `project_code` — system-generated reference in monospace font
- Stage badge (coloured by stage)
- `brand` chip (if set)
- `category` chip (with underscores replaced by spaces)
- `target_launch_date` — shown as "Target: YYYY-MM-DD"
- `estimated_cogs` — shown as "Est. COGS: KES {value}/unit"

Clicking a card navigates to the project detail page at `/dashboard/npd/{id}`.

Creating an NPD Project

Button: `+ New Project` (requires `npd.create`)

New NPD Project inline form — Product Name, Category, Brand, Target Market, Launch Date, Est. COGS/Unit, Description.

New Project Form Fields

Field	Label	Required	Default	Notes
<code>name</code>	Product Name	Yes	—	e.g. "New Mango Juice 500ml"
<code>category</code>	Category	Yes	<code>NEW_PRODUCT</code>	See category options below
<code>brand</code>	Brand	No	—	e.g. "POVU Naturals"
<code>target_market</code>	Target Market	No	—	e.g. "Kenya Urban"
<code>target_launch_date</code>	Target Launch Date	No	—	Date picker
<code>estimated_cogs</code>	Est. COGS/Unit (KES)	No	—	Decimal; e.g. <code>25.50</code>
<code>description</code>	Description	No	—	Brief summary

Submit disabled until `name` is filled. A `project_code` is auto-generated by the backend on creation.

New projects start in `IDEA` stage automatically.

Project Categories

Category	Display
NEW_PRODUCT	New Product
LINE_EXTENSION	Line Extension
REFORMULATION	Reformulation
PACK_SIZE_CHANGE	Pack Size Change
COST_REDUCTION	Cost Reduction

Stage Colours

Stage	Badge Style
IDEA	Grey background, grey text
CONCEPT	Blue-100 background, blue-700 text
DEVELOPMENT	Indigo-100 background, indigo-700 text
PILOT	Amber-100 background, amber-700 text
LAUNCH	Orange-100 background, orange-700 text
LAUNCHED	Green-100 background, green-700 text
CANCELLED	Red-100 background, red-600 text

NPD Project Detail Page

Route: `/dashboard/npd/{project_id}`

The detail page provides:

- Stage advancement controls (move project to next stage)
- Formulation trials linked to Recipes module
- BOM linkage (link the NPD project to a BOM once formula is approved)
- Cost modelling worksheets
- Gate review records (approval decisions at each stage gate)
- Document attachments (specifications, regulatory submissions)
- Task list for the R&D and commercialisation team

Stage Gate Rules

Each stage gate requires specific documentation and approvals before advancing:

Gate	From → To	Typical Requirements
Gate 1	IDEA → CONCEPT	Business case approved by NPD manager
Gate 2	CONCEPT → DEVELOPMENT	Preliminary formula exists; market size confirmed
Gate 3	DEVELOPMENT → PILOT	Formula approved; regulatory pre-check done
Gate 4	PILOT → LAUNCH	Pilot trial passed; full costing complete; supply chain ready
Gate 5	LAUNCH → LAUNCHED	First production run complete; trade launch confirmed

API Endpoints (Reference)

Method	Path	Description
GET	/api/v1/npd-workflow/projects	List projects (filter: <code>stage</code> , limit: 100 default)
POST	/api/v1/npd-workflow/projects	Create project
GET	/api/v1/npd-workflow/projects/{id}	Project detail
GET	/api/v1/npd-workflow/dashboard	Dashboard KPIs

Permissions

Action	Permission
View NPD project list	<code>npd.view</code>
Create new project	<code>npd.create</code>
Advance project stage	<code>npd.create</code> (or higher)
View project detail	<code>npd.view</code>

OOE, Downtime & Yield Reporting

Routes: /dashboard/production?tab=oeo , ?tab=downtime , ?tab=waste-yield , ?tab=reports , ?tab=variance

Permission required: production.view

Overall Equipment Effectiveness (OOE)

Tab key: oee **Route:** /dashboard/production/oeo/page

OOE ID	WORK CENTER	MACHINE ID	DATE	SHIFT	PLANNED (MIN)	DOWNTIME (MIN)	AVAILABILITY	PERFORMANCE	QUALITY	OOE	ALERTS
OOE-MC-PACK-01-20260604	Packaging & Labelling	—	2026-06-04	—	480	44	91.0%	92.0%	97.8%	81.9%	OK
OOE-MC-FILL-SAC-20260604	Sachet Filling Line	—	2026-06-04	—	480	77	84.0%	84.0%	97.6%	68.9%	OK
OOE-MC-FILL-BOT-20260604	Bottle Filling Line	—	2026-06-04	—	480	58	88.0%	87.0%	97.4%	74.6%	OK
OOE-MC-MIX-02-20260604	Mixing & Blending Line B	—	2026-06-04	—	480	101	79.0%	77.0%	97.2%	59.1%	OK
OOE-MC-MIX-01-20260604	Mixing & Blending Line A	—	2026-06-04	—	480	87	82.0%	79.0%	97.0%	62.8%	OK
OOE-MC-MIX-02-20260603	Mixing & Blending Line B	—	2026-06-03	—	480	104	78.5%	77.0%	97.2%	58.8%	OK
OOE-MC-PACK-01-	Packaging &	—	2026-06-	—	480	46	90.5%	92.0%	97.8%	81.4%	OK

OOE tab showing seeded Availability, Performance, and Quality metrics per work center from TASK-015.2 seed data.

OOE is the primary manufacturing performance metric. It measures how effectively production equipment is being used compared to its full potential.

OOE Formula

$$\text{OOE} = \text{Availability} \times \text{Performance} \times \text{Quality}$$

$$\text{Availability} = (\text{Planned time} - \text{Downtime}) / \text{Planned time}$$

$\text{Performance} = (\text{Actual output} / \text{Maximum possible output}) \times 100$
 $\text{Quality} = (\text{Good output} / \text{Total output}) \times 100$

OEE Benchmark Targets

OEE	World-Class	Good	Acceptable	Poor
%	≥ 85%	70–85%	60–70%	< 60%

OEE Dashboard Metrics

Metric	Description
OEE %	Composite OEE score per work center or line
Availability %	Time equipment was available vs. scheduled
Performance %	Speed efficiency vs. rated speed
Quality %	Good output vs. total output
Planned vs. Actual	Comparison of planned production vs. actual completed

OEE by Work Center

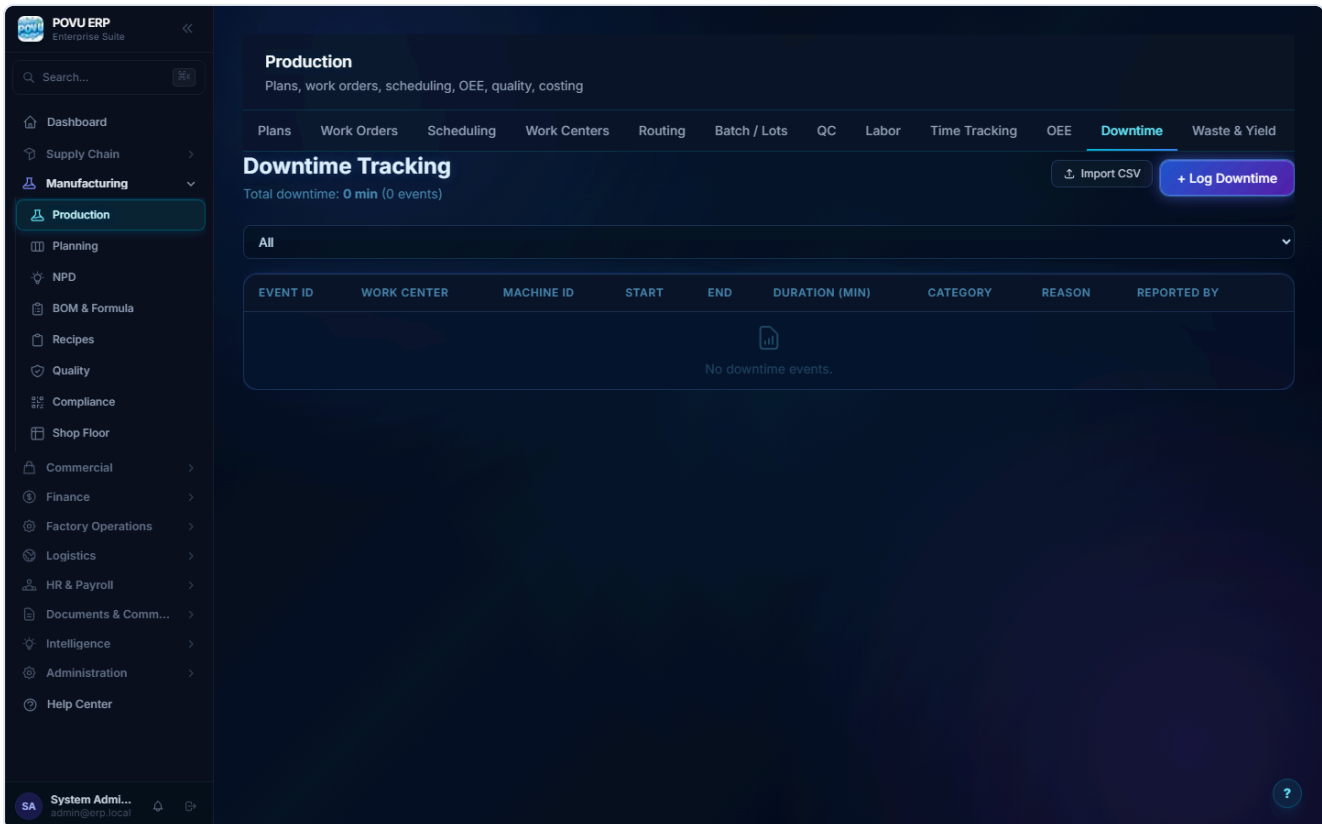
The OEE tab shows OEE broken down per work center for the selected period. Work centers below the target OEE threshold are highlighted.

OEE Trend Chart

Historical OEE trend for the selected work center and date range. Used to detect systematic decline (equipment wear, process drift) or improvement after maintenance interventions.

Downtime Analysis

Tab key: `downtime` **Route:** `/dashboard/production/downtime/page`



Downtime tab showing seeded downtime log with categories, durations, work center, and Pareto analysis.

Downtime Impact on OEE

All downtime events recorded in the Shop Floor module feed directly into OEE Availability calculations. Planned downtime (maintenance windows) is excluded from availability calculations; unplanned downtime reduces availability.

Downtime Report Metrics

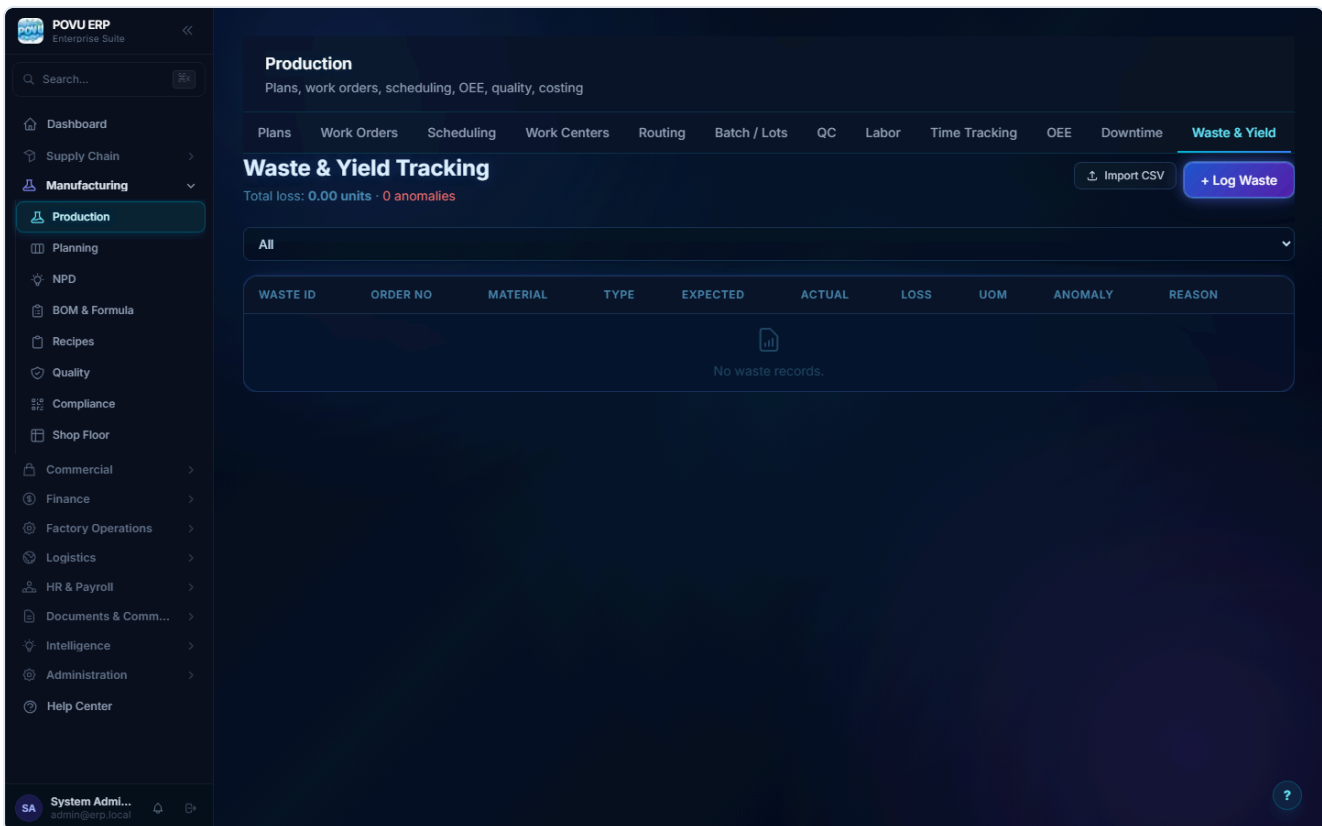
Metric	Description
Total downtime minutes	Sum of all downtime durations in the period
Downtime by category	Pareto chart — which categories cause most lost time
Mean Time Between Failures (MTBF)	Average operating time between breakdowns
Mean Time To Repair (MTTR)	Average repair duration
Top 3 machines by downtime	Identifies most problematic equipment

Pareto Analysis

The downtime tab provides a Pareto chart of downtime by category. The 80/20 rule applies: typically 20% of root causes drive 80% of lost time. Focus corrective actions on the top causes.

Waste & Yield

Tab key: waste-yield **Route:** /dashboard/production/waste-yield/page



Waste & Yield tab showing seeded scrap quantities, yield percentages, and trend analysis per work order.

Key Metrics

Metric	Formula	Notes
Yield %	$\text{Good Qty} / (\text{Good Qty} + \text{Scrap Qty}) \times 100$	Per work order or per period
Waste Qty	Sum of scrap quantities	In production UOM (KG, L, etc.)
Waste Cost	$\text{Waste Qty} \times \text{Material Cost}$	Material value of waste
Loss %	$100 - \text{Yield \%}$	Inverse of yield

Waste by Category

Waste is categorised at the work order level. Common waste categories:

- **Startup waste** — lost during line startup before process stabilises
- **Shutdown waste** — product remaining in lines at shutdown
- **Quality reject** — off-spec product removed at QC checks
- **Overflow / spill** — process losses from equipment

Yield vs. Recipe Standard

Each work order has a **standard yield** from the recipe's `loss_percentage` settings. The Waste & Yield tab compares:

- Actual yield vs. standard yield
- Yield variance (favourable if actual > standard; adverse if actual < standard)

Production Reports Tab

Tab key: `reports`

Route: `/dashboard/production/reports/page`

The screenshot shows the 'Production Reports' tab in the POVU ERP system. The interface includes a sidebar with navigation options like Dashboard, Supply Chain, Manufacturing, and Production. The main content area displays a 'Production Reports' section with a subtitle 'Planned vs actual, downtime, and OEE summary'. It features date filters for 'From' (04/30/2026 09:00 PM) and 'To' (05/19/2026 11:58 AM), a 'Run Report' button, and several summary cards showing metrics such as Total Orders (0), Completed (0), Avg OEE (—), Total Downtime (0 min), Planned Qty Total (0), Actual Qty Total (0), and Overall Variance (+0). At the bottom, there is an 'Order Detail' table with columns for DATE, ORDER NO, PRODUCT, BATCH, PLANNED, ACTUAL, VARIANCE, DOWNTIME, and STATUS.

Production Reports tab with report type selector, date filters, and export options.

Standard reports available:

Report	Description
Production Summary	Orders completed, quantities produced, by period and work center
Work Order Status	All open and closed work orders with progress
Material Consumption	Materials issued vs. BOM standard
Labour Efficiency	Actual hours vs. standard routing hours
OEE Report	OEE, Availability, Performance, Quality by work center
Waste & Yield Summary	Yield rates and waste costs by product and period

Reports can be filtered by date range, work center, product, and production plan. Export to CSV is available on all reports.

Variance Analysis

Tab key: variance

Route: /dashboard/production/variance/page

Variance Types

Variance	Favourable (F)	Adverse (A)
Material Usage	Used less material than standard	Used more than standard
Material Price	Materials cost less than standard	Materials cost more than standard
Labour Efficiency	Fewer hours than standard	More hours than standard
Labour Rate	Labour rates lower than standard	Labour rates higher than standard
Overhead Absorption	More overhead absorbed than expected	Under-absorbed (low production volume)

Variance Report

For each work order or production period, the variance report shows:

- Standard cost per unit (from BOM + Routing)
- Actual cost per unit (from materials issued + hours logged)
- Total variance (KES)
- Variance percentage
- Variance type breakdown

Variances above the configured threshold (e.g. $\pm 5\%$) are flagged for management review and root cause documentation.

Demo Data — OEE, Downtime, and Waste Seed (TASK-015.2)

The system ships with seeded OEE, downtime, and waste/yield data. Pages show real metrics, not empty states.

Seeded OEE Records

`OEERecord` entries seeded per work center with:

- `availability_pct`, `performance_pct`, `quality_pct`, `oee_pct` values
- `planned_time_min`, `downtime_min`, `planned_production`, `actual_production`, `good_units`
- Multiple work centers covered; one record per shift period per work center

OEE scores are representative FMCG line performance values. Some work centers are seeded below the 85% world-class threshold to demonstrate the alert highlighting.

Seeded Downtime Logs

`DowntimeLog` entries seeded with:

- `start_time`, `end_time`, `duration_min` per event
- `downtime_category` (UNPLANNED, PLANNED, SETUP, CHANGEOVER, BREAKDOWN)
- `root_cause` description
- Work center assignment

Downtime feeds directly into OEE Availability calculations — $\text{availability_pct} = (\text{planned_time} - \text{total_downtime}) / \text{planned_time}$.

Seeded Waste Records

`WasteRecord` entries seeded with:

- `scrap_qty`, `waste_category`
- `good_qty`, `total_qty`, `yield_pct` per work order
- Waste categories: STARTUP, SHUTDOWN, QUALITY_REJECT, OVERFLOW

Yield variance shown against the BOM standard loss percentage.

Idempotency

All TASK-015.2 seed helpers use `_get_or_create_*` patterns. Re-running Docker startup does not create duplicate records.

Compliance & Labelling

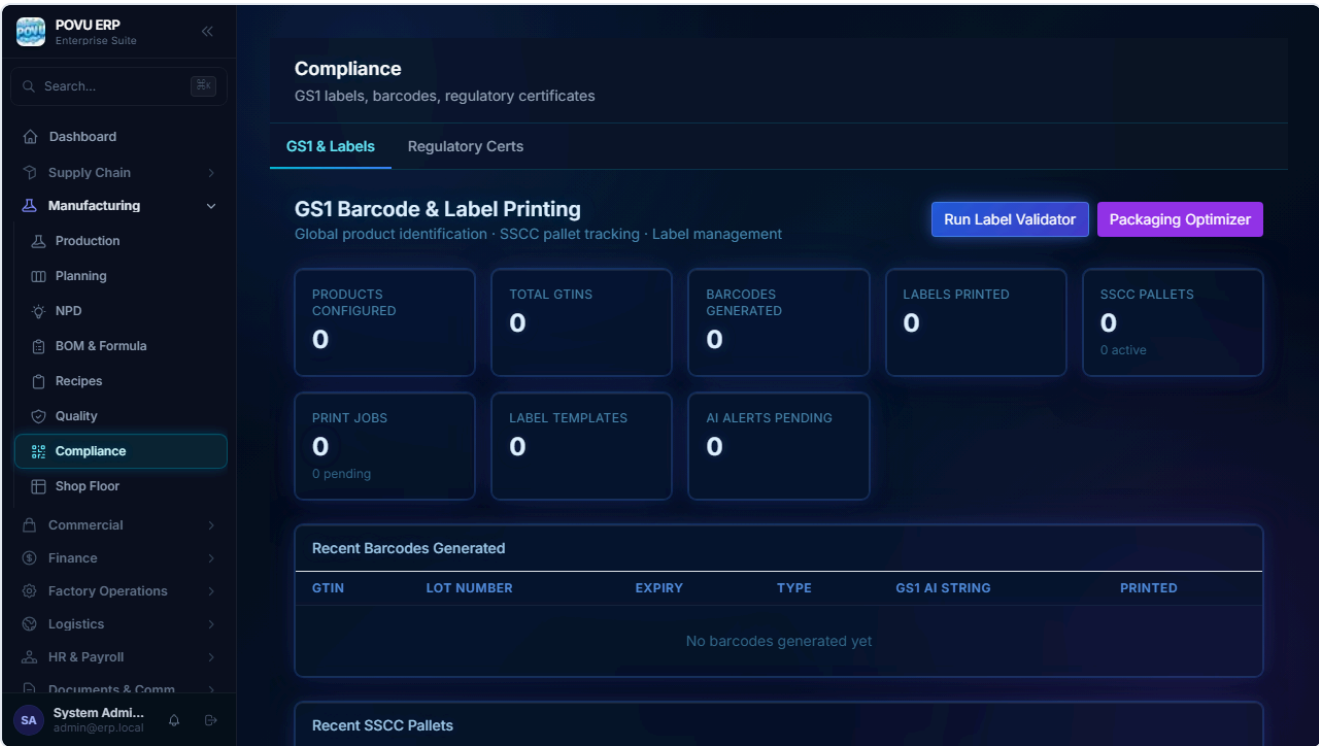
Route: /dashboard/compliance

Permission required: gs1.view

Workspace tabs: GS1 & Labels, Regulatory Certs

What It Does

The Compliance module manages product labelling and regulatory certification requirements. It covers GS1 barcode generation and label printing, and provides a consolidated view of all regulatory certificates linked to products and suppliers.



Compliance workspace showing the GS1 & Labels tab with label validator and packaging optimizer actions.

GS1 & Labels Tab

Tab key: gs1

Route: /dashboard/gs1/page (embedded)

GS1 Overview

GS1 is the global standard for product identification codes. The ERP uses GS1 identifiers for:

- **GTIN** (Global Trade Item Number) — 14-digit product barcode
- **SSCC** (Serial Shipping Container Code) — pallet and logistics unit identification
- **GLN** (Global Location Number) — identifies company locations (factory, warehouse)
- **GS1-128** — variable-length barcode for batch/lot and expiry date encoding on labels

GS1 barcodes are mandatory for products distributed through modern trade (supermarkets, export) that use barcode scanning at point of sale and goods receipt.

Label Types

Label Type	Usage
Product Label	Consumer-facing label on individual units
Case Label	Outer carton label (case GTIN + batch + expiry)
Pallet Label	SSCC pallet label for logistics
Shelf Label	Price and PLU label for trade
QC Sample Label	Laboratory sample identification

GS1 Label Fields

Each label record includes:

- `product_id` — linked product
- `gtin` — 14-digit GS1 product code
- `batch_no` — production batch number
- `expiry_date` — product expiry date
- `net_weight` — declared net weight (g or mL)
- `country_of_origin` — ISO country code
- `label_version` — label design version number

Label Printing Flow

1. Navigate to **GS1 & Labels** tab
2. Select product(s) and batch(es)
3. Choose label type and quantity
4. Click **Print** — sends to configured label printer (ZPL-compatible or PDF)

Labels can also be exported as PDFs for proofing before print.

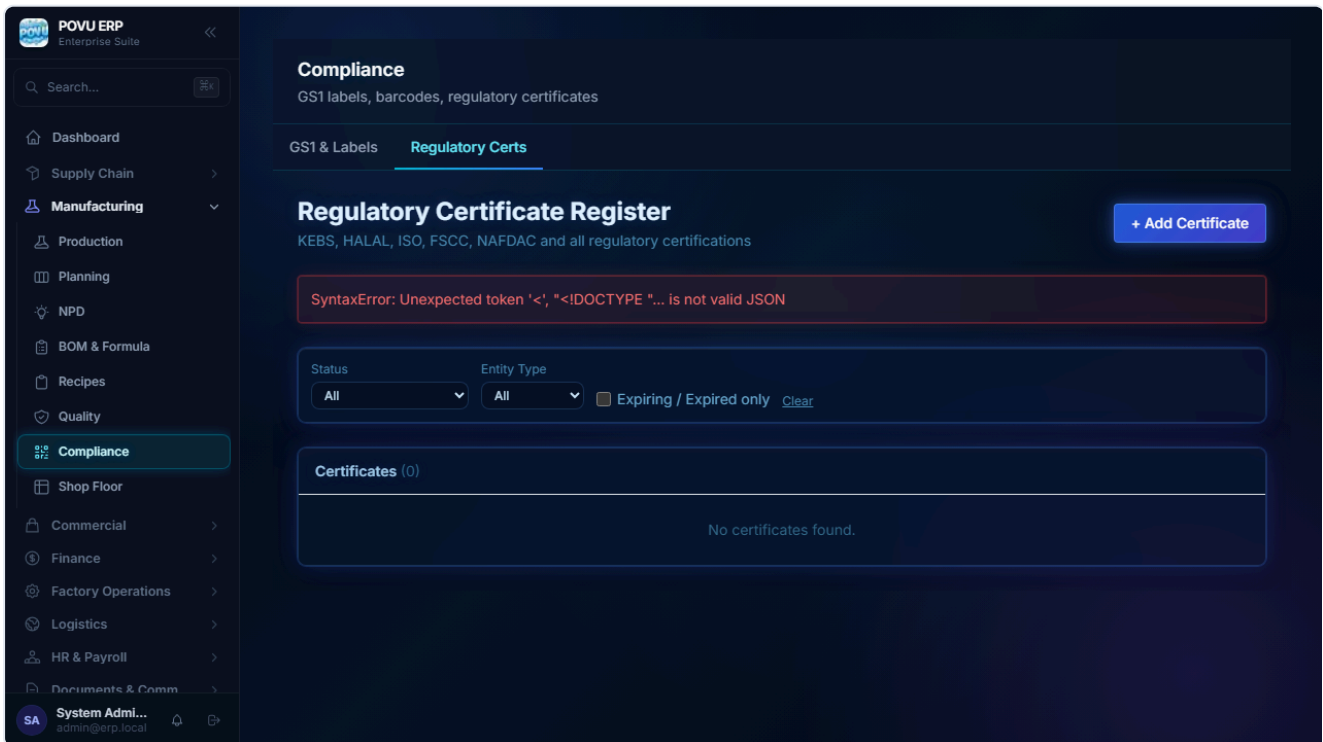
Allergen Declarations on Labels

Allergen information from the **Allergen** tab (Quality module) is pulled automatically into product labels. The ERP enforces that labels cannot be approved for printing until the allergen declaration has been reviewed for the current BOM version.

Regulatory Certs Tab

Tab key: `regulatory-certs`

Route: `/dashboard/quality/certificates/page` (embedded)



Regulatory Certs tab listing all certificates with type, status, expiry, and linked entity.

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Enterprise Suite

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Dashboard

Supply Chain

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Quality

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Finance

Factory Operations

Logistics

HR & Payroll

Documents & Comm

SA System Admin...
admin@erp.local

Compliance
GS1 labels, barcodes, regulatory certificates

GS1 & Labels **Regulatory Certs**

Regulatory Certificate Register
KEBS, HALAL, ISO, FSCC, NAFDAC and all regulatory certifications

+ Add Certificate

SyntaxError: Unexpected token '<', '<!DOCTYPE '... is not valid JSON

Status: All Entity Type: All ☐ Expiring / Expired only [Clear](#)

Register Certificate

Entity Name * e.g. Mango Juice 500ml or Plant A Authority * KEBS

Entity Type: PRODUCT Certificate Number: e.g. KEBS/SB/2026/123 Issued Date: mm/dd/yyyy

Expiry Date: mm/dd/yyyy Country: Kenya Status: ACTIVE

Document URL

Add Certificate form — fields for cert type, number, issuing body, issued/expiry dates, product or supplier.

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Compliance
GS1 labels, barcodes, regulatory certificates

GS1 & Labels **Regulatory Certs**

Regulatory Certificate Register
KEBS, HALAL, ISO, FSCC, NAFDAC and all regulatory certifications

+ Add Certificate

SyntaxError: Unexpected token '<', '<!DOCTYPE '... is not valid JSON

Status: All Entity Type: All ☐ Expiring / Expired only [Clear](#)

ACTIVE
PENDING RENEWAL
EXPIRED
SUSPENDED
REVOKED
PENDING INITIAL

No certificates found.

Status filter dropdown — ACTIVE, EXPIRED, SUSPENDED, PENDING options.



Entity Type filter dropdown — Product, Supplier, Material options.

The Regulatory Certs tab is the same content as **Quality** → **Certificates**. It is surfaced here in the Compliance workspace for users whose primary focus is regulatory compliance rather than QC operations.

Certificate Types

Certificate Type	Description
KEBS	Kenya Bureau of Standards product certification
KEPHIS	Kenya Plant Health Inspectorate Service (for plant-derived products)
KRA AEO	Kenya Revenue Authority Authorised Economic Operator
HALAL	Halal certification from an accredited body
KOSHER	Kosher certification
Organic	Organic certification from an accredited certifier
ISO 22000	Food safety management system certification
FSSC 22000	Food Safety System Certification 22000
GMP	Good Manufacturing Practice certificate
Export Certificate	Certificate of conformity for export markets

Certificate Record Fields

Field	Description
<code>cert_type</code>	Type of certificate (from list above)
<code>cert_number</code>	Certificate reference number
<code>issuing_body</code>	Organisation that issued the certificate
<code>issued_date</code>	Date of issue
<code>expiry_date</code>	Certificate expiry date
<code>product_id</code>	Product(s) covered (or supplier-level)
<code>supplier_id</code>	Supplier (for incoming material certs)
<code>status</code>	ACTIVE / EXPIRED / SUSPENDED / PENDING
<code>document</code>	Attached PDF of the certificate

Certificate Expiry Alerts

The system flags certificates expiring within 60 days (configurable). Expired certificates are shown with an EXPIRED badge in red. Quality managers receive dashboard alerts for approaching expiries so renewal can be initiated in time.

Certificate Linking

Certificates can be linked to:

- **Products** — for product-level certifications (e.g. KEBS mark)
- **Suppliers** — for supplier audit certificates (e.g. GMP, ISO 22000)
- **Materials** — for material-level compliance (e.g. Halal certificate for an ingredient)
- **QC Inspections** — attaching supplier COA (Certificate of Analysis) to an incoming inspection

Compliance Workflow

Label Approval Flow

1. R&D finalises BOM + Recipe (APPROVED status)
2. QA reviews allergen declaration
3. Regulatory submits KEBS / product registration if required
4. Marketing approves label artwork

5. Compliance officer approves label for print
6. Label is printed and barcode is verified at the scanner

Certificate Renewal Flow

1. Alert generated (60 days before expiry)
2. Quality manager assigns renewal task
3. Renewal submitted to issuing body
4. New certificate received → uploaded to ERP
5. Old certificate archived
6. `expiry\_date` updated on the new record